Fall 2026 Lecture Notes

# STAT 8561

## Linear Statistical Analysis I

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# Preface

The topic of this course includes statistical inference, Multivariate normal distribution, distribution of quadratic forms, linear models, regression models and experimental design models.

The course moves from Gaussian linear-model theory to regression, experimental comparisons, and model checking. A short final unit extends the same design-matrix framework to logistic and Poisson regression. Advanced extensions are collected under **Optional topic**.

**Prerequisites**

Math 4751/6751 Mathematical Statistics I and Math 4752/6752 Mathematics and Statistics II.

**Instructor**

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## Office Hours

Wednesdays, 9:00 a.m. to 12:00 p.m., or by appointment.

## Grade Distribution

- Homework – 40%
- Exam 1 – 20%
- Exam 2 – 20%
- Final Project – 20%
- Attendance – *7%

## Assignment

- ☒ Assignment 1: September 11, 2026
- ☒ Assignment 2: September 30, 2026
- ☐ Assignment 3: TBA

## Exam

☒ Exam 1: October 14, 2026
☐ Exam 2, November 18, 2026

## Project

**Final assessment decision:** The final assessment will be a **Final Project**, selected by a class vote of **20 to 5**.

See Instruction for the project description, requirements, and grading criteria.

☐ Report: Due on December 13, 2026 (20%)

## Topics and Corresponding Lectures

Those chapters are based on the lecture notes. This part will be updated frequently.

| Status | Chapter | Topic | Lecture |
|---|---|---|---|
| Done | | Welcome and Overview | 1 |
| Done | Ch. 1 | Math Notation | 2–3 |
| Done | Ch. 2 | Least Squares Estimation | 3–6 |
| Done | Ch. 3 | Inference on Least Square Estimation | 7–10 |
| Next | Ch. 4 | Model Comparison, Partial F Tests, and ANOVA | 11– |
| Done | | Exam 1 | 15 |
| Next | Ch. 5 | Multiple Regression and Categorical Predictors | TBA |
| Exam | | Exam 2 | 25 |

### Optional topic

Students interested in extensions can read Quantile Regression and Mixed-Effects Models. These readings include explanations, graphical examples, R code, and further reading; they are supplementary to the scheduled chapters.

## Recommended Textbooks

- Ding, Peng. *Linear Model and Extensions*. CRC Press, 2027. arXiv link
- Montgomery, Douglas C., Elizabeth A. Peck, and G. Geoffrey Vining. *Introduction to Linear Regression Analysis*. John Wiley & Sons, 2021. (Montgomery et al. 2021)
- Seber, George AF, and Alan J. Lee. *Linear Regression Analysis*. John Wiley & Sons, 2003. (Seber and Lee 2003)

# Side Readings

- Montgomery, Douglas C. (2017). *Design and Analysis of Experiments*. John Wiley & Sons. (Montgomery 2017)

# Introduction

This course develops a common language for statistical modelling: specify the research question, construct a design matrix, estimate a mean structure, quantify uncertainty, and check whether the assumptions are credible.

The core sequence covers matrix notation and least squares; normal distributions and quadratic forms; inference and model comparison; categorical predictors, contrasts, and experimental design; diagnostics, transformations, and weighting; and an introduction to generalized linear models for binary and count responses.

The final GLM unit keeps the same linear predictor while extending the response distribution and mean link. Logistic and Poisson regression provide practical examples. The **Optional topic** group contains supplementary material on rank-deficient models, automatic selection, correlated errors/GLS, GLM likelihood and computation, quantile regression, and mixed-effects models. Quantile regression and mixed-effects models are self-study readings for students who want to explore distributional features or grouped observations.

Lectures, homework, two in-class exams, and a final assessment develop both mathematical reasoning and reproducible analysis in R. Core tasks include interpreting estimates and intervals, identifying the experimental unit, and explaining the limits of an analysis.

To illustrate the concepts we will be covering in this course, let's consider a simple example. Suppose we have a data set of heights and weights of individuals. We want to understand the relationship between height and weight, and we can use linear regression to model this relationship.

### Example 1: First Example

```
# Load necessary libraries
library(ggplot2)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
# Create a sample data set

set.seed(8561)

theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

n <- 100
heights <- rnorm(n, mean = 170, sd = 10)
weights <- 0.5 * heights + rnorm(n, mean = 0, sd = 5)
df <- data.frame(heights, weights)
head(df, 5)
```

```
   heights  weights
1 179.2812 88.01599
2 168.0474 76.10017
3 178.3029 83.46016
4 178.7006 82.01763
5 172.6678 92.84592
```

```r
# Fit a linear regression model
model <- lm(weights ~ heights, data = df)
# Summarize the model
summary(model)
```

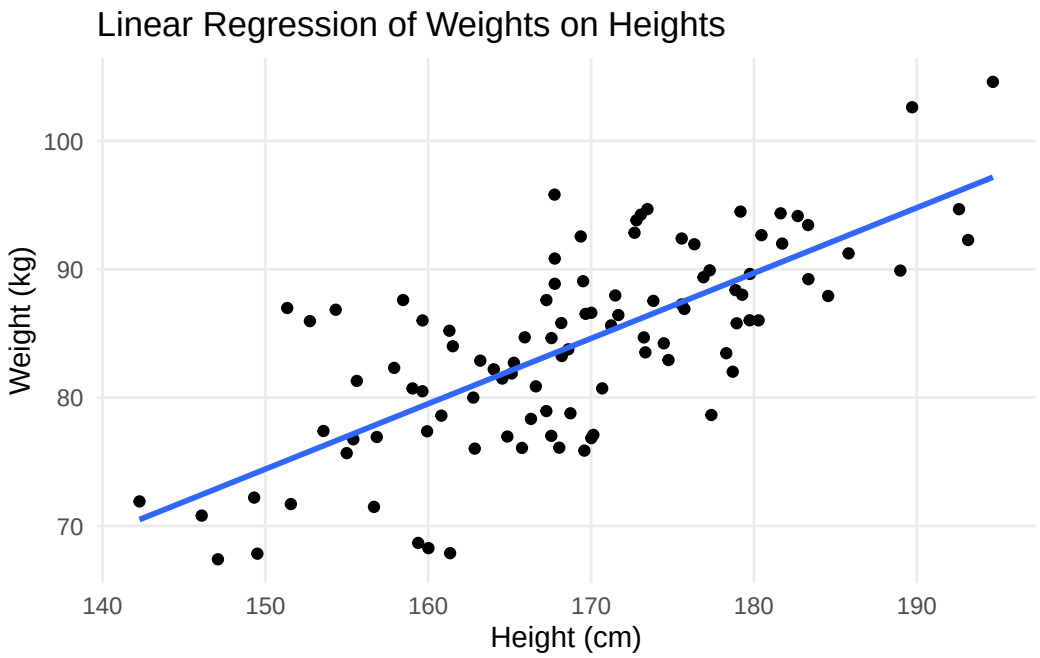
```
Call:
lm(formula = weights ~ heights, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-12.3303  -3.8998  -0.2191   3.2380  12.3478

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.92852    8.22226  -0.235    0.815
heights      0.50907    0.04857  10.481   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.251 on 98 degrees of freedom
Multiple R-squared:  0.5285,    Adjusted R-squared:  0.5237
F-statistic: 109.9 on 1 and 98 DF,  p-value: < 2.2e-16
```

```r
# Plot the data and the fitted line
ggplot(df, aes(x = heights, y = weights)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
  labs(title = "Linear Regression of Weights on Heights",
       x = "Height (cm)",
       y = "Weight (kg)")
```




In this example, we generated a data set of heights and weights, fitted a linear regression model to the data, and visualized the relationship between height and weight. This is just a simple example, but it illustrates the types of analyses we will be doing in this course. We will be covering much more complex models and data sets as we progress through the semester.

Throughout the course, we will be using **R** programming software.

## Section 1

# Matrix Representation and Notation

In this chapter, we introduce the mathematical language and statistical framework that will be used throughout the course. Our focus is on the notation of vectors and matrices, vectors of random variables, expectation and covariance operators, and the basic form of the linear regression model.

**Learning Objectives**

By the end of this chapter, students should be able to:

- use standard matrix notation for linear statistical models;
- distinguish between scalars, vectors, matrices, random variables, and random vectors;
- compute expectations and covariance matrices for random vectors;
- interpret the linear regression model in matrix form;
- understand why projection and least squares will play a central role in this course.

**Reading**

Recommended reading for this chapter:

- Seber and Lee (2003), Chapter 1:
  - 1.1 Notation
  - 1.2 Statistical Models
  - 1.3 Linear Regression Models
  - 1.4 Expectation and Covariance Operators
- Optional preview:
  - Chapter 2: Multivariate Normal Distribution

## Section 1.1 Why Linear Statistical Analysis?

Linear statistical analysis is one of the central foundations of graduate statistics. Many methods that at first look different are built on the same underlying structure:

- regression
- analysis of variance
- analysis of covariance
- prediction
- model comparison

- and parts of generalized linear modelling

A major goal of this course is to see these topics under a unified framework.

## Section 1.2 A unifying point of view

A large part of the course can be summarized by the model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

- $\mathbf{Y}$ is a response vector,
- $\mathbf{X}$ is a design matrix,
- $\boldsymbol{\beta}$ is an unknown parameter vector,
- $\boldsymbol{\varepsilon}$ is a random error vector.

This compact expression contains a great deal of statistical structure. Over the semester, we will study how to estimate $\boldsymbol{\beta}$, quantify uncertainty, test hypotheses, diagnose model failures, and make predictions.

## Section 1.3 Basic Notation

### Scalars, vectors, and matrices

We use the following conventions throughout the course:

- scalars are written in ordinary (nonbold) type, such as $a$, $b$, $n$;
- vectors are written in boldface, such as $\mathbf{x}$, $\mathbf{y}$, $\boldsymbol{\beta}$, and the random response vector $\mathbf{Y}$;
- matrices are written in bold uppercase letters, such as $\mathbf{X}$, $\mathbf{A}$;
- random variables are often written in uppercase letters, such as $Y$;
- realizations of random variables are written in lowercase letters, such as $y$.

Individual entries remain scalars: $Y_i$ is an entry of $\mathbf{Y}$, $x_{ij}$ is an entry of $\mathbf{X}$, and $\beta_j$ is an entry of $\boldsymbol{\beta}$. The scalar mean $\bar{Y}$ and variance $\sigma^2$ are not bold. We write the zero vector or matrix as $\mathbf{0}$ and the all-ones vector as $\mathbf{1}$; a scalar zero or one remains 0 or 1.

### Univariate case

A random variable is **univariate** if it takes values in one dimension. We write

$$X \in \mathbb{R}.$$

A realization, or observed value, of $X$ is denoted by $x$.

Some common quantities associated with a univariate random variable $X$ include:

- **Mean**
$$\mu = \mathbb{E}(X).$$
- **Variance**
$$\sigma^2 = \mathrm{Var}(X) = \mathbb{E}\left[(X-\mu)^2\right].$$
- **$k$th moment**
$$\mathbb{E}(X^k).$$
- **Cumulative distribution function (CDF)**
$$F(x) = P(X \le x).$$
- **Probability density function (PDF)**, for a continuous random variable,
$$f(x).$$
- **Probability mass function (PMF)**, for a discrete random variable,
$$p(x) = P(X = x).$$
- **Quantile**
$$q_\alpha = F^{-1}(\alpha),$$
where $0 < \alpha < 1$.

For a random sample of size $n$, we write

$$X_1, X_2, \dots, X_n,$$

with observed values

$$x_1, x_2, \dots, x_n.$$

The sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

and the sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

### Multivariate case

A column vector in $\mathbb{R}^n$ is written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \ldots, x_n).$$

An $n \times p$ matrix is written as

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}.$$

**Transpose, inverse, and rank**

If $\mathbf{A}$ is a matrix, then:

- $\mathbf{A}^\top$ denotes its transpose;
- $\mathbf{A}^{-1}$ denotes its inverse, when it exists;
- $\text{rank}(\mathbf{A})$ denotes its rank;
- $\mathbf{I}_n$ denotes the $n \times n$ identity matrix.

### 1.3.1 Inner product and norm

**Definition 1: inner product in Eucludean space**

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

**Definition 2: Euclidean norm**

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

These ideas are fundamental because least squares estimation is based on minimizing squared Euclidean distance.

**Definition 3: Random Vectors**

A random vector is a vector whose entries are random variables. For example,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

is an $n$-dimensional random vector.

In linear models, the response is naturally treated as a random vector.

**Definition 4: Mean vector**

The mean vector of $\mathbf{Y}$ is

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} \mathbb{E}[Y_1] \\ \mathbb{E}[Y_2] \\ \vdots \\ \mathbb{E}[Y_n] \end{pmatrix}.$$

We often write $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}$.

**Definition 5: Covariance matrix**

The covariance matrix of $\mathbf{Y}$ is

$$\mathrm{Var}(\mathbf{Y}) = \mathbb{E}\left[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top\right].$$

Equivalently,

$$\mathrm{Var}(\mathbf{Y}) = \mathbb{E}[\mathbf{Y}\mathbf{Y}^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top.$$

**Example 1**

Suppose $\mathbf{Y} = (Y_1, Y_2, Y_3)$.
Then the corresponding covariance matrix is

$$\mathrm{Var}(\mathbf{Y}) = \begin{pmatrix} \mathrm{Var}(Y_1) & \mathrm{Cov}(Y_1, Y_2) & \mathrm{Cov}(Y_1, Y_3) \\ \mathrm{Cov}(Y_2, Y_1) & \mathrm{Var}(Y_2) & \mathrm{Cov}(Y_2, Y_3) \\ \mathrm{Cov}(Y_3, Y_1) & \mathrm{Cov}(Y_3, Y_2) & \mathrm{Var}(Y_3) \end{pmatrix}.$$

### 1.3.2 Expectation and Covariance Operators

**Property 1: Linearity of expectation**

If $\mathbf{A}$ is a constant matrix and $\mathbf{b}$ is a constant vector, then

$$\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{Y}] + \mathbf{b}.$$

This is one of the most useful identities in the course.

**Property 2: Covariance of linear transformations**

If $\mathbf{A}$ is a constant matrix, then

$$\mathrm{Var}(\mathbf{A}\mathbf{Y}) = \mathbf{A}\,\mathrm{Var}(\mathbf{Y})\,\mathbf{A}^\top.$$

More generally, if $\mathbf{Y}$ and $\mathbf{Z}$ are random vectors, then

$$\mathrm{Cov}(\mathbf{AY}, \mathbf{BZ}) = \mathbf{A}\,\mathrm{Cov}(\mathbf{Y}, \mathbf{Z})\,\mathbf{B}^\top.$$

These formulas are essential for deriving the variance of estimators later.

**Example 2: independent components**

If $Y_1, \ldots, Y_n$ are independent and each has variance $\sigma^2$, then

$$\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

This is the most common starting assumption in classical linear regression.

## Section 1.4 Statistical Models

A statistical model is a set of probability distributions that may plausibly describe the data-generating mechanism.

### 1.4.1 General idea

Suppose we observe data $y$ from a random quantity $Y$. A model introduces assumptions about the distribution of $Y$, often indexed by an unknown parameter $\theta$.

For example:

$$Y \sim N(\mu, \sigma^2)$$

with unknown parameters $\mu$ and $\sigma^2$.

In regression, the model is not only about the marginal distribution of the response but also about how the mean changes with explanatory variables.

### 1.4.2 Deterministic part and random part

A useful way to think about a statistical model is:

$$\text{data} = \text{systematic part} + \text{random part}.$$

For linear regression, this becomes

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Here,

- $\beta_0 + \beta_1 x_i$ is the systematic part;
- $\varepsilon_i$ is the random part.

# Section 1.5 The Linear Regression Model

Perhaps the most simple model is the *simple linear regression* where there are only **one** independent variable and **one** response variable.

**Definition 6: Simple linear regression**

A simple linear regression model is defined as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

with the typical assumption as

$$\mathbb{E}[\varepsilon_i] = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2, \qquad \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j.$$

**Matrix form**

We can write the model compactly as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \qquad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

This notation will allow us to treat simple regression, multiple regression, ANOVA, and ANCOVA in one common language.

Under the same assumption,

$$\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0} \qquad \text{and} \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

we have the following.

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}, \quad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

These are immediate consequences of the expectation and covariance rules above.

## 1.5.1 Geometry Preview

A central idea in linear regression is projection.

The fitted values $\hat{\mathbf{Y}}$ will later be obtained by projecting $\mathbf{Y}$ onto the column space of $\mathbf{X}$.

**Definition 7: Column space**

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^p\}.$$

This is the set of all mean vectors that the model can represent.

## Section 1.6 Why projection matters

The least squares estimator chooses $\hat{\boldsymbol{\beta}}$ so that

$$\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$$

is minimized.

So regression is not only algebra. It is also geometry.

**Example 3**

Suppose we observe the following data:

| $x_i$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y_i$ | 1 | 3 | 3 | 5 |

Then

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

We will learn in Chapter 2 — Least Squares Estimation that the least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For now, the main point is to understand how the model is written and how the data are represented in matrix form.

**R code**

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)
mod <- lm(y ~ x)
summary(mod)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4
-0.2  0.6 -0.6  0.2

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.2000     0.5292   2.268   0.1515
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9, Adjusted R-squared:   0.85
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

```r
# Inspect the design matrix, fitted values, and residuals.
model.matrix(mod)
```

```
  (Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr(,"assign")
[1] 0 1
```

```r
fitted(mod)
```

```
  1   2   3   4
1.2 2.4 3.6 4.8
```

```r
residuals(mod)
```

```
   1    2    3    4
-0.2  0.6 -0.6  0.2
```

```r
ggplot(data.frame(x, y), aes(x, y)) +
  geom_point(size = 2.5) +
  geom_abline(intercept = coef(mod)[1], slope = coef(mod)[2],
              colour = "steelblue4", linewidth = 0.8) +
  labs(x = "x", y = "y")
```

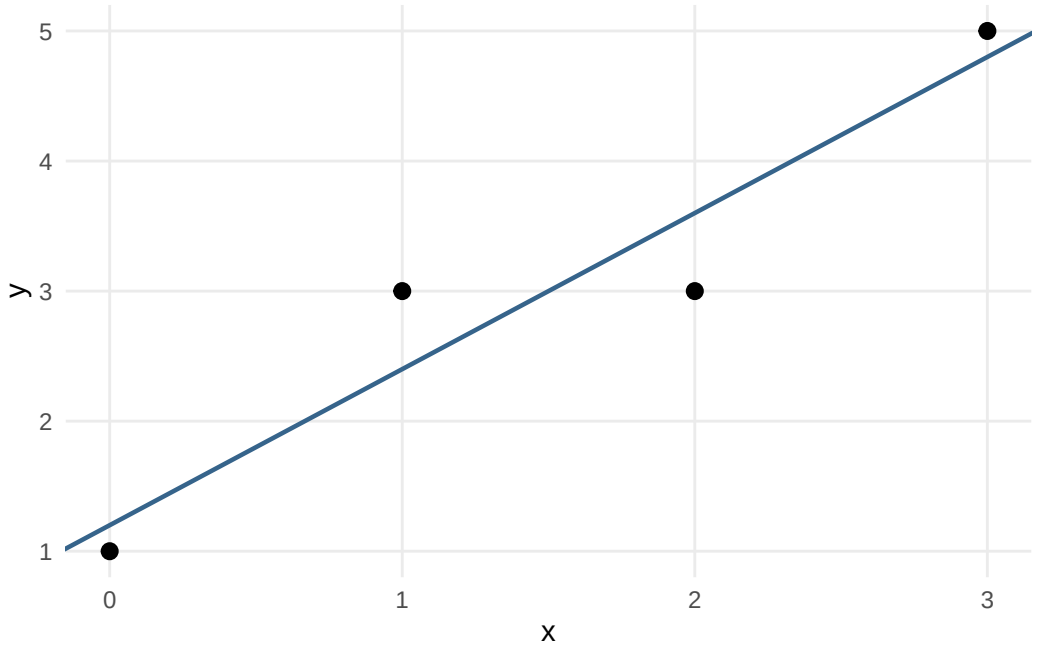



Figure 1.1: Four observations and their least-squares fitted line.

## Section 1.7 Discussion

1. Why is it helpful to write regression models in matrix form rather than only scalar notation?
2. What does the covariance matrix tell us that separate variances do not?
3. What is the interpretation of the column space of $\mathbf{X}$?
4. In what sense is regression a projection problem?

## Section 1.8 Practice Problems

### Conceptual

1. Explain the difference between a random variable and a random vector.
2. Explain why $\mathrm{Var}(\mathbf{Y})$ must be a symmetric matrix.
3. Give an example of a statistical model outside regression.

### Computational

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$$

with

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathrm{Var}(\mathbf{Y}) = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Compute:

1. $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$
2. $\mathrm{Var}(\mathbf{AY})$

**Regression setup**

For the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

write out the matrices $\mathbf{Y}$, $\mathbf{X}$, $\boldsymbol{\beta}$, and $\boldsymbol{\varepsilon}$ for $n = 5$ observations.

## Section 1.9 Take home Practice

Complete the following:

- review matrix multiplication and transpose rules;
- derive $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$ from first principles;
- derive $\mathrm{Var}(\mathbf{AY})$ using the definition of covariance;
- write the simple linear regression model in matrix form for a dataset of your choice;
- fit a simple regression in R and report:
- the estimated coefficients,
- fitted values,
- residuals,
- and a scatterplot with the fitted line.

## Section 1.10 Summary

This chapter introduced the notation and basic probabilistic tools needed for the rest of the course. We defined random vectors, mean vectors, covariance matrices, and the matrix form of the linear regression model. These ideas will support everything that follows.

Chapter 2 — Least Squares Estimation, we will study least squares estimation and the geometry of projection in more detail.

For some optional review of the matrix algebra, see Matrix Algebra Review.

## Section 2

# Least Squares Estimation

In this chapter, we study one of the fundamental ideas in linear models: **least squares estimation**.

Recall that the linear model is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

The unknown parameter vector $\boldsymbol{\beta}$ determines the mean structure of the response. Given observed data, our first goal is to estimate $\boldsymbol{\beta}$.

The central idea of least squares is simple:

> Choose the value of $\boldsymbol{\beta}$ for which the fitted response $\mathbf{X}\boldsymbol{\beta}$ is as close as possible to the observed response $\mathbf{Y}$.

This optimization problem leads to the **normal equations**, the **least squares estimator**, and a useful geometric interpretation in terms of **orthogonal projection**.

> Learning Objectives
>
> By the end of this chapter, students should be able to:
>
> - formulate least squares estimation as an optimization problem;
> - define the residual sum of squares;
> - derive the normal equations;
> - obtain the least squares estimator when $\mathbf{X}$ has full column rank;
> - explain the role of the column space $\mathcal{C}(\mathbf{X})$;
> - interpret fitted values as an orthogonal projection;
> - explain why residuals are orthogonal to the model space;
> - define and interpret the hat matrix and residual-maker matrix;
> - verify symmetry and idempotence of projection matrices;
> - explain the decomposition $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$;
> - distinguish the uniqueness of $\hat{\boldsymbol{\beta}}$ from the uniqueness of the fitted values.

**Reading**

Recommended reading for this week:

- Seber and Lee:
  - Chapter 3: sections on least squares estimation

- Montgomery, Peck, and Vining:
  - introductory sections on estimation in linear regression

**Recap of the Linear Model**

Recall from Week 1, the linear model can be written as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

- $\mathbf{Y}$ is an $n \times 1$ response vector;
- $\mathbf{X}$ is an $n \times p$ design matrix;
- $\boldsymbol{\beta}$ is a $p \times 1$ parameter vector;
- $\boldsymbol{\varepsilon}$ is an $n \times 1$ error vector.

The design matrix may be written as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix},$$

where $\mathbf{x}_i \in \mathbb{R}^p$ contains the predictors associated with observation $i$.
Thus,

$$Y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Under the classical linear model, we often assume

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n.$$

Consequently,

$$\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \qquad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

**Least Squares Does Not Require a Distributional Assumption**

The least squares estimator is defined through the optimization problem

$$\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 .$$

No distributional assumption on $\boldsymbol{\varepsilon}$ is required to define or compute the least squares estimator.
Assumptions such as

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$$

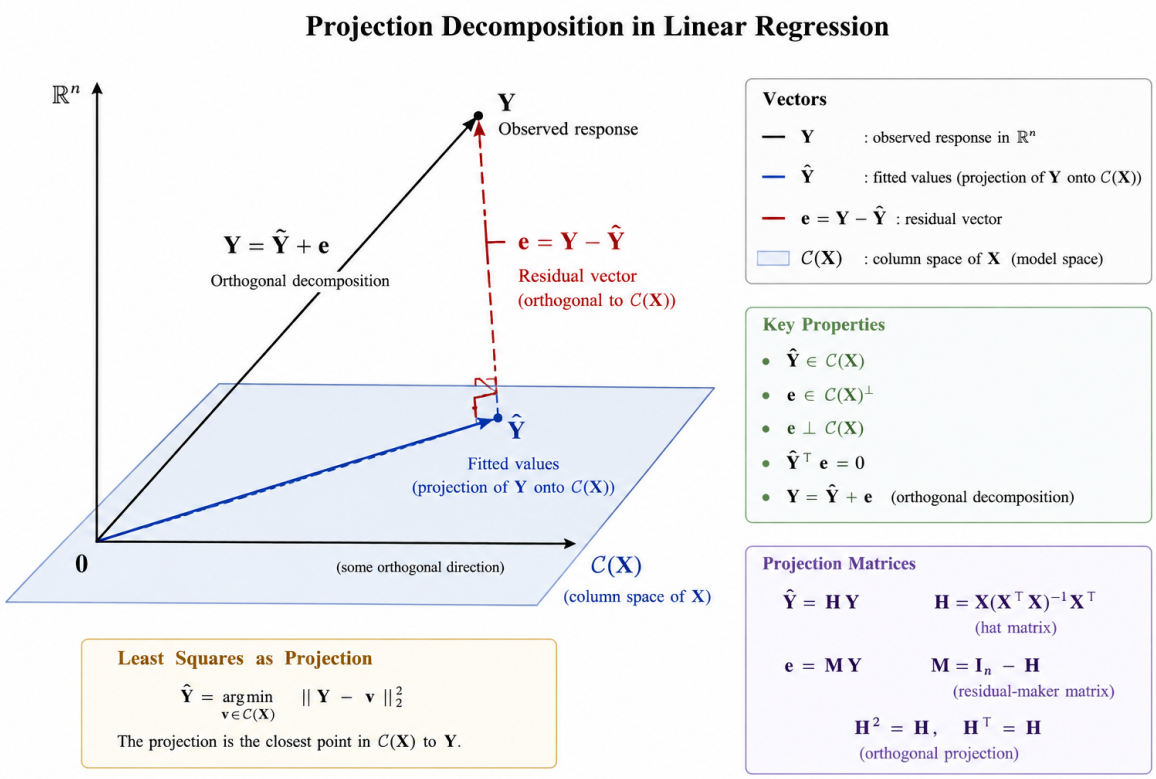



Figure 2.1: Geometric interpretation of least squares as an orthogonal projection.

are needed when we study the **statistical properties** of the estimator.
Normality will be introduced later when we study exact finite-sample inference.

## Section 2.1 The Least Squares Criterion

### Motivation

For any candidate value $\boldsymbol{\beta}$, the corresponding fitted vector is

$$\mathbf{X}\boldsymbol{\beta}.$$

The discrepancy between the observed response $\mathbf{Y}$ and this candidate fit is

$$\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}.$$

A natural idea is to choose $\boldsymbol{\beta}$ so that this discrepancy is as small as possible.

To measure its size, we use the squared Euclidean norm.

### Residual Sum of Squares

The **residual sum of squares** associated with a candidate $\boldsymbol{\beta}$ is

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 .$$

Equivalently,

$$S(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).$$

In scalar notation,

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left(Y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\right)^2 .$$

This last expression explains the name **least squares**: we choose $\boldsymbol{\beta}$ to make the sum of the squared discrepancies as small as possible.

We therefore define

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} S(\boldsymbol{\beta}).$$

**Definition 2.1** (Least Squares Estimator)
A **least squares estimator** is any value $\hat{\boldsymbol{\beta}}$ satisfying

$$S(\hat{\boldsymbol{\beta}}) \le S(\boldsymbol{\beta})$$

for every $\boldsymbol{\beta} \in \mathbb{R}^p$.

**Example 2.1** (LSE for Simple Linear Regression)
Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

where

$$\mathbb{E}[\varepsilon_i] = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2.$$

The fitted value at $x_i$ is

$$\hat{Y}_i = \beta_0 + \beta_1 x_i.$$

The corresponding residual is

$$e_i = Y_i - \beta_0 - \beta_1 x_i.$$

The least squares method chooses $\beta_0$ and $\beta_1$ to minimize the **sum of squared residuals**

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 x_i\right)^2 .$$

**Derivation of the Normal Equations**

To minimize $S(\beta_0, \beta_1)$, differentiate with respect to $\beta_0$ and $\beta_1$.
For $\beta_0$,

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^{n} \left( Y_i - \beta_0 - \beta_1 x_i \right).$$

Setting this equal to zero gives

$$\sum_{i=1}^{n} \left( Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right) = 0.$$

Therefore,

$$\sum_{i=1}^{n} Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i.$$

Dividing by $n$,

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x},$$

so

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}}.$$

For $\beta_1$,

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_i \left( Y_i - \beta_0 - \beta_1 x_i \right).$$

Setting this equal to zero gives

$$\sum_{i=1}^{n} x_i \left( Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i \right) = 0.$$

Hence,

$$\sum_{i=1}^{n} x_i Y_i = \hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2.$$

Substituting

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x},$$

we obtain

$$\sum_{i=1}^{n} x_i Y_i = \left( \bar{Y} - \hat{\beta}_1 \bar{x} \right) \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2.$$

Since

$$\sum_{i=1}^{n} x_i = n\bar{x},$$

we have

$$\sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y} = \hat{\beta}_1 \left( \sum_{i=1}^{n} x_i^2 - n\bar{x}^2 \right).$$

Using

$$\sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y}) = \sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y},$$

and

$$\sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2,$$

we obtain

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}}.$$

Therefore,

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}}.$$

**Alternative Notation**
It is common to define

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

and

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y}).$$

Then

$$\boxed{\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}}$$

and

$$\boxed{\hat{\beta}_0 = \bar{Y} - \frac{S_{xy}}{S_{xx}} \bar{x}}.$$

Thus, the fitted regression line is

$$\boxed{\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i}.$$

# Section 2.2 Derivation of the Least Squares Estimator in a matrix form

**Expansion of the Criterion** Starting from

$$S(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}),$$

we expand:

$$S(\boldsymbol{\beta}) = \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Because

$$\mathbf{Y}^\top \mathbf{X}\boldsymbol{\beta}$$

is a scalar,

$$\mathbf{Y}^\top \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y}.$$

Therefore,

$$\boxed{S(\boldsymbol{\beta}) = \mathbf{Y}^\top \mathbf{Y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.}$$

**Derivation of the Normal Equations**

Differentiate $S(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$:

$$\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

At a minimum, the gradient is zero, so

$$-2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}.$$

Thus,

$$\boxed{\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}.}$$

These are called the **normal equations**.

> **i** Why Are They Called the Normal Equations?
>
> The normal equations can be written as

$$\mathbf{X}^\top \left(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{0}.$$

Later we will define

$$\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Therefore,

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

This means that $\mathbf{e}$ is perpendicular, or **normal**, to the column space of $\mathbf{X}$.
Thus, the word *normal* is being used in its geometric sense.

**Why is the Stationary Point a Minimum?**

The Hessian of the least squares criterion is

$$\nabla^2 S(\boldsymbol{\beta}) = 2\mathbf{X}^\top \mathbf{X}.$$

For any $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X}\mathbf{a} = (\mathbf{X}\mathbf{a})^\top (\mathbf{X}\mathbf{a}) = \|\mathbf{X}\mathbf{a}\|_2^2 \geq 0.$$

Therefore, $\mathbf{X}^\top \mathbf{X}$ is positive semidefinite, and $S(\boldsymbol{\beta})$ is a convex function.

If $\mathbf{X}$ has full column rank, then for every $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{X}\mathbf{a} \neq \mathbf{0},$$

and hence

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X}\mathbf{a} > 0.$$

Thus, $\mathbf{X}^\top \mathbf{X}$ is positive definite and the least squares minimizer is unique.

**Theorem 1: Least Squares Estimator**
A vector $\hat{\boldsymbol{\beta}}$ minimizes

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$$

if and only if it satisfies the normal equations

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}.$$

If $\mathbf{X}$ has full column rank,

$$\operatorname{rank}(\mathbf{X}) = p,$$

then $\mathbf{X}^\top \mathbf{X}$ is invertible and the least squares estimator is unique:

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}.}$$

This is commonly called the **ordinary least squares estimator**, or **OLS estimator**.

**i When Is the Closed-Form Formula Valid?**

The expression

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

exists if and only if $\mathbf{X}$ has full column rank:

$$\text{rank}(\mathbf{X}) = p.$$

This means that the columns of $\mathbf{X}$ are linearly independent.
If they are linearly dependent, then $\mathbf{X}^\top \mathbf{X}$ is singular and the inverse does not exist.
We will return to this issue later.

**Example 2.2** (LSE for Simple Linear Regression continue)
As shown in Example 2.1, we know the expression of the LSE $\hat{\boldsymbol{\beta}}$. Here, we want to see its connection to the matrix Form.
For simple linear regression,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

The general least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}.$$

For simple linear regression,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix},$$

and

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n x_i Y_i \end{pmatrix}.$$

Therefore,

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y},$$

which gives exactly

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

and

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}}.$$

Hence,

The familiar simple linear regression formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ are special cases of the general matrix LSE

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

## Section 2.3 Geometric Interpretation of Least Squares

The algebraic derivation gives us the estimator. However, the geometry explains **why** least squares has its important properties.

**Recap: Column Space of the Design Matrix**

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^p\}.$$

Thus, $\mathcal{C}(\mathbf{X})$ is the set of all response vectors that can be represented exactly by the linear model.
If $\mathbf{X}$ has full column rank $p$, then

$$\dim\{\mathcal{C}(\mathbf{X})\} = p.$$

Because $\mathbf{X}$ has $n$ rows,

$$\mathcal{C}(\mathbf{X}) \subseteq \mathbb{R}^n.$$

### 2.3.1 Least Squares as Projection

The fitted value vector is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Because

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}),$$

least squares chooses the vector in $\mathcal{C}(\mathbf{X})$ that is closest to the observed vector $\mathbf{Y}$.

That is,

$$\hat{\mathbf{Y}} = \arg\min_{\mathbf{v}\in\mathcal{C}(\mathbf{X})} \|\mathbf{Y}-\mathbf{v}\|_2^2.$$

Therefore, $\hat{\mathbf{Y}}$ is the **orthogonal projection** of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

> So we can say that the fitted vector $\hat{\mathbf{Y}}$ is constrained to live in the model space generated by the predictors. Least squares finds the point in that space closest to the observed data vector $\mathbf{Y}$.

**Definition 2.2** (Orthogonality)
Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are said to be **orthogonal** if their inner product is zero:

$$\mathbf{u}^\top \mathbf{v} = 0.$$

We write

$$\mathbf{u} \perp \mathbf{v}.$$

Geometrically, two nonzero vectors are orthogonal if the angle between them is $90^\circ$.

**Definition 2.3** (Residual Vector)
After obtaining the LSE, define the **residual vector**

$$\boxed{\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}.}$$

The individual residuals are

$$e_i = Y_i - \hat{Y}_i, \qquad i = 1, \dots, n.$$

Geometrically, $\mathbf{e}$ is the component of $\mathbf{Y}$ that is orthogonal to the model space $\mathcal{C}(\mathbf{X})$.

**Proposition 2.1** (Orthogonality of the Residuals)
Recall that the LSE satisfies the normal equations $\mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top\mathbf{Y}$. If we rearrange, we have $\mathbf{X}^\top\left(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{0}$.
Since the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}},$$

we obtain the fundamental least squares property

$$\boxed{\mathbf{X}^\top\mathbf{e} = \mathbf{0}.}$$

To understand what this means, write the design matrix as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}^{(1)} & \mathbf{x}^{(2)} & \cdots & \mathbf{x}^{(p)} \end{pmatrix},$$

where $\mathbf{x}^{(j)}$ is the $j$th column of $\mathbf{X}$.
Then

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix} (\mathbf{x}^{(1)})^\top \mathbf{e} \\ (\mathbf{x}^{(2)})^\top \mathbf{e} \\ \vdots \\ (\mathbf{x}^{(p)})^\top \mathbf{e} \end{pmatrix} = \mathbf{0}.$$

Therefore,

$$(\mathbf{x}^{(j)})^\top \mathbf{e} = 0, \qquad j = 1, \dots, p.$$

Thus, the residual vector is orthogonal to every column of $\mathbf{X}$.
Because the column space $\mathcal{C}(\mathbf{X})$ consists of all linear combinations of the columns of $\mathbf{X}$, it follows that

$$\boxed{\mathbf{e} \perp \mathcal{C}(\mathbf{X}).}$$

**Intuition.** The columns of $\mathbf{X}$ represent all directions that the linear model is allowed to use. After least squares has chosen the best fitted value, the remaining error $\mathbf{e}$ has no component left in any of these directions. Otherwise, we could move the fitted value within $\mathcal{C}(\mathbf{X})$ and reduce the residual sum of squares further.

**Proposition 2.2** (Fundamental Orthogonal Decomposition)
The fitted response vector is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Therefore,

$$\boxed{\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}).}$$

In other words, $\hat{\mathbf{Y}}$ must lie in the space generated by the columns of $\mathbf{X}$.
Since

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}},$$

we can write

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.}$$

The two components satisfy

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$$

and

$$\mathbf{e} \in \mathcal{C}(\mathbf{X})^\perp.$$

Hence,

$$\boxed{\hat{\mathbf{Y}}^\top \mathbf{e} = 0.}$$

Geometrically, least squares projects $\mathbf{Y}$ onto the column space of $\mathbf{X}$:

$$\mathbf{Y} = \underbrace{\hat{\mathbf{Y}}}_{\text{part explained by the model}} + \underbrace{\mathbf{e}}_{\text{part orthogonal to the model space}}.$$

Because the two components are orthogonal, the Pythagorean theorem gives

$$\boxed{\|\mathbf{Y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\mathbf{e}\|^2.}$$

This geometric decomposition is the foundation of least squares regression and will later lead naturally to the decomposition of sums of squares in ANOVA.

**Exercise 2.1** (Computational Problems)
Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Compute $\mathbf{X}^\top \mathbf{X}$.
2. Compute $\mathbf{X}^\top \mathbf{Y}$.
3. Find $\hat{\boldsymbol{\beta}}$.
4. Compute $\hat{\mathbf{Y}}$.
5. Compute the residual vector $\mathbf{e}$.
6. Verify that $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$.

**Solution 2.1** (Solution for Exercise 2.1)
**Hand calculation**
We have

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}.$$

Also,

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Therefore,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}.$$

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \begin{pmatrix} 7/6 \\ 5/3 \\ 13/6 \end{pmatrix}.$$

Thus, the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \begin{pmatrix} -1/6 \\ 1/3 \\ -1/6 \end{pmatrix}.$$

**R calculation**

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

# Define X and Y
X <- matrix(
  c(
    1, 0,
    1, 1,
    1, 2
  ),
  nrow = 3,
  byrow = TRUE
)

Y <- matrix(
  c(1, 2, 2),
  ncol = 1
)

# 1. Compute X'X
XtX <- t(X) %*% X
XtX
```

```
     [,1] [,2]
[1,]    3    3
[2,]    3    5
```

```
# 2. Compute X'Y
XtY <- t(X) %*% Y
XtY
```

```
     [,1]
[1,]    5
[2,]    6
```

```r
# 3. Compute beta_hat
beta_hat <- solve(XtX) %*% XtY
beta_hat
```

```
         [,1]
[1,] 1.166667
[2,] 0.500000
```

```r
# 4. Compute fitted values
Y_hat <- X %*% beta_hat
Y_hat
```

```
         [,1]
[1,] 1.166667
[2,] 1.666667
[3,] 2.166667
```

```r
# 5. Compute residuals
e <- Y - Y_hat
e
```

```
           [,1]
[1,] -0.1666667
[2,]  0.3333333
[3,] -0.1666667
```

```r
# Optional checks
t(X) %*% e
```

```
              [,1]
[1,] -8.881784e-16
[2,] -8.881784e-16
```

```r
t(Y_hat) %*% e
```

```
              [,1]
[1,] -1.474129e-15
```

**Exercise 2.2**
Explain in words what the column space

$$\mathcal{C}(\mathbf{X})$$

represents in a linear regression model.

**Solution 2.2**
The column space of $\mathbf{X}$ is the set of all response patterns that can be represented by the linear model.
More intuitively,
$\mathcal{C}(\mathbf{X})$ describes everything the model is capable of fitting. Anything outside this space cannot be explained by the predictors in $\mathbf{X}$.

### 2.3.2 Consequence when an Intercept Is Included

If the model includes an intercept, then one column of $\mathbf{X}$ is

$$\mathbf{1}_n = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Since the residual vector is orthogonal to every column of $\mathbf{X}$,

$$\mathbf{1}_n^\top \mathbf{e} = 0.$$

Therefore,

$$\boxed{\sum_{i=1}^{n} e_i = 0.}$$

Since

$$e_i = Y_i - \hat{Y}_i,$$

we also have

$$\sum_{i=1}^{n} Y_i = \sum_{i=1}^{n} \hat{Y}_i.$$

Thus,

$$\boxed{\bar{\hat{Y}} = \bar{Y}.}$$

**Intuition for the Mean of the Fitted Values**

The intercept allows the fitted model to shift up or down. At the least squares solution, there can be no overall tendency for the fitted values to be systematically too high or too low. Therefore, the residuals sum to zero and the fitted values have the same mean as the observed

responses.

## Section 2.4 The Hat Matrix

In the full-column-rank case,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Substituting

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y},$$

we obtain

$$\hat{\mathbf{Y}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}.$$

**Definition 1: The Hat Matrix**

Define

$$\boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top.}$$

Then

$$\boxed{\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.}$$

The matrix $\mathbf{H}$ is called the **hat matrix** because it puts the "hat" on $\mathbf{Y}$:

$$\mathbf{Y} \quad \longrightarrow \quad \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

**Proposition 2.3** (Properties of the Hat Matrix)

When $\mathbf{X}$ has full column rank, the hat matrix satisfies:

**i) Symmetry**

$$\boxed{\mathbf{H}^\top = \mathbf{H}.}$$

**ii) Idempotence**

$$\boxed{\mathbf{H}^2 = \mathbf{H}.}$$

**iii) Projection of the design matrix**

$$\boxed{\mathbf{H}\mathbf{X} = \mathbf{X}.}$$

**iv) Rank**

$$\boxed{\text{rank}(\mathbf{H}) = p.}$$

**v) Trace**

$$\boxed{\text{tr}(\mathbf{H}) = p.}$$

A symmetric and idempotent matrix represents an orthogonal projection. Therefore, $\mathbf{H}$ is the orthogonal projection matrix onto

$$\mathcal{C}(\mathbf{X}).$$

**i** Why it is symmetric

We have

$$\begin{aligned}\mathbf{H}^\top &= \left[\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\right]^\top \\ &= \mathbf{X}\left[(\mathbf{X}^\top\mathbf{X})^{-1}\right]^\top\mathbf{X}^\top.\end{aligned}$$

Since $\mathbf{X}^\top\mathbf{X}$ is symmetric, its inverse is also symmetric. Therefore,

$$\mathbf{H}^\top = \mathbf{H}.$$

**i** Why it is idempotent

We have

$$\begin{aligned}\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ &= \mathbf{H}.\end{aligned}$$

Thus,

$$\mathbf{H}^2 = \mathbf{H}.$$

**i** What Does Idempotence Mean Geometrically?

If we project $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ once, we obtain

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.$$

Projecting the fitted vector again does nothing:

$$\mathbf{H}\hat{\mathbf{Y}} = \mathbf{H}^2\mathbf{Y} = \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

Once a vector is already inside the model space, projecting it onto the same space leaves it

unchanged.

Notice that, $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ and $\hat{\mathbf{Y}} = \mathbf{HY}$, we have

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}.$$

**Definition 2.4** (Residual-Maker Matrix)
Define

$$\boxed{\mathbf{M} = \mathbf{I}_n - \mathbf{H}.}$$

Then

$$\boxed{\mathbf{e} = \mathbf{MY}.}$$

The matrix $\mathbf{M}$ is called the **residual-maker matrix**.
It removes the component of $\mathbf{Y}$ lying in $\mathcal{C}(\mathbf{X})$.

**Proposition 2.4** (Properties of the Residual-Maker Matrix)
Since

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H},$$

we obtain

$$\mathbf{M}^\top = \mathbf{M}$$

and

$$\mathbf{M}^2 = \mathbf{M}.$$

Thus, $\mathbf{M}$ is also an orthogonal projection matrix.
While $\mathbf{H}$ projects onto

$$\mathcal{C}(\mathbf{X}),$$

$\mathbf{M}$ projects onto

$$\mathcal{C}(\mathbf{X})^\perp.$$

In addition,

$$\boxed{\mathbf{HM} = \mathbf{MH} = \mathbf{0}.}$$

Under full column rank,

$$\text{rank}(\mathbf{M}) = n - p$$

and

$$\text{tr}(\mathbf{M}) = n - p.$$

**Two Complementary Projection Matrices**

The matrices $\mathbf{H}$ and $\mathbf{M}$ split $\mathbb{R}^n$ into two orthogonal components:

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^{\perp}.$$

For the observed response,

$$\mathbf{Y} = \underbrace{\mathbf{H}\mathbf{Y}}_{\hat{\mathbf{Y}}} + \underbrace{\mathbf{M}\mathbf{Y}}_{\mathbf{e}}.$$

Thus,

$$\boxed{\mathbf{H} + \mathbf{M} = \mathbf{I}_n.}$$

## Section 2.5 The common name to refer to what we have done – Ordinary Least Squares (OLS)

Now that we have seen both the algebraic and geometric views of least squares, we can explain the name **ordinary least squares**, or **OLS**.

- **Squares:** we measure the discrepancy using squared residuals,

$$\sum_{i=1}^{n} e_i^2 = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2.$$

- **Least:** we choose $\boldsymbol{\beta}$ to make this quantity as small as possible.
- **Ordinary:** we use the usual unweighted sum of squared residuals, so each observation enters the criterion in the same way.

Thus,

$$\boxed{\text{OLS} = \text{minimize the ordinary sum of squared residuals.}}$$

Geometrically, OLS finds the point in $\mathcal{C}(\mathbf{X})$ that is closest to $\mathbf{Y}$ using ordinary Euclidean distance.

Later, we will consider extensions such as weighted least squares and generalized least squares.

**Exercise 2.3** (Quick Check)
What do **ordinary**, **least**, and **squares** each mean in OLS?

# Section 2.6 Sum of Squares Decomposition

## 2.6.1 Uncentered Geometric Decomposition

Since

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

and

$$\hat{\mathbf{Y}}^\top \mathbf{e} = 0,$$

we have

$$\begin{aligned}\mathbf{Y}^\top \mathbf{Y} &= (\hat{\mathbf{Y}} + \mathbf{e})^\top (\hat{\mathbf{Y}} + \mathbf{e}) \\ &= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + 2\hat{\mathbf{Y}}^\top \mathbf{e} + \mathbf{e}^\top \mathbf{e} \\ &= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + \mathbf{e}^\top \mathbf{e}.\end{aligned}$$

Therefore,

$$\boxed{\|\mathbf{Y}\|_2^2 = \|\hat{\mathbf{Y}}\|_2^2 + \|\mathbf{e}\|_2^2.}$$

This is a direct application of the Pythagorean theorem.

## 2.6.2 Centered Sum of Squares Decomposition

When the regression model contains an intercept,

$$\bar{\hat{Y}} = \bar{Y}.$$

Therefore,

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i).$$

The two components are orthogonal, giving

$$\boxed{\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.}$$

These quantities will later be called

- **total sum of squares**
$$\text{SST} = \sum_{i=1}^{n}(Y_i - \bar{Y})^2;$$
- **regression sum of squares**
$$\text{SSR} = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2;$$
- **error sum of squares**
$$\text{SSE} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2.$$

Thus,

$$\boxed{\text{SST} = \text{SSR} + \text{SSE}.}$$

**!** Centered vs. Uncentered Decomposition

The geometric identity

$$\mathbf{Y}^\top\mathbf{Y} = \hat{\mathbf{Y}}^\top\hat{\mathbf{Y}} + \mathbf{e}^\top\mathbf{e}$$

is an **uncentered** decomposition.
The familiar regression decomposition

$$\text{SST} = \text{SSR} + \text{SSE}$$

is a **centered** decomposition and relies on the model containing an intercept.
These two decompositions should not be confused.

### 2.6.3 Example: Simple Linear Regression

**Example 1: Hand Calculation**

Consider the simple regression dataset

| $x_i$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y_i$ | 1 | 3 | 3 | 5 |

.

We fit the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The response vector is

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix},$$

and the design matrix is

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

**Step 1: Compute $\mathbf{X}^\top\mathbf{X}$**
We have

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}.$$

**Step 2: Compute $\mathbf{X}^\top\mathbf{Y}$**
We obtain

$$\mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 12 \\ 24 \end{pmatrix}.$$

**Step 3: Compute $\hat{\boldsymbol{\beta}}$**
Since

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{20}\begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix},$$

the least squares estimator is

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} \\ &= \frac{1}{20}\begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}\begin{pmatrix} 12 \\ 24 \end{pmatrix} \\ &= \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix}. \end{aligned}$$

Thus,

$$\hat{\beta}_0 = 1.2, \qquad \hat{\beta}_1 = 1.2,$$

and the fitted regression line is

$$\boxed{\hat{Y} = 1.2 + 1.2x.}$$

**Step 4: Compute the Fitted Values**
The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

**Step 5: Compute the Residuals**

The residual vector is

$$\begin{aligned}\mathbf{e} &= \mathbf{Y} - \hat{\mathbf{Y}} \\ &= \begin{pmatrix}1\\3\\3\\5\end{pmatrix} - \begin{pmatrix}1.2\\2.4\\3.6\\4.8\end{pmatrix} \\ &= \begin{pmatrix}-0.2\\0.6\\-0.6\\0.2\end{pmatrix}.\end{aligned}$$

Notice that

$$\sum_{i=1}^{4} e_i = -0.2 + 0.6 - 0.6 + 0.2 = 0.$$

This occurs because the model contains an intercept.

**Step 6: Verify Orthogonality**

We compute

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix}1 & 1 & 1 & 1\\0 & 1 & 2 & 3\end{pmatrix} \begin{pmatrix}-0.2\\0.6\\-0.6\\0.2\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}.$$

Thus,

$$\boxed{\mathbf{X}^\top \mathbf{e} = \mathbf{0}.}$$

We can also verify that
$\hat{\mathbf{Y}}^\top \mathbf{e} = 0$.

**Step 7: Compute the Hat Matrix**

The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

For this example,

$$\mathbf{H} = \begin{pmatrix}0.7 & 0.4 & 0.1 & -0.2\\0.4 & 0.3 & 0.2 & 0.1\\0.1 & 0.2 & 0.3 & 0.4\\-0.2 & 0.1 & 0.4 & 0.7\end{pmatrix}.$$

Multiplying by $\mathbf{Y}$ gives

$$\mathbf{HY} = \begin{pmatrix}1.2\\2.4\\3.6\\4.8\end{pmatrix} = \hat{\mathbf{Y}}.$$

Thus, the hat matrix projects the observed response onto the model space.

### Example 2: R Demonstration

We can verify the previous calculations in R.

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

df <- data.frame(
  x = x,
  y = y
)

fit <- lm(y ~ x, data = df)

coef(fit)
```

```
(Intercept)           x
        1.2         1.2
```

The estimated coefficients are

$$\hat{\beta}_0 = 1.2, \qquad \hat{\beta}_1 = 1.2.$$

We can also compute the estimator directly using matrix operations.

```
X <- model.matrix(fit)
X
```

```
  (Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr(,"assign")
[1] 0 1
```

```
Y <- matrix(y, ncol = 1)

beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y

beta_hat
```

```
            [,1]
(Intercept)  1.2
x            1.2
```

Compute the fitted values and residuals:

```
y_hat <- X %*% beta_hat
e <- Y - y_hat

y_hat
```

```
  [,1]
1  1.2
2  2.4
3  3.6
4  4.8
```

```
e
```

```
  [,1]
1 -0.2
2  0.6
3 -0.6
4  0.2
```

Check that the residuals sum to zero:

```
sum(e)
```

```
[1] -3.774758e-15
```

Compute the hat matrix:

```
H <-
  X %*%
  solve(t(X) %*% X) %*%
  t(X)

round(H, 4)
```

```
     1   2   3    4
1  0.7 0.4 0.1 -0.2
2  0.4 0.3 0.2  0.1
3  0.1 0.2 0.3  0.4
4 -0.2 0.1 0.4  0.7
```

Verify symmetry:

```
round(H - t(H), 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify idempotence:

```
round(H %*% H - H, 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify the normal equations geometrically:

```
round(t(X) %*% e, 10)
```

```
            [,1]
(Intercept)    0
x              0
```

Verify that fitted values and residuals are orthogonal:

```
round(t(y_hat) %*% e, 10)
```

```
     [,1]
[1,]    0
```

Construct the residual-maker matrix:

```
M <- diag(nrow(X)) - H

round(M, 4)
```

```
     1    2    3    4
1  0.3 -0.4 -0.1  0.2
2 -0.4  0.7 -0.2 -0.1
3 -0.1 -0.2  0.7 -0.4
4  0.2 -0.1 -0.4  0.3
```

```
round(M %*% M - M, 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify that

$$\mathbf{e} = \mathbf{MY}.$$

```
round(M %*% Y - e, 10)
```

```
  [,1]
1    0
2    0
3    0
4    0
```

Finally, plot the fitted regression line:

```
ggplot(df, aes(x, y)) +
  geom_point(size = 2.5) +
  geom_abline(intercept = coef(fit)[1], slope = coef(fit)[2],
              colour = "steelblue4", linewidth = 0.8) +
  labs(x = "x", y = "y")
```

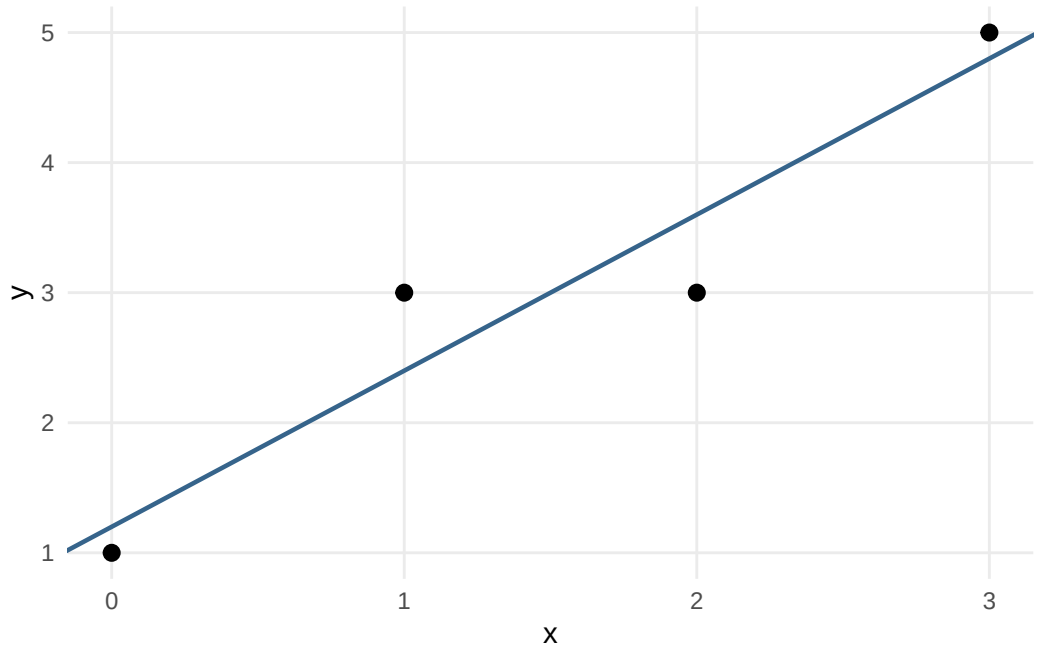



Figure 2.2: The least-squares line for the four-observation example.

## Section 2.7 What Happens Without Full Column Rank?

Suppose

$$\text{rank}(\mathbf{X}) < p.$$

Then the columns of $\mathbf{X}$ are linearly dependent, and

$$\mathbf{X}^\top \mathbf{X}$$

is singular.

Therefore,

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

does not exist.

**!** A Subtle but Important Point

When $\mathbf{X}$ does not have full column rank,

- a least squares solution still exists;
- $\hat{\boldsymbol{\beta}}$ may not be unique;
- the fitted vector $\hat{\mathbf{Y}}$ is still unique;
- the residual vector $\mathbf{e}$ is still unique.

Thus,

$$\boxed{\hat{\boldsymbol{\beta}} \text{ may not be unique, but } \hat{\mathbf{Y}} \text{ is unique.}}$$

The reason is geometric: even if several coefficient vectors represent the same point in $\mathcal{C}(\mathbf{X})$, the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ is unique.
We will discuss rank deficiency, generalized inverses, and estimability in more detail later.

## Section 2.8 Interpretation of the Geometry

The observed vector

$$\mathbf{Y} \in \mathbb{R}^n.$$

When $\mathbf{X}$ has full column rank $p$, the model space

$$\mathcal{C}(\mathbf{X})$$

is a $p$-dimensional subspace of $\mathbb{R}^n$.

Least squares finds the point in this subspace that is closest to $\mathbf{Y}$.

The central relationships are

$$\boxed{\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2}$$

$$\Downarrow$$

$$\boxed{\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}}$$

$$\Downarrow$$

$$\boxed{\mathbf{e} \perp \mathcal{C}(\mathbf{X})}$$

$$\Downarrow$$

$$\boxed{\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}}$$

$$\Downarrow$$

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.}$$

Thus,

- fitted values are orthogonal projections;
- residuals are orthogonal to the model space;
- the hat matrix is a projection matrix;
- the residual-maker matrix projects onto the orthogonal complement;
- sum-of-squares decompositions are consequences of the Pythagorean theorem.

# Section 2.9 Looking Ahead: Statistical Properties of OLS

**Looking Ahead**

Everything we have done so far is primarily **algebraic and geometric**.
We defined $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ by solving an optimization problem.
However, under the statistical model, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, the response $\mathbf{Y}$ is random. Therefore, $\hat{\boldsymbol{\beta}}$ is also a random vector.
In Chapter 3 — Distribution Theory of OLS and Inference, we will study quantities such as

$$\mathbb{E}(\hat{\boldsymbol{\beta}}), \quad \text{and} \quad \mathrm{Var}(\hat{\boldsymbol{\beta}}),$$

followed by estimation of $\sigma^2$, sampling distributions, confidence intervals, and hypothesis testing.

**Summary**

In this chapter, we introduced the least squares estimator through the optimization problem

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 .$$

Differentiating the residual sum of squares leads to the normal equations

$$\boxed{\mathbf{X}^\top\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top\mathbf{Y}.}$$

When $\mathbf{X}$ has full column rank,

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}.}$$

The fitted response is

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$$

is the hat matrix.
Geometrically,

$$\hat{\mathbf{Y}}$$

is the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.
The residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$$

lies in the orthogonal complement of the model space, so

$$\mathbf{X}^\top\mathbf{e} = \mathbf{0}.$$

Therefore,

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}}$$

is an orthogonal decomposition.
In Chapter 3 — Distribution Theory of OLS and Inference, we will treat $\hat{\boldsymbol{\beta}}$ as a random vector and study its expectation, variance, sampling distribution, and the foundations of statistical inference for linear models.

## Section 2.10 Practice at Home

1. Explain why

   $$\mathbf{X}^\top \mathbf{e} = \mathbf{0}$$

   means that the residual vector is orthogonal to the column space $\mathcal{C}(\mathbf{X})$.

2. Explain why

   $$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

   is an orthogonal projection, and why

   $$\mathbf{H}^2 = \mathbf{H}.$$

3. If the model contains an intercept, explain why

   $$\sum_{i=1}^{n} e_i = 0$$

   and hence

   $$\bar{\hat{Y}} = \bar{Y}.$$

4. Let

   $$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

   Compute

   $$\hat{\boldsymbol{\beta}}, \qquad \hat{\mathbf{Y}}, \qquad \mathbf{e},$$

and verify that

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

5. Show that the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$$

is symmetric and idempotent.

---

Side Fact: Matrix Calculus Facts Used in this chapter

**Theorem 2: Matrix Calculus Facts**

Let $\boldsymbol{\beta}$ be a column vector.
For a constant vector $\mathbf{a}$,

$$\frac{\partial}{\partial \boldsymbol{\beta}} (\mathbf{a}^\top \boldsymbol{\beta}) = \mathbf{a}.$$

For a constant matrix $\mathbf{A}$,

$$\frac{\partial}{\partial \boldsymbol{\beta}} (\boldsymbol{\beta}^\top \mathbf{A} \boldsymbol{\beta}) = (\mathbf{A} + \mathbf{A}^\top)\boldsymbol{\beta}.$$

If $\mathbf{A}$ is symmetric,

$$\mathbf{A}^\top = \mathbf{A},$$

and therefore

$$\boxed{\frac{\partial}{\partial \boldsymbol{\beta}} (\boldsymbol{\beta}^\top \mathbf{A} \boldsymbol{\beta}) = 2\mathbf{A}\boldsymbol{\beta}.}$$

These identities justify the derivative used in the least squares criterion.

## Section 3

# Distribution Theory of OLS and Inference

In Chapter 2 — Least Squares Estimation, we studied least squares primarily from an **algebraic and geometric** point of view.

We obtained

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

and showed that

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \qquad \mathbf{e} = \mathbf{M}\mathbf{Y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

and

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

We now move from **estimation** to **statistical inference**.

Since $\mathbf{Y}$ is random, quantities such as

$$\hat{\boldsymbol{\beta}}, \qquad \hat{\mathbf{Y}}, \qquad \mathbf{e}, \qquad \text{SSE}$$

are also random.

Our goal is to understand their sampling distributions and use them to quantify uncertainty.

> Learning Objectives
>
> By the end of this chapter, students should be able to:
>
> - derive the expectation and variance of the OLS estimator;
> - state the normal linear model;
> - derive the distribution of the OLS estimator under normality;
> - explain the distribution of the residual sum of squares;
> - explain why the residual degrees of freedom are $n - p$;

- estimate $\sigma^2$ using the mean squared error;
- explain why $\hat{\boldsymbol{\beta}}$ and SSE are independent;
- construct confidence intervals for regression coefficients;
- perform hypothesis tests for regression coefficients;
- perform inference for linear combinations of regression coefficients;
- distinguish inference for a mean response from prediction of a future observation;
- interpret standard regression output from R.

### Reading

Recommended reading for this chapter:

- Seber and Lee:
  - sections on distribution theory of least squares estimators;
  - estimation of the error variance;
  - inference for regression coefficients.
- Montgomery, Peck, and Vining:
  - sections on confidence intervals;
  - hypothesis testing;
  - estimation and prediction in linear regression.

**Recap from Least Squares**

From Chapter 2 — Least Squares Estimation, when $\mathbf{X}$ has full column rank,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

We also have

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \qquad \mathbf{e} = \mathbf{M}\mathbf{Y},$$

with

$$\mathbf{H} + \mathbf{M} = \mathbf{I}_n$$

and

$$\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}.$$

Geometrically,

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

is an orthogonal decomposition.
In this chapter, we will see that this geometric decomposition also provides the foundation for regression inference.

> **!** Throughout This Chapter
>
> We treat the design matrix $\mathbf{X}$ as fixed.
> If the predictors are random, the results can instead be interpreted conditionally on the observed design matrix $\mathbf{X}$.

## Section 3.1 Mean and Variance of the OLS Estimator

Before introducing normality, we can already study the expectation and variance of the OLS estimator.

Assume

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n.$$

Recall that

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

Substituting this into the OLS estimator gives

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}.\end{aligned}$$

Thus,

$$\boxed{\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}.}$$

This representation will be useful throughout the chapter.

**Proposition 3.1** (Mean and Variance of OLS)
Under

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

the OLS estimator satisfies

$$\boxed{\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}}$$

and

$$\boxed{\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.}$$

### 3.1.1 Why Is OLS Unbiased?

Using

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon},$$

we obtain

$$\begin{aligned}\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) &= \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbb{E}\left(\boldsymbol{\varepsilon}\right)\\ &= \boldsymbol{\beta}.\end{aligned}$$

Thus,

$$\boxed{\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}.}$$

OLS is therefore an **unbiased estimator** of $\boldsymbol{\beta}$.

Intuitively, if we repeated the experiment many times using the same design matrix, the OLS estimates would fluctuate around the true parameter vector.

### 3.1.2 Variance of OLS

Similarly,

$$\begin{aligned}\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) &= \mathrm{Var}\left[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}\right]\\ &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\,\mathrm{Var}(\boldsymbol{\varepsilon})\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\\ &= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.\end{aligned}$$

**!** Normality Is Not Needed Yet

The results

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}$$

and

$$\text{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$$

do **not** require normally distributed errors.
They follow from the first two moment assumptions

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n.$$

Normality becomes important when we want exact finite sample distributions and inference.

**Exercise 3.1** (Which Assumptions Are Needed?)
Suppose

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n.$$

Do we need to assume that the errors are normally distributed to conclude that

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}?$$

**Solution 3.1**
No.
Normality is **not** required for unbiasedness.
The result follows from

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}$$

and

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}.$$

Normality will be needed later for exact finite sample inference.

## Section 3.2 The Normal Linear Model

To obtain exact sampling distributions, we now strengthen the assumptions.

**Definition 3.1** (Normal Linear Model)
The **normal linear model** assumes

$$\boxed{\mathbf{Y} \sim N_n\left(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}_n\right).}$$

Equivalently,

$$\boxed{\boldsymbol{\varepsilon} \sim N_n\left(\mathbf{0}, \sigma^2\mathbf{I}_n\right).}$$

Under this assumption:

- each response has a normal distribution;
- the errors have common variance $\sigma^2$;
- the errors are independent;
- exact finite sample distributions can be obtained for many regression statistics.

**i** Why Does Normality Matter?

Least squares itself does not require normality.
Normality is introduced because it gives us exact distributions for

$$\hat{\boldsymbol{\beta}}, \qquad \text{SSE}, \qquad T, \qquad F.$$

These distributions allow us to construct exact finite sample confidence intervals and hypothesis tests.

**Exercise 3.2** (What Changes When We Add Normality?)
Suppose the errors satisfy

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \qquad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n,$$

but are not normally distributed.
Which of the following can still hold?

1. OLS can still be computed.
2. OLS can still be unbiased.
3. We can still have
$$\text{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$
4. Exact finite sample $t$ and $F$ distributions automatically hold.

**Solution 3.2**
Statements 1, 2, and 3 can still hold.
Statement 4 does not generally hold.

Normality is not required to define OLS or derive its mean and variance. Its main role here is to provide exact finite sample distributions for inference.

## Section 3.3 Distribution of the OLS Estimator

Recall that

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Thus, $\hat{\boldsymbol{\beta}}$ is a **linear transformation** of $\mathbf{Y}$.

A fundamental property of the multivariate normal distribution is that a linear transformation of a multivariate normal vector is again multivariate normal.

**Theorem 3.1** (Distribution of the OLS Estimator)
Under the normal linear model,

$$\boxed{\hat{\boldsymbol{\beta}} \sim N_p\left(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right).}$$

Let

$$\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Then

$$\operatorname{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2 \mathbf{C}.$$

If $c_{jj}$ denotes the $j$th diagonal element of $\mathbf{C}$, then

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

Therefore,

$$\boxed{\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).}$$

**Exercise 3.3** (Reading the Covariance Matrix)
Let

$$\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

If $c_{jj}$ is the $j$th diagonal element of $\mathbf{C}$:

1. What is $\text{Var}(\hat{\beta}_j)$?
2. What happens to the uncertainty of $\hat{\beta}_j$ if $c_{jj}$ becomes larger?

**Solution 3.3**
The variance is

$$\boxed{\text{Var}(\hat{\beta}_j) = \sigma^2 c_{jj}.}$$

Therefore, a larger $c_{jj}$ means greater sampling variability and less precise estimation of $\beta_j$. Thus, the precision of OLS depends not only on $\sigma^2$, but also on the design matrix $\mathbf{X}$.

There is still one difficulty.

The quantity

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}}$$

depends on the unknown parameter $\sigma$.

Before performing inference, we need to estimate $\sigma^2$.

**Exercise 3.4** (Why Can We Not Use the Normal Statistic Directly?)
We have

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

Why can we usually not use this statistic directly for inference?

**Solution 3.4**
Because the error standard deviation $\sigma$ is unknown.
We therefore need to estimate $\sigma$ from the data.
Replacing $\sigma$ with an estimator will lead naturally to the $t$ distribution.

## Section 3.4 Residual Sum of Squares and Estimation of $\sigma^2$

Recall from Chapter 2 — Least Squares Estimation that

$$\mathbf{e} = \mathbf{MY},$$

where

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

Since

$$\mathbf{MX} = \mathbf{0},$$

we have

$$\begin{aligned}\mathbf{e} &= \mathbf{MY} \\ &= \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= \mathbf{M}\boldsymbol{\varepsilon}.\end{aligned}$$

Thus,

$$\boxed{\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}.}$$

The residual vector therefore contains the part of the random error that lies outside the model space.

### 3.4.1 Residual Sum of Squares

The residual sum of squares is

$$\text{SSE} = \mathbf{e}^\top \mathbf{e}.$$

Since

$$\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon},$$

and $\mathbf{M}$ is symmetric and idempotent,

$$\begin{aligned}\text{SSE} &= \boldsymbol{\varepsilon}^\top \mathbf{M}^\top \mathbf{M}\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\top \mathbf{M}\boldsymbol{\varepsilon}.\end{aligned}$$

Therefore,

$$\boxed{\text{SSE} = \boldsymbol{\varepsilon}^\top \mathbf{M}\boldsymbol{\varepsilon}.}$$

### 3.4.2 Distribution of SSE

Under the normal linear model,

$$\frac{\boldsymbol{\varepsilon}}{\sigma} \sim N_n\left(\mathbf{0}, \mathbf{I}_n\right).$$

Let

$$\mathbf{Z} = \frac{\boldsymbol{\varepsilon}}{\sigma}.$$

Then

$$\frac{\text{SSE}}{\sigma^2} = \mathbf{Z}^\top \mathbf{M} \mathbf{Z}.$$

Recall that $\mathbf{M}$ is symmetric and idempotent and

$$\text{rank}(\mathbf{M}) = n - p.$$

Therefore,

**Theorem 3.2** (Distribution of the Residual Sum of Squares)
Under the normal linear model,

$$\boxed{\frac{\text{SSE}}{\sigma^2} \sim \chi^2_{n-p}.}$$

### 3.4.3 Why Are the Degrees of Freedom $n - p$?

Recall that

$$\text{rank}(\mathbf{H}) = p$$

and

$$\text{rank}(\mathbf{M}) = n - p.$$

The fitted vector lies in

$$\mathcal{C}(\mathbf{X}),$$

which has dimension $p$.

The residual vector lies in

$$\mathcal{C}(\mathbf{X})^{\perp},$$

which has dimension

$$n - p.$$

Therefore, the residual variation has

$$\boxed{n - p}$$

degrees of freedom.

**Geometric Interpretation of Degrees of Freedom**

The response space can be decomposed as

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^{\perp}.$$

The fitted component occupies $p$ dimensions.
The remaining

$$n - p$$

dimensions are available for residual variation.
This provides a geometric explanation for why estimating $p$ regression parameters costs $p$ degrees of freedom.

**Exercise 3.5** (Degrees of Freedom)
Suppose a regression model has

$$n = 20$$

observations and

$$p = 3$$

regression parameters.

1. What is the residual degrees of freedom?
2. Why is it not 20?

**Solution 3.5**
The residual degrees of freedom are

$$n - p = 20 - 3 = 17.$$

It is not 20 because fitting the regression model uses $p = 3$ dimensions of the response space. The residual vector therefore lies in a space of dimension

$$20 - 3 = 17.$$

### 3.4.4 Estimating $\sigma^2$

Since a chi-square random variable with $\nu$ degrees of freedom has expectation $\nu$,

$$\mathbb{E}\left(\frac{\text{SSE}}{\sigma^2}\right) = n - p.$$

Therefore,

$$\mathbb{E}(\text{SSE}) = (n - p)\sigma^2.$$

This gives us an unbiased estimator of the error variance.

**Definition 3.2** (Mean Squared Error)
The **mean squared error** is defined as

$$\boxed{\text{MSE} = \hat{\sigma}^2 = \frac{\text{SSE}}{n - p}.}$$

It satisfies

$$\mathbb{E}\left(\hat{\sigma}^2\right) = \sigma^2.$$

The estimated error standard deviation is

$$\boxed{\hat{\sigma} = \sqrt{\text{MSE}}.}$$

**Exercise 3.6** (Estimate the Error Variance)
Suppose

$$n = 20, \qquad p = 3, \qquad \text{SSE} = 34.$$

Compute

$$\hat{\sigma}^2$$

and

$$\hat{\sigma}.$$

**Solution 3.6**
The residual degrees of freedom are

$$n - p = 17.$$

Therefore,

$$\hat{\sigma}^2 = \frac{34}{17} = 2.$$

Hence,

$$\boxed{\hat{\sigma} = \sqrt{2} \approx 1.414.}$$

## Section 3.5 Independence of OLS and Residual Variation

A crucial result under the normal linear model is

$$\boxed{\hat{\boldsymbol{\beta}} \quad \text{and} \quad \text{SSE} \quad \text{are independent.}}$$

This result is closely connected to the projection geometry from Chapter 2 — Least Squares Estimation.

### 3.5.1 Why Does Independence Occur?

Recall that

$$\hat{\mathbf{Y}} = \mathbf{HY}$$

and

$$\mathbf{e} = \mathbf{MY}.$$

We also know that

$$\mathbf{HM} = \mathbf{0}.$$

Thus, the fitted component and the residual component come from orthogonal parts of the response vector.

> **Geometry Becomes Probability**
>
> In Chapter 2 — Least Squares Estimation, we learned that

$$\hat{\mathbf{Y}} \perp \mathbf{e}.$$

This is a geometric statement.
Under the normal linear model, these components are jointly normal.
For jointly normal random vectors, zero covariance implies independence.
Thus, under normality, geometric orthogonality leads to probabilistic independence.

More formally,

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$$

and

$$\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}.$$

Therefore,

$$\begin{aligned} \operatorname{Cov}\left(\hat{\boldsymbol{\beta}}, \mathbf{e}\right) &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{M} \\ &= \mathbf{0}, \end{aligned}$$

because

$$\mathbf{X}^\top \mathbf{M} = \mathbf{0}.$$

Since the two random vectors are jointly normal,

$$\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \mathbf{e}.$$

Since

$$\text{SSE} = \mathbf{e}^\top \mathbf{e}$$

is a function of $\mathbf{e}$,

$$\boxed{\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \text{SSE}.}$$

**Exercise 3.7** (Does Orthogonal Always Mean Independent?)
We know that

$$\mathbf{H}\mathbf{M} = \mathbf{0}.$$

Does the orthogonality between the fitted and residual components automatically imply that

they are independent?
Why is normality important?

**Solution 3.7**
No.
In general, zero covariance or orthogonality does **not** imply independence.
Under the normal linear model, the fitted and residual components are jointly normal.
For jointly normal random vectors, zero covariance implies independence.
Thus, normality is what allows us to move from orthogonality to independence.

## Section 3.6 Inference for a Single Regression Coefficient

For the $j$th regression coefficient,

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

If $\sigma$ were known, we could use

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

In practice, $\sigma$ is unknown.

We replace it with

$$\hat{\sigma} = \sqrt{\frac{\text{SSE}}{n-p}}.$$

### 3.6.1 Standard Error

The estimated standard deviation of $\hat{\beta}_j$ is

**Definition 3.3** (Standard Error of an OLS Coefficient)
The standard error of $\hat{\beta}_j$ is

$$\boxed{\text{SE}\left(\hat{\beta}_j\right) = \hat{\sigma}\sqrt{c_{jj}}.}$$

### 3.6.2 Why Does the $t$ Distribution Appear?

We know that

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

We also know that

$$U = \frac{\text{SSE}}{\sigma^2} \sim \chi^2_{n-p}.$$

In addition,

$$Z \quad \text{and} \quad U$$

are independent.

Therefore,

$$\frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}.$$

Substituting the regression quantities gives

**Theorem 3.3** ($t$ Distribution for an OLS Coefficient)
Under the normal linear model,

$$\boxed{T = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{c_{jj}}} \sim t_{n-p}.}$$

**i** Normal vs. t Distribution

If $\sigma$ were known, we would use a normal distribution.
Because $\sigma$ is unknown and must be estimated from the data, we introduce additional uncertainty.
The $t$ distribution accounts for this additional uncertainty.
As the degrees of freedom increase,

$$t_\nu$$

approaches

$$N(0,1).$$

**Exercise 3.8** (Why Do We Use a $t$ Distribution?)
Suppose

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

Why do we use a $t$ distribution rather than a standard normal distribution when performing inference for $\beta_j$?

**Solution 3.8**
Because $\sigma$ is unknown.
We replace $\sigma$ with the estimate $\hat{\sigma}$ obtained from the residual sum of squares.
This additional estimation uncertainty changes the standardized statistic from a normal distribution to a $t$ distribution.

### 3.6.3 Confidence Interval for $\beta_j$

A $100(1-\alpha)\%$ confidence interval for $\beta_j$ is

$$\boxed{\hat{\beta}_j \pm t_{1-\alpha/2,\, n-p}\, \mathrm{SE}\left(\hat{\beta}_j\right).}$$

Equivalently,

$$\boxed{\hat{\beta}_j \pm t_{1-\alpha/2,\, n-p} \hat{\sigma} \sqrt{c_{jj}}.}$$

### 3.6.4 Hypothesis Test for $\beta_j$

Suppose we want to test

$$H_0 : \beta_j = \beta_{j,0}$$

against

$$H_1 : \beta_j \neq \beta_{j,0}.$$

The test statistic is

$$\boxed{T = \frac{\hat{\beta}_j - \beta_{j,0}}{\mathrm{SE}(\hat{\beta}_j)}.}$$

Under $H_0$,

$$T \sim t_{n-p}.$$

For a two sided test at significance level $\alpha$, reject $H_0$ if

$$|T| > t_{1-\alpha/2,\, n-p}.$$

**Exercise 3.9** (Confidence Interval and Hypothesis Test)
Suppose we test

$$H_0 : \beta_j = 0$$

against

$$H_1 : \beta_j \neq 0$$

at significance level

$$\alpha = 0.05.$$

What should happen if the corresponding 95% confidence interval for $\beta_j$ does not contain zero?

**Solution 3.9**
We reject

$$H_0 : \beta_j = 0$$

at the 5% significance level.
A two sided hypothesis test at level $\alpha$ and a $100(1-\alpha)\%$ confidence interval give equivalent conclusions.

## Section 3.7 Inference for Linear Combinations

Sometimes the quantity of interest is not a single coefficient. Suppose we are interested in $\mathbf{a}^\top \boldsymbol{\beta}$, where $\mathbf{a}$ is a fixed $p \times 1$ vector. Since

$$\hat{\boldsymbol{\beta}} \sim N_p\left(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right),$$

we have

$$\mathbf{a}^\top \hat{\boldsymbol{\beta}} \sim N\left(\mathbf{a}^\top \boldsymbol{\beta}, \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}\right).$$

Therefore,

$$\boxed{\frac{\mathbf{a}^\top\hat{\boldsymbol{\beta}} - \mathbf{a}^\top\boldsymbol{\beta}}{\hat{\sigma}\sqrt{\mathbf{a}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{a}}} \sim t_{n-p}.}$$

A $100(1-\alpha)\%$ confidence interval for $\mathbf{a}^\top\boldsymbol{\beta}$ is

$$\boxed{\mathbf{a}^\top\hat{\boldsymbol{\beta}} \pm t_{1-\alpha/2,\,n-p}\hat{\sigma}\sqrt{\mathbf{a}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{a}}.}$$

**Why Is This Useful?**

Many regression questions can be expressed as linear combinations of the regression coefficients. Examples include:

- a single coefficient;
- the difference between two coefficients;
- the mean response at a particular predictor value.

Thus, inference for a single regression coefficient is a special case of inference for a linear combination.

**Exercise 3.10** (A Single Coefficient as a Linear Combination)
Suppose

$$\boldsymbol{\beta} = \begin{pmatrix}\beta_0\\ \beta_1\\ \beta_2\end{pmatrix}.$$

What vector $\mathbf{a}$ would you choose so that $\mathbf{a}^\top\boldsymbol{\beta} = \beta_1$?

**Solution 3.10**
Choose

$$\boxed{\mathbf{a} = \begin{pmatrix}0\\ 1\\ 0\end{pmatrix}.}$$

Then $\mathbf{a}^\top\boldsymbol{\beta} = \beta_1$. Thus, inference for an individual coefficient is a special case of inference for a linear combination.

## Section 3.8 General Linear Hypotheses

Sometimes we want to test several linear restrictions simultaneously. Suppose $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$, where $\mathbf{C}$ is an $r \times p$ matrix of rank $r$ and $\mathbf{d}$ is an $r \times 1$ vector.

**Definition 3.4** (General Linear Hypothesis)
A hypothesis of the form

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$$

is called a **general linear hypothesis**. It allows several linear restrictions on the regression coefficients to be tested simultaneously.

The corresponding test statistic is

$$F = \frac{(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})^\top \left[\mathbf{C}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{C}^\top\right]^{-1} (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})}{r\hat{\sigma}^2}.$$

Under $H_0$,

$$\boxed{F \sim F_{r,\, n-p}.}$$

For the distribution background, see Section 3.15; the full derivations are in the Basic Mathematical Statistics appendix.

### Relationship Between $t$ and $F$

When only one restriction is being tested, $r = 1$. In this case,

$$\boxed{F = T^2.}$$

Thus, the familiar $t$ test for one coefficient is a special case of the general $F$ test. We will study $F$ tests more thoroughly in Chapter 4 — Model Comparison, Partial F Tests, and ANOVA when we develop the ANOVA decomposition and compare nested regression models.

**Exercise 3.11** ($t$ Test or $F$ Test?)
Suppose we want to test only $H_0 : \beta_2 = 0$. Could we use either a $t$ test or an $F$ test? How are the two test statistics related?

**Solution 3.11**
Yes. Because there is only one linear restriction, either test can be used. The corresponding statistics satisfy

$$\boxed{F = T^2.}$$

Therefore, the two tests produce the same conclusion.

## Section 3.9 Inference for the Mean Response

Suppose we want to estimate the mean response at a predictor vector $\mathbf{x}_0$. The true mean response is

$$\mu(\mathbf{x}_0) = \mathbf{x}_0^\top \boldsymbol{\beta}.$$

Its estimator is

$$\hat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}.$$

Since this is a linear combination of $\hat{\boldsymbol{\beta}}$,

$$\operatorname{Var}\left(\hat{\mu}(\mathbf{x}_0)\right) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0.$$

Thus,

$$\frac{\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} - \mathbf{x}_0^\top \boldsymbol{\beta}}{\hat{\sigma}\sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}} \sim t_{n-p}.$$

Therefore, a $100(1-\alpha)\%$ confidence interval for the mean response is

$$\boxed{\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \pm t_{1-\alpha/2,\, n-p} \hat{\sigma} \sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.}$$

## Section 3.10 Prediction for a New Observation

Now suppose we want to predict one new observation at the same predictor value $\mathbf{x}_0$:

$$Y_{\text{new}} = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_{\text{new}},$$

where

$$\varepsilon_{\text{new}} \sim N(0, \sigma^2)$$

and is independent of the original data. The predictor is

$$\hat{Y}_{\text{new}} = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}.$$

The prediction error is

$$Y_{\text{new}} - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}.$$

Its variance is

$$\operatorname{Var}\left(Y_{\text{new}} - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}\right) = \sigma^2 \left[1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0\right].$$

Therefore, a $100(1-\alpha)\%$ prediction interval is

$$\boxed{\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \pm t_{1-\alpha/2,\, n-p} \hat{\sigma} \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.}$$

**!** Mean Response vs. New Observation

The two intervals answer different questions.

| Goal | Source of uncertainty |
|---|---|
| Estimate a mean response | Uncertainty in estimating the regression mean |
| Predict a new observation | Uncertainty in the regression mean plus new random variation |

The additional 1 in $1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$ represents the variability of the new observation itself. Therefore, at the same $\mathbf{x}_0$ and confidence level, the prediction interval is wider than the confidence interval for the mean response.

**Exercise 3.12** (Mean Response or Prediction?)
Consider the following questions.

1. What is the **average stopping distance** of all cars travelling at 50 mph?
2. What stopping distance should we expect for **one particular future car** travelling at 50 mph?

Which question requires a confidence interval for the mean response, and which requires a prediction interval?

**Solution 3.12**
The first question concerns the **mean response**, so we use a confidence interval for the regression mean. The second question concerns **one future observation**, so we use a prediction interval. The prediction interval is wider because it also includes the random variation associated with the future observation.

## Section 3.11 Worked Example

Consider again the simple regression dataset from Chapter 2 — Least Squares Estimation:

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline Y_i & 1 & 3 & 3 & 5 \end{array}.$$

The design matrix and response vector are

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}.$$

From Chapter 2 — Least Squares Estimation,

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix},$$

and

$$\hat{\mathbf{Y}} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

Therefore,

$$\mathbf{e} = \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix}.$$

**Estimate the Error Variance**

The residual sum of squares is

$$\begin{aligned} \text{SSE} &= \mathbf{e}^\top \mathbf{e} \\ &= (-0.2)^2 + 0.6^2 + (-0.6)^2 + 0.2^2 \\ &= 0.8. \end{aligned}$$

Since $n = 4$ and $p = 2$, the residual degrees of freedom are $n - p = 2$. Thus,

$$\boxed{\hat{\sigma}^2 = \frac{0.8}{2} = 0.4.}$$

Therefore,

$$\boxed{\hat{\sigma} = \sqrt{0.4} \approx 0.632.}$$

### Standard Error of the Slope

We have

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}.$$

For the slope,

$$c_{22} = \frac{4}{20} = 0.2.$$

Therefore,

$$\text{SE}(\hat{\beta}_1) = \sqrt{0.4(0.2)} = \sqrt{0.08} \approx 0.283.$$

### Test the Slope

Consider $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$. The test statistic is

$$T = \frac{1.2}{0.283} \approx 4.243.$$

The residual degrees of freedom are 2. For a 5% two sided test, $t_{0.975,2} \approx 4.303$. Since $|4.243| < 4.303$, we fail to reject $H_0$ at the 5% significance level.

**i** What Does This Example Show?

The estimated slope $\hat{\beta}_1 = 1.2$ is relatively large. However, there are only four observations, leaving only two residual degrees of freedom. Therefore, there is substantial uncertainty in the estimate. A large estimated effect does not automatically imply strong evidence against the null hypothesis.

### Confidence Interval for the Slope

A 95% confidence interval is $1.2 \pm 4.303(0.283)$, which gives approximately

$$\boxed{(-0.017,\ 2.417).}$$

The interval contains zero, which agrees with the hypothesis test.

**Exercise 3.13** (Check the Connection)
Without performing another hypothesis test, what conclusion can you draw from the fact that the 95% confidence interval $(-0.017,\ 2.417)$ contains zero?

**Solution 3.13**
We fail to reject $H_0 : \beta_1 = 0$ at the 5% significance level. This agrees with the previous $t$ test.

**Mean Response at $x_0 = 2$**

For simple linear regression with an intercept,

$$\mathbf{x}_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The estimated mean response is

$$\hat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}} = 3.6.$$

Also,

$$\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 = 0.3.$$

Therefore,

$$\text{SE}\left(\hat{\mu}(\mathbf{x}_0)\right) = \sqrt{0.4(0.3)} \approx 0.346.$$

The 95% confidence interval for the mean response is approximately

$$\boxed{(2.110,\ 5.090).}$$

**Prediction at $x_0 = 2$**

For one new observation,

$$\text{SE}_{\text{pred}} = \sqrt{0.4(1 + 0.3)} \approx 0.721.$$

Therefore, the 95% prediction interval is approximately

$$\boxed{(0.497,\ 6.703).}$$

Notice that the prediction interval is substantially wider than the confidence interval for the mean response.

## Section 3.12 R Demonstration

We can reproduce the previous calculations using R.

```r
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

fit <- lm(y ~ x)

summary(fit)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4
-0.2  0.6 -0.6  0.2

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.2000     0.5292   2.268   0.1515
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9, Adjusted R-squared:   0.85
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

### Estimated Coefficients

```r
coef(fit)
```

```
(Intercept)           x
        1.2         1.2
```

### Residual Degrees of Freedom and Error Variance

```r
df.residual(fit)
```

```
[1] 2
```

```r
sigma(fit)
```

```
[1] 0.6324555
```

```r
sigma(fit)^2
```

```
[1] 0.4
```

### Covariance Matrix and Standard Errors

```r
vcov(fit)
```

```
            (Intercept)     x
(Intercept)        0.28 -0.12
x                 -0.12  0.08
```

```r
sqrt(diag(vcov(fit)))
```

```
(Intercept)           x
  0.5291503   0.2828427
```

### Verify the Formulas Using Matrix Operations

```r
X <- model.matrix(fit)
Y <- matrix(y, ncol = 1)

n <- nrow(X)
p <- ncol(X)

beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y

e <- Y - X %*% beta_hat

SSE <- as.numeric(t(e) %*% e)

sigma2_hat <- SSE / (n - p)

V_beta_hat <-
  sigma2_hat *
  solve(t(X) %*% X)

SE_beta_hat <-
  sqrt(diag(V_beta_hat))

beta_hat
```

```
            [,1]
(Intercept)  1.2
x            1.2
```

```
SSE
```

```
[1] 0.8
```

```
sigma2_hat
```

```
[1] 0.4
```

```
V_beta_hat
```

```
            (Intercept)     x
(Intercept)        0.28 -0.12
x                 -0.12  0.08
```

```
SE_beta_hat
```

```
(Intercept)           x
  0.5291503   0.2828427
```

### Hypothesis Tests and Confidence Intervals

R automatically reports coefficient level $t$ tests:

```
summary(fit)$coefficients
```

```
            Estimate Std. Error  t value  Pr(>|t|)
(Intercept)      1.2  0.5291503 2.267787 0.1514719
x                1.2  0.2828427 4.242641 0.0513167
```

Confidence intervals for the regression coefficients can be obtained using

```
confint(fit)
```

```
                2.5 %   97.5 %
(Intercept) -1.07674982 3.476750
x           -0.01697397 2.416974
```

### Mean Response and Prediction

At $x_0 = 2$, use

```r
newdat <- data.frame(x = 2)

predict(
  fit,
  newdata = newdat,
  interval = "confidence"
)
```

```
  fit      lwr      upr
1 3.6 2.109517 5.090483
```

```r
predict(
  fit,
  newdata = newdat,
  interval = "prediction"
)
```

```
  fit      lwr      upr
1 3.6 0.497313 6.702687
```

The first interval estimates the **mean response**. The second interval predicts **one future observation**.

## Section 3.13 Reading Standard Regression Output

For

```r
summary(fit)
```

the coefficient table contains four important columns:

| Column | Interpretation |
|---|---|
| `Estimate` | Estimated regression coefficient |
| `Std. Error` | Estimated standard deviation of the coefficient estimator |
| `t value` | $t$ statistic for testing whether the coefficient equals zero |
| `Pr(>\|t\|)` | Two sided $p$ value for the test |

The output also includes:

- the residual standard error;
- the residual degrees of freedom;

- $R^2$;
- adjusted $R^2$;
- an overall $F$ test.

We will study $R^2$, the ANOVA decomposition, and the overall $F$ test more carefully in Chapter 4 — Model Comparison, Partial F Tests, and ANOVA.

**Exercise 3.14** (Reading an R Regression Table)
Suppose R reports

| | Estimate | Std. Error | t value | Pr(> |
|---|---|---|---|---|
| $x_1$ | 2.40 | 0.60 | 4.00 | 0.001 |

Answer the following.

1. What is $\hat{\beta}_1$?
2. What is $\text{SE}(\hat{\beta}_1)$?
3. What null hypothesis is being tested by default?

**Solution 3.14**
The estimate is $\hat{\beta}_1 = 2.40$. The standard error is $\text{SE}(\hat{\beta}_1) = 0.60$. The default coefficient level test is $H_0 : \beta_1 = 0$. The reported $t$ statistic is

$$\frac{2.40}{0.60} = 4.$$

## Section 3.14 Practice at Home

1. Starting from

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon},$$

derive

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}$$

and

$$\text{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

2. Explain why normality is not required for OLS to be unbiased but is useful for exact finite sample inference.

3. Explain geometrically why the residual degrees of freedom are $n - p$.
4. Explain why $\hat{\boldsymbol{\beta}}$ and SSE are independent under the normal linear model.
5. Derive the $t$ statistic for testing $H_0 : \beta_j = \beta_{j,0}$.
6. Explain the connection between a two sided hypothesis test and the corresponding confidence interval.
7. Explain the difference between a confidence interval for $\mathbf{x}_0^\top \boldsymbol{\beta}$ and a prediction interval for a future observation at $\mathbf{x}_0$.
8. Fit a simple regression model in R and identify:
    - the coefficient estimates;
    - their standard errors;
    - the $t$ statistics;
    - the $p$ values;
    - the residual standard error;
    - the residual degrees of freedom.

**Summary**

This chapter moved from the **geometry of least squares** to the **probability of least squares**. Under the first two moment assumptions,

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}$$

and

$$\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

Under the additional normality assumption,

$$\boxed{\hat{\boldsymbol{\beta}} \sim N_p\left(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\right).}$$

The residual sum of squares satisfies

$$\boxed{\frac{\mathrm{SSE}}{\sigma^2} \sim \chi^2_{n-p}.}$$

This leads to the unbiased estimator

$$\boxed{\hat{\sigma}^2 = \frac{\mathrm{SSE}}{n-p}.}$$

The independence of $\hat{\boldsymbol{\beta}}$ and SSE allows us to replace the unknown $\sigma$ with $\hat{\sigma}$ and obtain exact $t$ inference. The same framework extends naturally to:

- linear combinations of regression coefficients;
- general linear hypotheses;
- confidence intervals for the mean response;

- prediction intervals for future observations.

In Chapter 4 — Model Comparison, Partial F Tests, and ANOVA, we will use these results to develop the ANOVA decomposition, the overall $F$ test, and comparisons of nested regression models.

# Section 3.15 Side Fact

These three reference distributions are built from normal variables:

- **Chi-square** measures squared variation: if $Z_1, \dots, Z_r$ are independent $N(0,1)$ variables, then $U = \sum_{j=1}^{r} Z_j^2 \sim \chi_r^2$.
- **Student's** $t$ standardizes a normal variable by an estimated standard deviation: if $Z \sim N(0,1)$ and $V \sim \chi_\nu^2$ are independent, then $T = Z/\sqrt{V/\nu} \sim t_\nu$.
- $F$ compares two independent variance quantities: if $U \sim \chi_r^2$ and $V \sim \chi_\nu^2$ are independent, then $F = (U/r)/(V/\nu) \sim F_{r,\nu}$.

For the general linear hypothesis, $r$ is the number of independent restrictions and $\nu = n - p$ is the residual degrees of freedom. The normal model gives the required independence between the numerator quadratic form and the residual sum of squares.

The full explanations, density derivations, independence proof, regression applications, and reference books are in the Basic Mathematical Statistics appendix.

## Section 4

# Model Comparison, Partial F Tests, and ANOVA

In this chapter, we study the ANOVA decomposition for linear regression, the overall significance test, and the comparison of nested models. These ideas connect the geometry of least squares with the inferential tools developed in Chapter 2 — Least Squares Estimation and Chapter 3 — Distribution Theory of OLS and Inference.

**Learning Objectives**

By the end of this chapter, students should be able to:

- define the total, regression, and error sums of squares;
- explain the ANOVA decomposition in regression with an intercept;
- interpret the degrees of freedom associated with SST, SSR, and SSE;
- perform the overall $F$ test for regression;
- compare nested linear models using extra sums of squares;
- interpret ANOVA tables produced by statistical software.

**Reading**

Recommended reading for this chapter:

- Seber and Lee:
  - sections on analysis of variance in regression
  - sums of squares and decomposition of variability
  - tests for nested models
- Montgomery, Peck, and Vining:
  - sections on the ANOVA table
  - overall model significance
  - partial and sequential sums of squares

**i** Motivation: does measuring tree height add useful information?

Suppose a forestry team wants to estimate timber volume before felling a tree. Diameter is already available; measuring height requires additional effort. Does height contribute information beyond diameter?
The real `trees` dataset in R lets us explore this question with measurements on 31 felled black cherry trees. Consider three candidate mean models:

| Model | Information used | Mean volume |
|---|---|---|
| $M_0$ | No tree measurements | $\beta_0$ |
| $M_1$ | Diameter | $\beta_0 + \beta_1$ Diameter |
| $M_2$ | Diameter and height | $\beta_0 + \beta_1$ Diameter $+ \beta_2$ Height |

**Discuss before calculating:** Which comparison answers whether height adds information? Why is choosing the smallest training SSE insufficient? Keep these questions in mind; the optional data example below returns to them with fitted models.

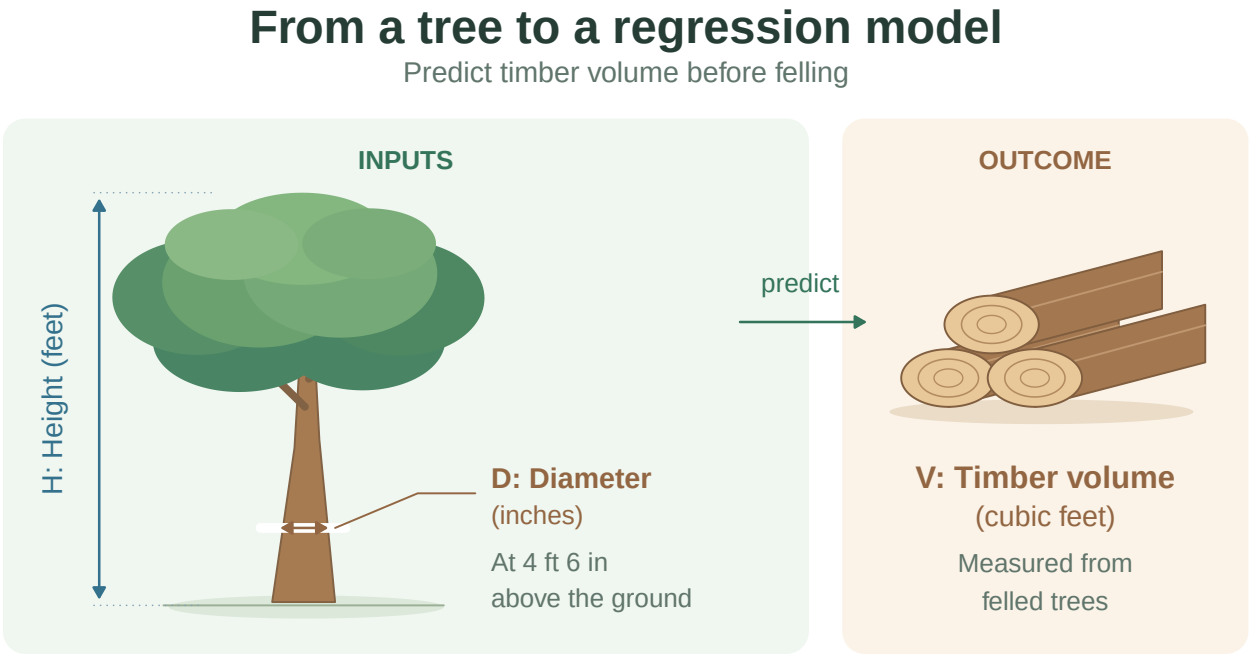



Figure 4.1: From measurements to a prediction question. Diameter and height are predictors; measured timber volume is the response. The tree and timber drawings are conceptual, not to scale.

**Recap of the Linear Model**

Recall the normal linear model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

When $\mathbf{X}$ has full column rank, the ordinary least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

and the fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

From Chapter 2 — Least Squares Estimation, we know that $\hat{\mathbf{Y}}$ and $\mathbf{e}$ are orthogonal. From Chapter 3 — Distribution Theory of OLS and Inference, we know that this leads to useful distributional results for inference. In this chapter, we organize these ideas into the ANOVA framework.

**Vector notation.** Following the convention used in this course, $\mathbf{a} = (a_1, a_2, a_3)$ denotes a column vector, while $\mathbf{a}^\top = (a_1\ a_2\ a_3)$ denotes a row vector. Commas distinguish the column-vector tuple; spaces separate entries of the row vector.

## Section 4.1 Total, Explained, and Unexplained Variation

A central question in regression is: How much of the variation in the response can be explained by the model? To answer this, we decompose the total variation in $\mathbf{Y}$ into:

- variation explained by regression;
- variation left unexplained by the model.

When the model includes an intercept, this decomposition takes a particularly simple and important form.

**Definition 4.1** (Regression Sums of Squares)
For an OLS fit with an intercept, let $\bar{Y} = n^{-1}\sum_{i=1}^n Y_i$, $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, and $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$. Define the **total**, **error**, and **regression sums of squares** by

$$\begin{aligned}
\mathrm{SST} &= \sum_{i=1}^n (Y_i - \bar{Y})^2 = \|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2, \\
\mathrm{SSE} &= \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \mathbf{e}^\top \mathbf{e}, \\
\mathrm{SSR} &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \|\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}\|^2.
\end{aligned}$$

SST measures variation around the sample mean; SSE measures deviations from the fitted model; SSR measures how much the fitted values vary around the sample mean. These are definitions. Their additive relationship below follows from OLS orthogonality and the inclusion of an intercept.

**!** A deviation may be negative; a sum of squares cannot

The subtraction inside each definition produces a **signed deviation**, which can be negative, zero, or positive. We then **square that deviation** and add the squared values. Thus

$$\text{SST} \geq 0, \qquad \text{SSR} \geq 0, \qquad \text{SSE} \geq 0.$$

For example, $y_i - \hat{y}_i = -0.6$ is a residual, while its contribution to SSE is $(-0.6)^2 = 0.36$. A negative residual is not a negative SSE. Also, $(a-b)^2$ is not $a^2 - b^2$. Reversing the direction of a deviation changes its sign but leaves its square unchanged.

**Theorem 4.1** (ANOVA Decomposition with an Intercept)
For an ordinary least squares fit whose design matrix contains an intercept,

$$\text{SST} = \text{SSR} + \text{SSE}.$$

The centered fitted vector $\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}$ is orthogonal to the residual vector $\mathbf{e}$. Thus total variation around the sample mean splits into explained variation and residual variation. This geometric identity does not require normal errors.

### Why the Decomposition Holds

The key reason is orthogonality. We can write

$$\mathbf{Y} - \bar{Y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) + (\mathbf{Y} - \hat{\mathbf{Y}}).$$

That is,

$$\mathbf{Y} - \bar{Y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) + \mathbf{e}.$$

Because the model contains an intercept, the vector $\bar{Y}\mathbf{1}$ lies in the column space of $\mathbf{X}$. Hence both $\hat{\mathbf{Y}}$ and $\bar{Y}\mathbf{1}$ lie in the model space, so their difference also lies in the model space. Since the residual vector $\mathbf{e}$ is orthogonal to the model space, we have

$$(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1})^\top \mathbf{e} = 0.$$

Therefore, by the Pythagorean theorem,

$$\|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2 = \|\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}\|^2 + \|\mathbf{e}\|^2,$$

which is exactly

$$\text{SST} = \text{SSR} + \text{SSE}.$$

**Exercise 4.1** (In-Class Discussion: Why Does the Intercept Matter?)
Why does the usual centered decomposition $\text{SST} = \text{SSR} + \text{SSE}$ require the model space to contain the constant vector? Which step could fail without an intercept? Does this identity require normal errors?

**Solution 4.1**
With an intercept, $\mathbf{1}$ and therefore $\bar{Y}\mathbf{1}$ belong to the model space. Thus the centered fitted vector $\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}$ is orthogonal to $\mathbf{e}$, and the cross term vanishes when its squared norm is expanded. Without a constant vector in the model space, $\sum_i e_i$ need not be zero, so the cross term $\mathbf{e}^\top(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) = -\bar{Y}\sum_i e_i$ need not vanish. The identity follows from OLS geometry, not normality.

**Exercise 4.2** (In Class: One Observation, Three Deviations)
For one observation, $y_i = 1$, $\hat{y}_i = 1.2$, and $\bar{y} = 3$. In one minute, identify the total, explained, and residual deviations. Does the squared total deviation equal the sum of the other two squared deviations for this observation?

**Solution 4.2**
The deviations are $y_i - \bar{y} = -2$, $\hat{y}_i - \bar{y} = -1.8$, and $y_i - \hat{y}_i = -0.2$. They add as signed deviations: $-2 = -1.8 - 0.2$. Their squares do not add: $4 \neq 3.24 + 0.04$. The ANOVA identity holds after summing over all observations because the fitted and residual vectors are orthogonal; it is not a point-by-point identity for squared deviations.

### Degrees of Freedom

The ANOVA decomposition is accompanied by a decomposition of degrees of freedom. When the model includes an intercept and $\mathbf{X}$ has rank $p$, we have:

- total degrees of freedom: $n - 1$;
- regression degrees of freedom: $p - 1$;
- error degrees of freedom: $n - p$.

Thus,

$$n - 1 = (p - 1) + (n - p).$$

These match the sum of squares decomposition:

$$\text{SST} = \text{SSR} + \text{SSE}.$$

**Definition 4.2** (Regression and Error Mean Squares)
A **mean square** is a sum of squares divided by its associated degrees of freedom. For a full-rank regression design with an intercept and $1 < p < n$, define

$$\text{MSR} = \frac{\text{SSR}}{p - 1}, \qquad \text{MSE} = \frac{\text{SSE}}{n - p}.$$

Here $p$ includes the intercept coefficient.

Dividing by degrees of freedom accounts for the different dimensions contributing to each sum of squares. Under the linear-model moment assumptions from Chapter 3 — Distribution Theory of

OLS and Inference, MSE is an unbiased estimator of $\sigma^2$. MSR estimates the same noise level under the overall null hypothesis, which motivates their ratio.

## Section 4.2 The Overall F Test

A major inferential question is whether the regression model provides any explanatory power beyond the intercept-only model. Suppose the model includes an intercept and $p-1$ additional predictors. The null hypothesis is

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0,$$

where $\beta_0$ is the intercept and $p$ counts all regression coefficients. Equivalently, under $H_0$, the mean response does not depend on the predictors. The alternative is that at least one non-intercept coefficient is nonzero.

**Theorem 4.2** (Overall F Test)
Under the normal linear model with a fixed, full-rank design containing an intercept and $1 < p < n$, the null hypothesis that all non-intercept coefficients are zero gives

$$F = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} = \frac{\text{MSR}}{\text{MSE}} \sim F_{p-1,n-p}.$$

Large values provide evidence against the intercept-only mean model. Exact finite-sample inference requires the normal, common-variance error assumptions; the sum-of-squares identity alone does not supply this distribution.

### Interpretation of the Overall F Test

The numerator measures explained variation per regression degree of freedom. The denominator measures unexplained variation per residual degree of freedom. So the $F$ statistic compares:

- how much signal the model explains;
- how much noise remains in the residuals.

If the predictors have no effect, then both quantities should be of similar size, and the ratio should not be unusually large. If the predictors explain substantial variation, then the numerator should be much larger than the denominator.

**Exercise 4.3** (In Class: What Does an Overall Rejection Mean?)
An intercept model with two predictors is fitted to $n = 25$ observations using a full-rank design. Write the overall null hypothesis and the F reference degrees of freedom under the normal linear model. How does this joint test differ from the individual t tests? If it rejects, must both individual slope tests also reject?

**Solution 4.3**
Here $p = 3$, so $H_0 : \beta_1 = \beta_2 = 0$ and the reference distribution is $F_{2,22}$. Rejection gives

evidence that the two slopes are not jointly zero. It does not show that both are nonzero, and it does not require either individual test to reject at the same level. Each individual t test asks whether one slope is zero while retaining the other predictor; the overall F test asks whether both slopes are jointly zero. These are different questions, especially when predictors are correlated.

**Relationship to the Intercept-Only Model**

The overall $F$ test compares two models:

- the reduced model: intercept only;
- the full model: intercept plus predictors.

Thus the ANOVA decomposition provides the basis for formal model comparison. This leads naturally to the idea of nested models.

## Section 4.3 Nested Models

Before comparing models, we need to say what their regression parameters mean.

**Definition 4.3** (Regression Coefficients and the Mean Response)
For fixed predictor values, consider the additive model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \qquad \mathbb{E}(\varepsilon_i) = 0.$$

The **mean response function** is $\mu(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. If predictors are random, read this as a conditional mean, with $\mathbb{E}(\varepsilon \mid x_1, x_2) = 0$.

| Parameter | Interpretation in this additive model |
|---|---|
| $\beta_0$ | The intercept: mean response when both $x_1 = 0$ and $x_2 = 0$. |
| $\beta_1$ | Difference in mean response for a one-unit increase in $x_1$, holding $x_2$ fixed. |
| $\beta_2$ | Difference in mean response for a one-unit increase in $x_2$, holding $x_1$ fixed. |

For example,

$$\mu(x_1 + h, x_2) - \mu(x_1, x_2) = h\beta_1.$$

Thus a positive $\beta_1$ means a higher mean response at larger $x_1$ when $x_2$ is fixed; a negative value means a lower mean. Its units are **response units per unit of** $x_1$. A zero coefficient means no change in this model's mean as $x_1$ varies with $x_2$ fixed; it does not rule out an unadjusted association between $x_1$ and $Y$.

**Parameter, estimate, and observation.** The coefficients $\beta_0, \beta_1, \beta_2$ are unknown model parameters. The fitted coefficients $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ are estimates calculated from a sample and vary across samples. Substituting them gives a fitted mean $\hat{\mu}$. An individual observed response also contains its error term, so a slope does not prescribe the exact difference between two observations. These mean comparisons do not by themselves establish a causal effect.

**Example 4.1** (Read a Fitted Equation in Its Units)
For the trees introduced earlier in this chapter, a fitted equation is

$$\widehat{\text{Volume}} = -57.988 + 0.339\,\text{Height} + 4.708\,\text{Diameter}.$$

Volume is in cubic feet, height in feet, and diameter in inches.

- $\hat{\beta}_{\text{Height}} = 0.339$: at the **same diameter**, trees one foot apart in height differ by an estimated 0.339 cubic feet in mean volume. A five-foot difference corresponds to $5(0.339) = 1.695$ cubic feet, using the rounded slope.
- $\hat{\beta}_{\text{Diameter}} = 4.708$: at the **same height**, a one-inch diameter difference corresponds to an estimated 4.708-cubic-foot difference in mean volume.
- $\hat{\beta}_0 = -57.988$: the fitted mean at height zero and diameter zero. Those values are outside the observed data, so this intercept is an extrapolated baseline, not a useful prediction for an actual tree.

The larger numerical diameter coefficient does not by itself make diameter the more important predictor: the predictors have different units and ranges, and the two coefficients describe different comparisons. The fitted coefficients also do not tell us their own uncertainty; that requires the standard errors and inference from Chapter 3.

We can now ask how these interpretations change when a predictor is added.

**Definition 4.4** (Nested Linear Models)
For the same response vector and observations, let the reduced and full models have mean spaces $\mathcal{C}(\mathbf{X}_R)$ and $\mathcal{C}(\mathbf{X}_F)$. The models are **nested** when

$$\mathcal{C}(\mathbf{X}_R) \subseteq \mathcal{C}(\mathbf{X}_F).$$

Thus every mean vector allowed by the reduced model is also allowed by the full model. A strict comparison requires ranks $p_R < p_F$; adding columns that are linear combinations of existing columns does not enlarge the mean space.

**Example 4.2** (Adding a Predictor)
Use different parameter symbols to keep the two candidate mean models distinct:

$$\mu_R(x_1) = \alpha_0 + \alpha_1 x_1,$$
$$\mu_F(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2.$$

| Coefficient | Mean comparison described by its model |
|---|---|
| $\alpha_1$ | A one-unit difference in $x_1$, without adjusting for $x_2$. |
| $\beta_1$ | A one-unit difference in $x_1$, holding $x_2$ fixed. |
| $\beta_2$ | A one-unit difference in $x_2$, holding $x_1$ fixed. |

The reduced intercept is the mean at $x_1 = 0$ under the reduced model; the full intercept is the mean at $x_1 = x_2 = 0$ under the full model. Each interpretation depends on the specified mean function and on whether its baseline is supported by the data.
Setting $\beta_2 = 0$ recovers the reduced mean function, with $\alpha_0 = \beta_0$ and $\alpha_1 = \beta_1$ **under that restriction**. If the $x_2$ column is not in the span of the intercept and $x_1$, the full model adds one degree of freedom.
When we fit the two models separately, OLS re-estimates every free coefficient. Nestedness therefore does **not** require $\hat{\alpha}_1 = \hat{\beta}_1$. Even if the reduced model is true, fitting the unrestricted full model can give different estimates in a finite sample. Chapter 5 illustrates this with the same height measurements before and after adjustment for diameter.

**i** Interpret the restriction in the full model

For the models above, the partial F comparison tests $H_0 : \beta_2 = 0$: does $x_2$ contribute to the mean after accounting for $x_1$? It does not directly test equality of the two fitted $x_1$ coefficients. A change in the size, sign, or p-value of the $x_1$ coefficient can accompany the change in adjustment; compare the scientific questions before comparing printed numbers.

**Exercise 4.4** (In Class: Which Parameter Is Being Tested?)
Compare `Volume ~ Height` with `Volume ~ Height + Diameter` on the same trees. Which null hypothesis does their partial F test address?
A. The two fitted height coefficients are equal.
B. The diameter coefficient in the full model is zero, after accounting for height.
C. The height coefficient in the full model is zero, after accounting for diameter.

**Solution 4.4**
**B.** Diameter is the added predictor. Statement C would instead compare `Volume ~ Diameter` with `Volume ~ Diameter + Height`. Neither comparison is a direct test that the height estimates from two different models are equal.

**Definition 4.5** (Extra Sum of Squares)
For nested models fitted by OLS to the same response and observations, define the **extra sum of squares** for the full model relative to the reduced model as

$$\mathrm{SS}_{\mathrm{extra}} = \mathrm{SSE}_R - \mathrm{SSE}_F.$$

Its associated degrees of freedom are $p_F - p_R$, where $p_R$ and $p_F$ are the design-matrix ranks.

**Exercise 4.5** (In-Class Discussion: Can the Larger Model Fit Worse?)
Why must $\mathrm{SSE}_F \leq \mathrm{SSE}_R$ when nested models are fitted by OLS to the same observations? Must the inequality be strict whenever a genuinely new predictor direction is added?

**Solution 4.5**
Every fitted mean vector allowed by the reduced model is also available to the full model. Minimizing the same residual sum of squares over this larger set cannot give a larger minimum. Equality is possible even if the model space grows: for the observed response, the added directions may provide no further reduction in residual variation. Thus greater flexibility guarantees a weak improvement in training fit, not a strictly positive improvement or a useful predictor.

A positive extra sum of squares alone is not evidence against the reduced model: the F test scales that improvement by its degrees of freedom and the estimated noise variance.

**Theorem 4.3** (Partial F Test for Nested Normal Linear Models)
Fit both models to the **same response and the same observations**. Suppose their fixed design matrices have nested column spaces, with ranks $p_R < p_F < n$, and the full model has independent $N(0, \sigma^2)$ errors. Under the null hypothesis that the mean belongs to the reduced model,

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_F)/(p_F - p_R)}{\mathrm{SSE}_F/(n - p_F)} \sim F_{p_F - p_R, n - p_F}.$$

Large values give evidence against the reduced mean model. The numerator measures improvement per added degree of freedom; the denominator estimates the unexplained variance under the full model.

The appendix at the end of this chapter derives $\mathrm{SSE}_R - \mathrm{SSE}_F$ geometrically and shows how Theorem 4.2 is the special case of Theorem 4.3 in which the reduced model contains only an intercept.

**!** Compare the models, not just the printed p-values

Different missing-value patterns can cause two R fits to use different observations. Construct a common analysis dataset before comparing them. For one added coefficient, the partial F statistic equals the square of its two-sided t-test statistic from the full model.

**Exercise 4.6** (In Class: A Better Fit, but Enough Improvement?)
Two nested models use the same $n = 30$ observations. The reduced model has rank $p_R = 2$ and $\mathrm{SSE}_R = 90$; the full model has rank $p_F = 4$ and $\mathrm{SSE}_F = 78$. Compute the extra sum of squares, its degrees of freedom, the full-model MSE, and the partial F statistic. Does a reduction in SSE by itself justify rejecting the reduced model?

**Solution 4.6**
The improvement is $90 - 78 = 12$ on $4 - 2 = 2$ degrees of freedom. The full-model MSE is $78/(30 - 4) = 3$, so $F = (12/2)/3 = 2$. Under the normal-model null, compare this with $F_{2,26}$; the upper-tail p-value is about 0.156. The improvement is insufficient for rejection at the 5% level. More flexibility always weakly reduces training SSE, including when the added coefficients are zero.

### Connection with General Linear Hypotheses

Nested column-space models correspond to linear restrictions on the full-model coefficients. More generally, Chapter 3 — Distribution Theory of OLS and Inference allows a specified affine restriction of the form

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}.$$

So the ANOVA comparison of nested models is another way of expressing the general $F$ test from Chapter 3 — Distribution Theory of OLS and Inference. In practice, this is one of the most common uses of regression ANOVA tables.

**Definition 4.6** (Sequential and Partial Sums of Squares)
For a term in a specified regression model, both quantities are extra sums of squares from a nested comparison on the same observations:

- **Sequential sum of squares:** the reduction in SSE when the term is added after the preceding terms in a specified order.
- **Partial sum of squares:** the increase in SSE when the specified term is removed from a full model while the other specified terms are retained.

When predictors are correlated, sequential sums of squares can depend on their order. Partial sums of squares answer a comparison conditional on the terms retained in the reduced model. With interactions or rank deficiency, state the exact restriction and respect model hierarchy rather than relying only on a Type I, II, or III label.

**Definition 4.7** (Coefficient of Determination)
For an OLS fit with an intercept and $\mathrm{SST} > 0$, the **coefficient of determination** is

$$R^2 = \frac{\mathrm{SSR}}{\mathrm{SST}} = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}.$$

By the ANOVA decomposition, $R^2$ lies between 0 and 1 and measures the proportion of centered variation explained by the fitted model. If every observed response is identical, $\mathrm{SST} = 0$ and this ratio is undefined.

### Interpretation of R Squared

- If $R^2$ is close to 1, the model explains a large proportion of the variation in the response.
- If $R^2$ is close to 0, the model explains little of the variation.

However, $R^2$ alone does not guarantee that the model is appropriate. A high $R^2$ does not ensure that assumptions are satisfied, and a low $R^2$ does not necessarily imply the model is useless.

**Definition 4.8** (Adjusted Coefficient of Determination)
For an OLS fit with an intercept, $\mathrm{SST} > 0$, and design rank $p < n$, define the **adjusted coefficient of determination** as

$$R^2_{\text{adj}} = 1 - \frac{\text{SSE}/(n-p)}{\text{SST}/(n-1)}.$$

Ordinary $R^2$ cannot decrease when the model space is enlarged on the same data. Adjusted $R^2$ accounts for the loss of residual degrees of freedom and can decrease or become negative. It compares residual mean square with the sample variance of the response; it is not a literal proportion of explained variation or an estimate of out-of-sample predictive performance.

**Example 4.3** (A Small ANOVA Calculation)
For the column vectors $\mathbf{x} = (0, 1, 2, 3)$ and $\mathbf{y} = (1, 3, 3, 5)$, the fitted line is $\hat{y} = 1.2 + 1.2x$. Therefore $\hat{\mathbf{y}} = (1.2, 2.4, 3.6, 4.8)$, $\mathbf{e} = (-0.2, 0.6, -0.6, 0.2)$, and $\bar{y} = 3$.

$$\begin{aligned}
\text{SST} &= (1-3)^2 + (3-3)^2 + (3-3)^2 + (5-3)^2 = 8,\\
\text{SSE} &= (-0.2)^2 + 0.6^2 + (-0.6)^2 + 0.2^2 = 0.8,\\
\text{SSR} &= \text{SST} - \text{SSE} = 8 - 0.8 = 7.2.
\end{aligned}$$

Here $n = 4$ and $p = 2$, so the regression, residual, and total degrees of freedom are 1, 2, and 3, respectively.

| Source | Sum of squares | df | Mean square | F |
|---|---|---|---|---|
| Regression | 7.2 | 1 | 7.2 | 18 |
| Error | 0.8 | 2 | 0.4 | |
| Total | 8.0 | 3 | | |

The model has $R^2 = 7.2/8 = 0.9$, yet the F-test p-value is $P(F_{1,2} \geq 18) \approx 0.0513$. With only two residual degrees of freedom, a high training $R^2$ does not automatically give evidence at the 5% significance level.

**Exercise 4.7** (In Class: Can Two Fit Summaries Disagree?)
For the same $n = 10$ observations with SST $= 100$, an intercept model has rank 2 and SSE 20. Adding a predictor whose column is linearly independent of the existing columns increases the rank to 3 and reduces SSE to 19. Compute $R^2$ and adjusted $R^2$ for both fits. Did the one-unit reduction in SSE improve both summaries? Why can relying on $R^2$ alone be misleading?

**Solution 4.7**
Ordinary $R^2$ increases from $1 - 20/100 = 0.80$ to $1 - 19/100 = 0.81$. Adjusted $R^2$ decreases from $1 - (20/8)/(100/9) = 0.775$ to $1 - (19/7)/(100/9) \approx 0.756$. The extra predictor reduces SSE but increases residual mean square, from 2.5 to about 2.714. Training $R^2$ cannot decrease when the model space grows, even when the new term only fits noise. A high $R^2$ also does not establish a correct mean function, appropriate error assumptions, or causality. Neither summary alone establishes better prediction on new observations.

### ANOVA Table Structure

A standard regression ANOVA table has the following structure:

| Source | Sum of Squares | Degrees of Freedom | Mean Square | F |
|---|---|---|---|---|
| Regression | SSR | $p-1$ | MSR | MSR/MSE |
| Error | SSE | $n-p$ | MSE | |
| Total | SST | $n-1$ | | |

Students should learn to move fluently between:

- formulas;
- geometric interpretation;
- software output.

## Section 4.4 R Demonstration

### Fit a simple regression model

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

fit <- lm(y ~ x)
summary(fit)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4
-0.2  0.6 -0.6  0.2

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.2000     0.5292   2.268   0.1515
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9, Adjusted R-squared:   0.85
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

### Obtain the ANOVA table

```
anova(fit)
```

```
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value  Pr(>F)
x          1    7.2     7.2      18 0.05132 .
Residuals  2    0.8     0.4
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

### Verify sums of squares manually

```
ybar <- mean(y)
yhat <- fitted(fit)
e <- resid(fit)

SST <- sum((y - ybar)^2)
SSE <- sum(e^2)
SSR <- sum((yhat - ybar)^2)

c(SST = SST, SSR = SSR, SSE = SSE)
```

```
SST SSR SSE
8.0 7.2 0.8
```

```
SST - SSR - SSE
```

```
[1] 3.330669e-16
```

**From deviations to sums of squares.** Figure 4.2 separates three operations: **subtract, square, then add**. Each column is one of the four observations. Read downward from a signed difference to its nonnegative squared contribution, then add across the row to obtain SS. The minus signs in the first row describe deviations; the boxed totals are the sums of squares.

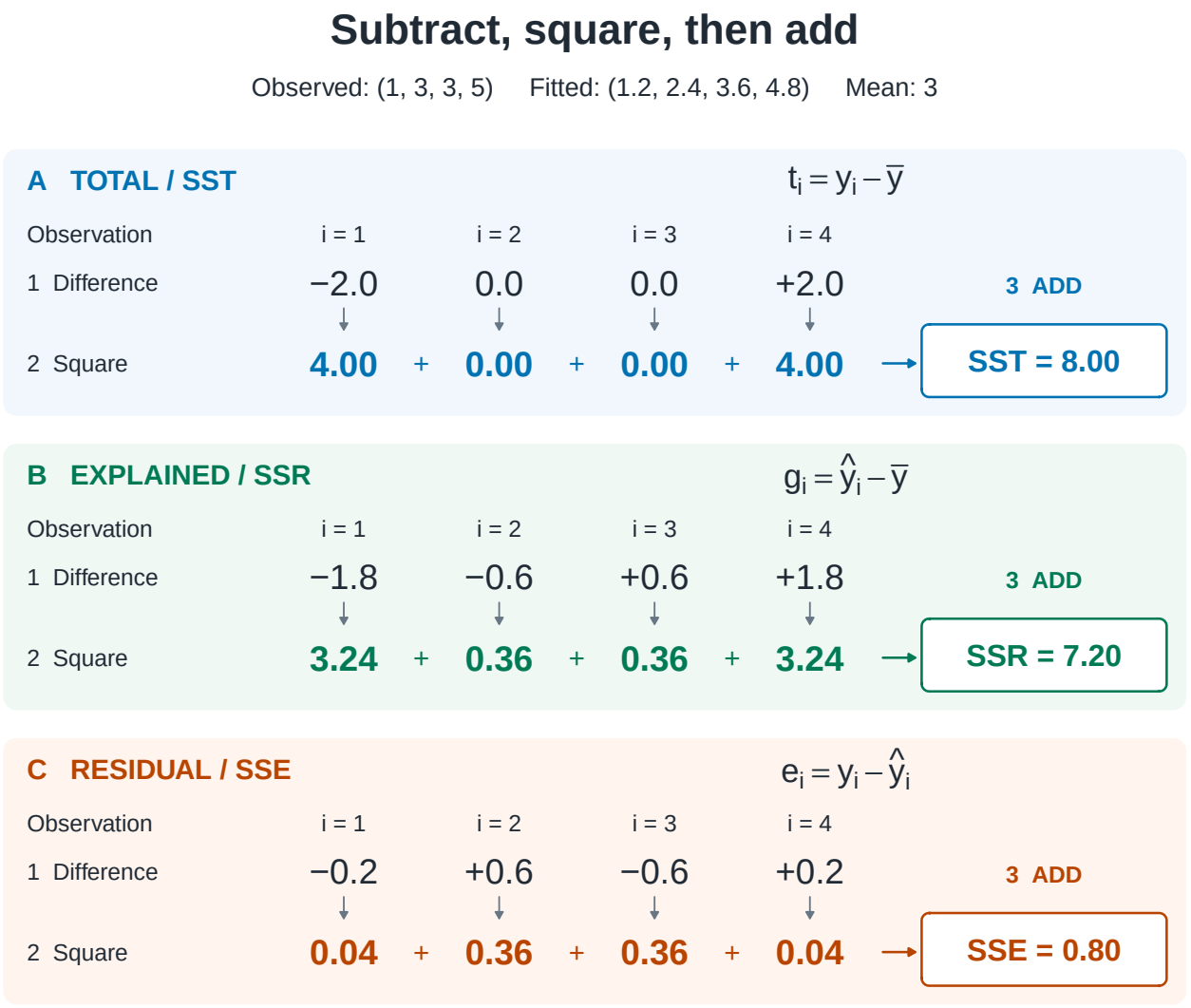


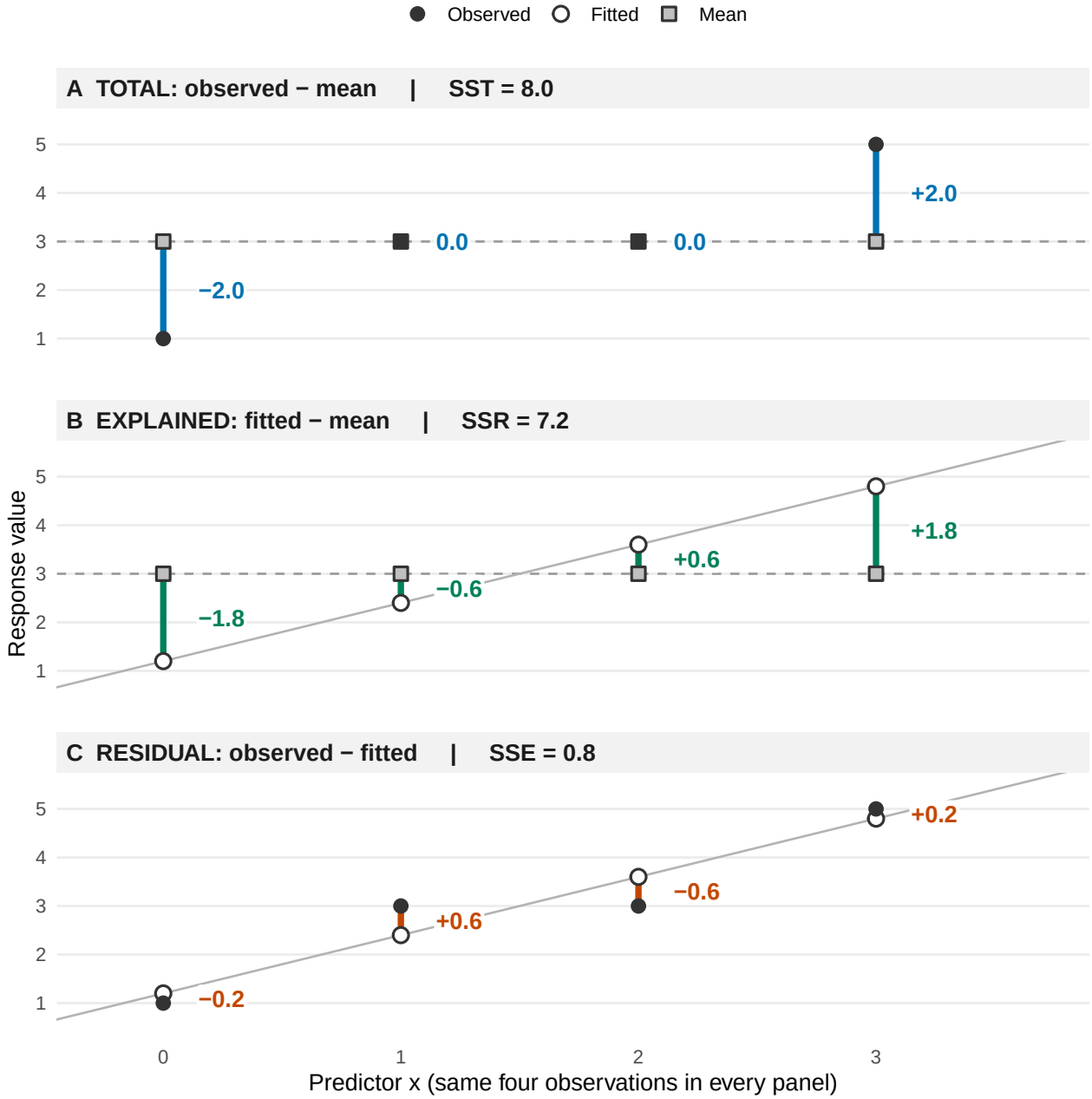

Figure 4.2: How to compute SST, SSR, and SSE from the same four observations. The first row in each panel contains signed deviations, the second contains their squares, and the box at the right contains the sum of those squares. Negative deviations become positive contributions when squared. The resulting sums are SST = 8.00, SSR = 7.20, and SSE = 0.80; none is negative.

**Adding deviations is different from adding their squares.** For these residuals,

$$\sum_i e_i = -0.2 + 0.6 - 0.6 + 0.2 = 0,$$
$$\text{SSE} = \sum_i e_i^2 = 0.04 + 0.36 + 0.36 + 0.04 = 0.80.$$

The signed residuals sum to zero because the OLS model includes an intercept. Their squared values do not cancel. Here all three SS values are strictly positive; in general an SS can equal zero when every corresponding deviation is zero.

**The same deviations in the $x, y$ plane.** Each point represents one of the four observations. The three panels use identical axes to show total, explained, and residual deviations. Squaring and adding the colored vertical differences gives the SS values in Figure 4.2.




Companion $x, y$ view for Figure 4.2. Filled circles are observed responses, open circles are fitted values, and squares mark the sample mean. The dashed line is the mean; the thin gray line is the fitted regression. Colored vertical segments show total (blue), explained (green), or residual (orange) deviations; their labels are signed differences, not sums of squares. Squaring and adding within each panel gives SST = 8.0, SSR = 7.2, and SSE = 0.8.

For example, at $x = 1$, the observed value equals the mean, so the **total deviation is zero**. Nevertheless, the explained deviation is $2.4 - 3 = -0.6$ and the residual is $3 - 2.4 = +0.6$. They

cancel as signed differences: $0 = -0.6 + 0.6$. Their squares do not cancel. Only after summing across all four observations do we obtain $8.0 = 7.2 + 0.8$.

### Compute R squared manually

```
R2 <- SSR / SST
R2
```

```
[1] 0.9
```

```
summary(fit)$r.squared
```

```
[1] 0.9
```

```
summary(fit)$adj.r.squared
```

```
[1] 0.85
```

```
pf(18, df1 = 1, df2 = 2, lower.tail = FALSE)
```

```
[1] 0.0513167
```

### Compare nested models

```
dat <- data.frame(
  y = c(4, 5, 7, 10, 8, 12, 13, 14),
  x1 = c(1, 2, 3, 4, 5, 6, 7, 8),
  x2 = c(2, 1, 3, 2, 5, 4, 6, 5)
)

fit_reduced <- lm(y ~ x1, data = dat)
fit_full <- lm(y ~ x1 + x2, data = dat)

anova(fit_reduced, fit_full)
```

```
Analysis of Variance Table

Model 1: y ~ x1
Model 2: y ~ x1 + x2
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1      6 6.8214
2      5 4.6613  1    2.1601 2.3171 0.1884
```

### Inspect the two fitted models

```r
summary(fit_reduced)
```

```
Call:
lm(formula = y ~ x1, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-1.85714 -0.30357  0.03571  0.33036  1.60714

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.5357     0.8308   3.052 0.022453 *
x1            1.4643     0.1645   8.900 0.000112 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.066 on 6 degrees of freedom
Multiple R-squared:  0.9296,    Adjusted R-squared:  0.9178
F-statistic: 79.21 on 1 and 6 DF,  p-value: 0.0001121
```

```r
summary(fit_full)
```

```
Call:
lm(formula = y ~ x1 + x2, data = dat)

Residuals:
      1       2       3       4       5       6       7       8
 0.4032 -1.0403  0.3306  0.8871 -1.1371  0.4194  0.7903 -0.6532

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.9677     0.8041   3.691  0.01413 *
x1            1.8387     0.2876   6.394  0.00139 **
x2           -0.6048     0.3973  -1.522  0.18845
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9655 on 5 degrees of freedom
Multiple R-squared:  0.9519,    Adjusted R-squared:  0.9326
F-statistic: 49.46 on 2 and 5 DF,  p-value: 0.0005079
```

### Interpretation of Software Output

For a fitted model in R:

- `anova(fit)` gives sequential sums of squares in the order terms enter a single model;
- `anova(fit_reduced, fit_full)` compares nested models;
- `summary(fit)` reports the overall $F$ statistic, $R^2$, and adjusted $R^2$.

Students should understand that these outputs are not separate topics. They are all built from the same least squares geometry and distribution theory.

**Example 4.4** (Optional Real-Data Example: Is Tree Height Useful Beyond Diameter?)
Return to the forestry question. R's `trees` data contain timber volume in cubic feet, height in feet, and a variable named `Girth` that is actually **diameter in inches**, measured 4 feet 6 inches above the ground. We rename it `Diameter` to avoid that labeling trap. Each row represents one of 31 felled black cherry trees. See the official dataset documentation for the source and measurement definitions.
**State the question first.** Comparing $M_0$ with $M_2$ asks whether diameter and height together improve on a constant mean. Comparing $M_1$ with $M_2$ asks whether height contributes after adjusting for diameter. For the second question, $H_0 : \beta_{\text{Height}} = 0$. These are distinct questions even though both use an F test.

```
tree_dat <- datasets::trees
tree_dat$Diameter <- tree_dat$Girth
tree_dat <- tree_dat[, c("Volume", "Diameter", "Height")]
stopifnot(!anyNA(tree_dat))
fit_tree0 <- lm(Volume ~ 1, data = tree_dat)
fit_tree_d <- lm(Volume ~ Diameter, data = tree_dat)
fit_tree_dh <- lm(Volume ~ Diameter + Height, data = tree_dat)
knitr::kable(data.frame(
  Model = c("Constant mean", "Diameter", "Diameter + height"),
  Rank = c(fit_tree0$rank, fit_tree_d$rank, fit_tree_dh$rank),
  SSE = c(deviance(fit_tree0), deviance(fit_tree_d),
          deviance(fit_tree_dh))), digits = 2)
```

| Model | Rank | SSE |
|---|---|---|
| Constant mean | 1 | 8106.08 |
| Diameter | 2 | 524.30 |
| Diameter + height | 3 | 421.92 |

```
anova(fit_tree_d, fit_tree_dh)
```

```
Analysis of Variance Table

Model 1: Volume ~ Diameter
Model 2: Volume ~ Diameter + Height
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1     29 524.30
2     28 421.92  1    102.38 6.7943 0.01449 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Read the comparison.** The SSE drops from about 524.30 to 421.92. The added term uses one degree of freedom, and the full model has $31 - 3 = 28$ residual degrees of freedom. Thus

$$F = \frac{(524.30 - 421.92)/1}{421.92/28} \approx 6.79, \qquad p \approx 0.0145.$$

Under the stated normal, common-variance error model, this provides evidence against a zero height coefficient after accounting for diameter. It is not the overall test of both slopes.

```r
library(ggplot2)
tree_comparison <- rbind(
  transform(tree_dat, prediction = fitted(fit_tree_d),
    model = "Diameter only"),
  transform(tree_dat, prediction = fitted(fit_tree_dh),
    model = "Diameter + height"))
tree_comparison$model <- factor(tree_comparison$model,
                                levels = c("Diameter only",
                                  "Diameter + height"))
ggplot(tree_comparison, aes(Diameter, Volume)) +
  geom_segment(aes(xend = Diameter, yend = prediction, colour = "Residual"),
    linewidth = 0.65, show.legend = TRUE) +
  geom_point(aes(colour = "Observed"), shape = 16, size = 1.8,
    show.legend = TRUE) +
  geom_point(aes(y = prediction, colour = "Fitted"), shape = 1, size = 2,
    show.legend = TRUE) +
  scale_colour_manual(
    values = c(Observed = "black", Fitted = "#0072B2", Residual = "#B84400"),
    breaks = c("Observed", "Fitted", "Residual"),
    labels = c("Observed volume", "Fitted volume",
               "Residual (observed minus fitted)")) +
  guides(colour = guide_legend(nrow = 3, byrow = TRUE,
    override.aes = list(shape = c(16, 1, NA), linetype = c(0, 0, 1),
                        size = 2.8, linewidth = 0.8))) +
  facet_wrap(~ model, nrow = 1) +
  labs(x = "Diameter (inches)", y = "Timber volume (cubic feet)",
       colour = NULL) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", legend.text = element_text(size = 11),
        panel.grid.minor = element_blank(),
        legend.key.width = grid::unit(1.3, "lines"),
        legend.key.height = grid::unit(1, "lines"))
```

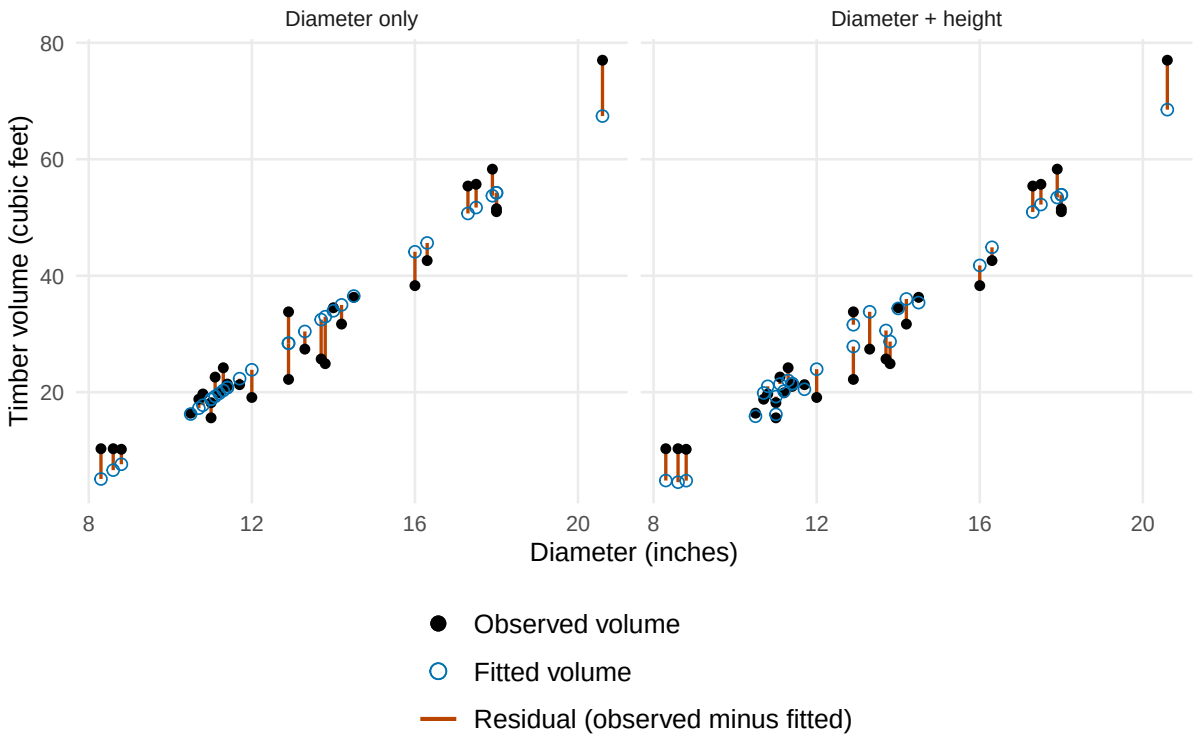



Figure 4.3: The same 31 trees under two mean models. The legend identifies black filled dots as observed volumes, blue open circles as fitted volumes, and orange vertical segments as residuals (observed minus fitted). Each segment connects the observation and prediction for one tree. Adding height reduces the sum of squared residual lengths, but the F test asks whether that improvement is large relative to the estimated noise.

**Check your understanding.** Why use 28 rather than 29 in the variance estimate? What would change if some trees had missing heights? The denominator comes from the full model. With missing heights, first form one common dataset and refit both models on it; comparing fits on different observations confounds the added predictor with a changed sample.

**What this example does not settle.** These are observational measurements, so the comparison is not a causal effect of making a tree taller. A statistically significant height term also does not decide whether measuring height is worth its cost or how much it helps predict new trees. Those decisions require practical error tolerances and validation. Inspect residuals and the mean function before using the raw-scale linear model operationally; transformations and model assessment appear later in the course.

## Section 4.5 Summary

In this chapter, we developed the ANOVA framework for linear regression. We defined:

$$\text{SST}, \qquad \text{SSR}, \qquad \text{SSE},$$

and showed that, with an intercept,

$$\text{SST} = \text{SSR} + \text{SSE}.$$

This decomposition led to:

- the ANOVA table;
- the overall $F$ test for regression;
- the comparison of nested models through extra sums of squares;
- the interpretation of $R^2$ and adjusted $R^2$.

## Section 4.6 Practice Problems

### Conceptual

1. Explain the meaning of SST, SSR, and SSE in words.
2. Explain why the regression degrees of freedom are $p-1$ when the model includes an intercept.
3. Explain the difference between the overall $F$ test and a test for a single coefficient.

### Computational

Suppose a regression model with intercept has:

- $n = 20$,
- $p = 4$,
- $\text{SST} = 100$,
- $\text{SSE} = 40$.

Compute:

1. SSR,
2. the degrees of freedom for regression and error,
3. MSR,
4. MSE,
5. the overall $F$ statistic,
6. $R^2$.

### Nested Model Problem

A reduced model has $\text{SSE}_R = 120$ with $p_R = 3$, and a full model has $\text{SSE}_F = 90$ with $p_F = 5$. If $n = 30$, compute the nested-model $F$ statistic.

### Suggested Take home exercise

Complete the following tasks:

- prove the decomposition $\text{SST} = \text{SSR} + \text{SSE}$ when the model includes an intercept;
- derive the overall $F$ statistic from the ANOVA decomposition;
- fit a regression model in R and reproduce the ANOVA table by hand;
- compare two nested models using an extra sum of squares test;
- interpret both $R^2$ and adjusted $R^2$ for a chosen dataset.

# Appendix: From the SSE Difference to the Overall F Test

Theorem 4.2 is the intercept-only special case of Theorem 4.3: the extra sum of squares $\mathrm{SSE}_R - \mathrm{SSE}_F$ then equals SSR.

## A geometric picture of the extra sum of squares

Each vertex in Figure 4.4 represents a vector of $n$ values. The green edge joins the reduced and full fitted vectors; its **squared length** is $\mathrm{SSE}_R - \mathrm{SSE}_F$.

For an intercept-only reduced model, $\hat{\mathbf{Y}}_R = (\bar{Y}, \ldots, \bar{Y})^\top = \bar{Y}\,\mathbf{1}_n$, where $\mathbf{1}_n$ is the all-ones vector. Panel B uses the same four observations as Figure 4.2, with $\bar{Y} = 3$.

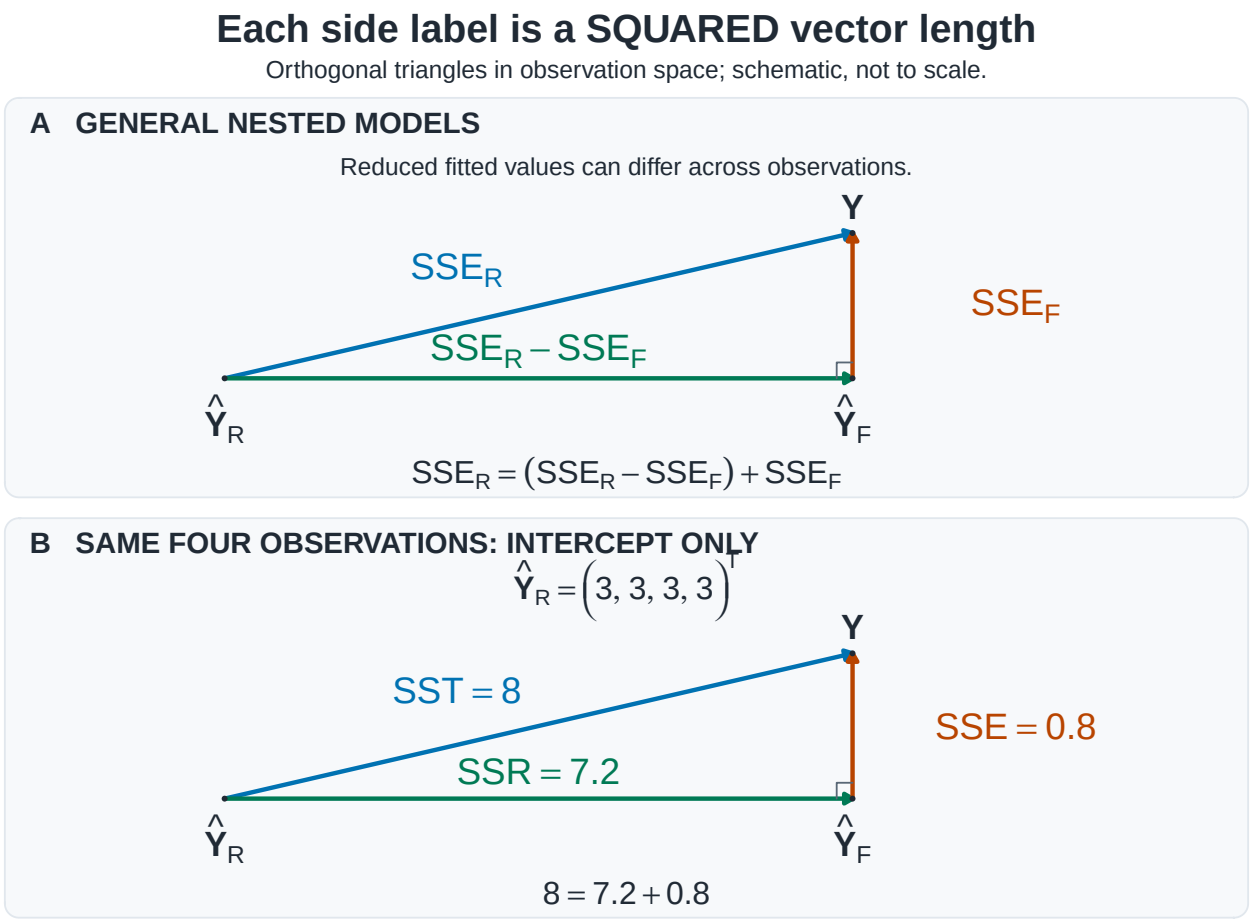



Figure 4.4: Squared lengths for nested OLS fits. Panel B uses the data in Figure 4.2: $\mathrm{SSE}_R = \mathrm{SST} = 8$, $\mathrm{SSE}_F = \mathrm{SSE} = 0.8$, and the green squared length is $\mathrm{SSE}_R - \mathrm{SSE}_F = \mathrm{SSR} = 7.2$. The diagrams are schematic, not to scale.

## Why subtract the two residual sums of squares?

*Proof.* Fit nested OLS models to the same response vector and observations. Write $\mathbf{P}_R$ and $\mathbf{P}_F$ for the orthogonal projections onto their mean spaces, so

$$\hat{\mathbf{Y}}_R = \mathbf{P}_R\mathbf{Y}, \qquad \hat{\mathbf{Y}}_F = \mathbf{P}_F\mathbf{Y}, \qquad \mathbf{e}_R = \mathbf{Y} - \hat{\mathbf{Y}}_R, \qquad \mathbf{e}_F = \mathbf{Y} - \hat{\mathbf{Y}}_F.$$

The reduced-model residual splits into two parts:

$$\underbrace{\mathbf{Y} - \hat{\mathbf{Y}}_R}_{\mathbf{e}_R} = \underbrace{\hat{\mathbf{Y}}_F - \hat{\mathbf{Y}}_R}_{\text{change in fitted values}} + \underbrace{\mathbf{Y} - \hat{\mathbf{Y}}_F}_{\mathbf{e}_F}.$$

Both fitted vectors belong to the full-model space. Their difference therefore lies in that space and is orthogonal to $\mathbf{e}_F$. Taking squared lengths gives

$$\mathrm{SSE}_R = \|\hat{\mathbf{Y}}_F - \hat{\mathbf{Y}}_R\|^2 + \mathrm{SSE}_F,$$

and hence

$$\boxed{\mathrm{SSE}_R - \mathrm{SSE}_F = \|\hat{\mathbf{Y}}_F - \hat{\mathbf{Y}}_R\|^2 = \mathbf{Y}^\top(\mathbf{P}_F - \mathbf{P}_R)\mathbf{Y}.}$$

□

The difference is nonnegative and measures the part of the reduced-model residual that the full model can fit. This identity requires nested OLS fits on the same observations; normal errors and an intercept are not needed.

## Recovering Theorem 4.2 from Theorem 4.3

*Proof.* Now choose the reduced model to be **intercept only**, with rank $p_R = 1$. Its fitted value for every observation is $\bar{Y}$, so

$$\hat{\mathbf{Y}}_R = \bar{Y}\,\mathbf{1}_n, \qquad \mathrm{SSE}_R = \sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \mathrm{SST}.$$

Let the full model contain an intercept and have rank $p_F = p$. Then

$$\mathrm{SSE}_F = \mathrm{SSE}, \qquad \mathrm{SSE}_R - \mathrm{SSE}_F = \mathrm{SST} - \mathrm{SSE} = \mathrm{SSR}, \qquad p_F - p_R = p - 1.$$

Substituting these quantities into Theorem 4.3 gives exactly Theorem 4.2:

$$\underbrace{\frac{(\mathrm{SSE}_R - \mathrm{SSE}_F)/(p_F - p_R)}{\mathrm{SSE}_F/(n - p_F)}}_{\text{partial F statistic}} = \underbrace{\frac{\mathrm{SSR}/(p-1)}{\mathrm{SSE}/(n-p)}}_{\text{overall F statistic}}.$$

The null hypothesis also becomes the overall null: the full-model mean is constant, or equivalently, all its non-intercept coefficients are zero. □

For the four observations in Figure 4.2, the intercept-only fit has $\mathrm{SSE}_R = 8.0$, and the fitted line has $\mathrm{SSE}_F = 0.8$. Thus

$$F = \frac{(8.0 - 0.8)/(2-1)}{0.8/(4-2)} = \frac{7.2}{0.4} = 18,$$

which is the same calculation as MSR/MSE in Example 4.3.

**i** When the reduced model already has predictors

If both models contain an intercept, they share the same centered SST. Therefore

$$\text{SSE}_R - \text{SSE}_F = (\text{SST} - \text{SSR}_R) - (\text{SST} - \text{SSR}_F) = \text{SSR}_F - \text{SSR}_R.$$

The numerator uses the **additional** explained variation beyond the reduced model. It is generally not the full model's entire $\text{SSR}_F$. For example, adding height to diameter explains about $524.30 - 421.92 = 102.38$ additional squared-volume units; the variation already explained by diameter is excluded.

## Why does the ratio have an F distribution?

*Proof.* Let $q = p_F - p_R$ and assume the normal, common-variance model in Theorem 4.3. Under $H_0$, the mean vector $\boldsymbol{\mu}$ lies in the reduced-model space. Nestedness implies

$$\mathbf{P}_F\mathbf{P}_R = \mathbf{P}_R\mathbf{P}_F = \mathbf{P}_R.$$

Consequently, $\mathbf{A} = \mathbf{P}_F - \mathbf{P}_R$ and $\mathbf{B} = \mathbf{I} - \mathbf{P}_F$ are symmetric idempotent matrices of ranks $q$ and $n - p_F$. Also, $\mathbf{AB} = \mathbf{0}$, $\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$, and $\mathbf{B}\boldsymbol{\mu} = \mathbf{0}$. The normal projection results from Chapter 3 therefore give

$$\frac{\text{SSE}_R - \text{SSE}_F}{\sigma^2} = \frac{\mathbf{Y}^\top\mathbf{AY}}{\sigma^2} \sim \chi^2_q, \qquad \frac{\text{SSE}_F}{\sigma^2} = \frac{\mathbf{Y}^\top\mathbf{BY}}{\sigma^2} \sim \chi^2_{n-p_F},$$

independently. Their ratio, after dividing each by its degrees of freedom, has distribution $F_{q,n-p_F}$. □

Under $H_0$, both mean squares estimate $\sigma^2$. The full-model residual provides an independent noise estimate; $\text{SSE}_R$ includes the improvement being tested. The geometric identity alone does not establish the F distribution: normal, common-variance errors are also required.

## Section 5

# Multiple Regression and Categorical Predictors

Chapter 4 asked whether additional columns of a design matrix improve a model. This chapter asks what those columns mean. We connect a scientific comparison to a mean function, construct the corresponding design matrix, and interpret the fitted coefficients.

**Learning Objectives**

By the end of this chapter, students should be able to:

- distinguish a marginal association from an adjusted association;
- translate a factor into indicator columns and identify the reference group;
- count model parameters and recognize redundant columns;
- write group-specific mean functions for additive and interaction models;
- explain how centering and reference coding change coefficient meanings;
- match a scientific question to a nested-model comparison;
- use `model.matrix()`, `lm()`, and predictions to check an interpretation.

**Reading.** Read the sections on multiple regression, qualitative predictors, and interaction terms in Montgomery, Peck, and Vining (Montgomery et al. 2021) and Seber and Lee (Seber and Lee 2003). The R examples use built-in datasets and require no additional data downloads.

> **i** Two class meetings, one modelling workflow
>
> In the first meeting, focus on adjusted interpretation, indicator coding, and additive models. In the second, study interactions, centering, and the model comparisons that distinguish different scientific questions. Chapter 6 will develop contrasts and inference for selected comparisons in more detail.
> Our workflow is **question → mean function → design matrix → interpretation → comparison**. The least squares and inference tools remain those from Chapters 2–4.

## Section 5.1 Multiple regression and adjusted interpretation

For $k$ continuous predictors, an additive mean model is

$$\mu(\mathbf{x}_i) = \mathbb{E}(Y_i \mid \mathbf{x}_i) = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij}, \qquad Y_i = \mu(\mathbf{x}_i) + \varepsilon_i.$$

With an intercept and no redundant columns, $p = k+1$. Throughout this course, $p$ **counts columns of the design matrix, including the intercept**. A factor or an interaction can contribute several columns, so the number of named variables need not equal the number of parameters.

The design matrix is

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix}.$$

Full column rank gives the familiar estimator $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$. Least squares itself does not require normal errors. For the exact $t$ and $F$ procedures in this chapter, we use the normal linear model from Chapter 3: fixed full-rank $\mathbf{X}$ and independent $N(0, \sigma^2)$ errors.

**Definition 5.1** (Adjusted Regression Coefficient)
In an additive linear mean model, $\beta_j$ is the difference in mean response associated with a one-unit increase in $x_j$, **holding the other predictors fixed**:

$$\mu(x_1, \ldots, x_j + 1, \ldots, x_k) - \mu(x_1, \ldots, x_j, \ldots, x_k) = \beta_j.$$

The intercept is the mean response when $x_1 = \cdots = x_k = 0$. It may describe a useful baseline, or a point outside the range supported by the data.
**Parameter versus estimate.** $\beta_j$ is an unknown parameter of the specified mean model; $\hat{\beta}_j$ is its estimate from the sample. Replacing $\beta_j$ by $\hat{\beta}_j$ describes a difference in **fitted means**, not an exact difference between two individual observations. The slope has units of response units per unit of $x_j$; a change of $h$ units corresponds to $h\beta_j$ in this additive model, with the other predictors fixed.
A useful interpretation states the response, its units, the predictor change, and the variables held fixed. For example: "At the same diameter, a one-foot increase in height is associated with an estimated 0.339-cubic-foot increase in mean volume." The adjustment set is part of the statement.

**Why an added predictor can change a coefficient.** A simple regression of $Y$ on $x_1$ compares observations with different $x_1$ values while leaving their other characteristics unadjusted. A regression on both $x_1$ and $x_2$ describes the association with $x_1$ at fixed $x_2$. If $x_1$ and $x_2$ vary together, these are different comparisons. Neither slope should be interpreted without naming its model.

**Example 5.1** (The Same Height Measurement, Two Comparisons)
Return to the 31 black cherry trees from Chapter 4. `Volume` is timber volume in cubic feet, `Height` is height in feet, and R's variable `Girth` actually records diameter in inches. Each row is one tree. We rename that variable `Diameter`. Dataset documentation.

```r
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))
ch5_trees <- transform(datasets::trees, Diameter = Girth)
ch5_tree_simple <- lm(Volume ~ Height, data = ch5_trees)
ch5_tree_adjusted <- lm(Volume ~ Height + Diameter, data = ch5_trees)
knitr::kable(data.frame(
  Model = c("Height alone", "Height adjusted for diameter"),
  Height_coefficient = c(coef(ch5_tree_simple)["Height"],
                         coef(ch5_tree_adjusted)["Height"])),
  digits = 3, col.names = c("Model", "Height coefficient"),
  row.names = FALSE)
```

| Model | Height coefficient |
|---|---|
| Height alone | 1.543 |
| Height adjusted for diameter | 0.339 |

The unadjusted height coefficient is about **1.543** cubic feet per foot. The adjusted coefficient is about **0.339**: among trees of the same diameter, one additional foot of height is associated with 0.339 more cubic feet of fitted mean volume. The fitted multiple regression equation is

$$\widehat{\text{Volume}} = -57.988 + 0.339\,\text{Height} + 4.708\,\text{Diameter}.$$

These slopes answer different questions. The multiple-regression height coefficient is also the coefficient tested when Chapter 4 compared the diameter-only model with the diameter-plus-height model.

**Why can the height estimate change?** For same-data OLS fits with intercepts and full column rank, write the reduced coefficients as $\hat{\alpha}_0, \hat{\alpha}_1$ and the full coefficients as $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$. Let $\hat{\delta}_1$ be the slope from an auxiliary regression of $x_2$ on $x_1$. The normal equations from Chapter 2 imply the exact sample identity

$$\boxed{\hat{\alpha}_1 = \hat{\beta}_1 + \hat{\beta}_2\hat{\delta}_1.}$$

Here $x_1$ is height and $x_2$ is diameter. The diameter-on-height slope is about 0.25575 inches per foot, so

$$1.54335 \approx 0.33925 + 4.70816(0.25575).$$

Taller trees in this sample also tend to have larger diameters. The height-only fit incorporates that accompanying diameter variation; the adjusted height coefficient describes a comparison at fixed diameter. The smaller adjusted estimate is not by itself evidence that height is unimportant or that either fit is a calculation error.

The two fitted slopes are equal if $\hat{\beta}_2\hat{\delta}_1 = 0$. For example, centered orthogonality of $x_1$ and $x_2$ gives $\hat{\delta}_1 = 0$. Adding a predictor therefore need not change the numerical slope, even though the stated adjustment differs. This identity concerns fitted OLS coefficients and does not require normal errors.

**!** Adjustment, overlap, and causal interpretation

"Holding diameter fixed" describes a comparison in the mean model; it does not mean an experiment made the same tree taller. An adjusted association is not automatically causal. A causal interpretation needs a defensible design and additional assumptions.

Also inspect which predictor combinations occur. If two predictors nearly move together, the data provide little information about changing one while holding the other fixed. Exact linear dependence causes rank deficiency; strong but imperfect dependence can inflate coefficient uncertainty. Detailed diagnostics come later.

**Exercise 5.1** (In Class: Interpret 0.339)
Which interpretation matches the adjusted height coefficient?
A. Increasing any tree's height by one foot causes its volume to increase by 0.339 cubic feet.
B. At the same diameter, a one-foot height difference corresponds to a 0.339-cubic-foot difference in fitted mean volume.
C. Every tree one foot taller has exactly 0.339 more cubic feet of observed volume.

**Solution 5.1**
**B.** The coefficient describes an adjusted mean association. It is neither an automatic causal effect nor an exact difference between individual responses.

## Section 5.2 Categorical predictors are columns of a design matrix

A categorical variable records group membership. Its labels do not supply numerical distances. Coding three unordered groups as 1, 2, and 3 and fitting one slope would impose equally spaced means; that is a modelling restriction, not a neutral relabeling.

**Definition 5.2** (Reference-Group Indicator Coding)
For a factor with $g$ observed levels, choose one reference level. With an intercept, use $g-1$ indicators for the remaining levels. For levels A, B, and C with A as reference, let

$$D_{Bi} = \mathbb{1}\{G_i = \mathrm{B}\}, \qquad D_{Ci} = \mathbb{1}\{G_i = \mathrm{C}\}.$$

Then

$$\mathbb{E}(Y_i \mid G_i) = \beta_0 + \beta_B D_{Bi} + \beta_C D_{Ci}.$$

| Group | Row of $\mathbf{X}$ | Mean |
|---|---|---|
| A | (1 0 0) | $\beta_0$ |
| B | (1 1 0) | $\beta_0 + \beta_B$ |
| C | (1 0 1) | $\beta_0 + \beta_C$ |

Thus $\beta_B = \mu_B - \mu_A$ and $\beta_C = \mu_C - \mu_A$. A comparison of B with C is $\beta_B - \beta_C$, not either coefficient alone.

For two observations from each group, ordered A, A, B, B, C, C,

$$\mathbf{X} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad p = 3.$$

```
ch5_groups <- data.frame(
  group = factor(rep(c("A", "B", "C"), each = 2),
                 levels = c("A", "B", "C")))
contrasts(ch5_groups$group) <-
  contr.treatment(levels(ch5_groups$group), base = 1)
knitr::kable(model.matrix(~ group, data = ch5_groups),
             row.names = FALSE)
```

| (Intercept) | groupB | groupC |
|---|---|---|
| 1 | 0 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |
| 1 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 0 | 1 |

**Why not use all indicators with an intercept?** For every observation,

$$D_{Ai} + D_{Bi} + D_{Ci} = 1.$$

The intercept column is therefore the sum of the three indicator columns. Four written coefficients would represent only three independent columns. One alternative is to remove the intercept and use all three indicators:

$$\mathbb{E}(Y_i \mid G_i) = \mu_A D_{Ai} + \mu_B D_{Bi} + \mu_C D_{Ci}.$$

This **cell-means parameterization** describes the same set of mean vectors. In R it is `y ~ 0 + group`. Removing the intercept here does not force the group means to zero; the indicator columns already span the constant vector.

**Example 5.2** (Different Coefficients, the Same Group Means)
Suppose the fitted means for A, B, and C are 18, 22, and 19.

| Reference | Intercept | Other coefficients |
|---|---|---|
| A | 18 | B minus A: 4; C minus A: 1 |
| B | 22 | A minus B: $-4$; C minus B: $-3$ |

Changing the reference changes the coordinate system for the same fitted means. It does not change fitted values, residuals, SSE, or model rank. A coefficient test can nevertheless change because its named comparison has changed: B versus A is a different null hypothesis from C versus B.

**Exercise 5.2** (In Class: Count Columns, Not Variable Names)
A model contains an intercept, two continuous predictors, and one four-level factor using reference coding. Assume all columns are independent and $n = 40$.

1. How many regression parameters are there?
2. What are the residual degrees of freedom?
3. Would replacing the three factor indicators by all four indicators while retaining the intercept add an identifiable parameter?

**Solution 5.2**
There are $p = 1 + 2 + 3 = 6$ parameters and $40 - 6 = 34$ residual degrees of freedom. **No:** all four indicators together with the intercept introduce a redundant column.

## Section 5.3 A continuous predictor and a group indicator

We now follow one dataset through several models. R's `ToothGrowth` data record an odontoblast-length response from 60 guinea pigs. The predictors are vitamin C dose (0.5, 1, or 2 mg/day) and delivery method: orange juice (`OJ`) or ascorbic acid (`VC`). Each row is one animal. Dataset documentation.

**Question.** How does the mean response vary with dose, and is the OJ-minus-VC mean difference the same at every dose? We first describe fitted associations. The dataset documentation alone is not a complete account of randomization or of all conditions needed for a causal conclusion.

```
ch5_tooth <- datasets::ToothGrowth
ch5_tooth$supp <- factor(ch5_tooth$supp, levels = c("VC", "OJ"))
ch5_tooth$dose_c <- ch5_tooth$dose - 1
ch5_tooth$dose_f <- factor(ch5_tooth$dose, levels = c(0.5, 1, 2))
contrasts(ch5_tooth$supp) <-
  contr.treatment(levels(ch5_tooth$supp), base = 1)
contrasts(ch5_tooth$dose_f) <-
  contr.treatment(levels(ch5_tooth$dose_f), base = 1)
stopifnot(!anyNA(ch5_tooth))
ch5_cell_summary <- aggregate(
  len ~ supp + dose, data = ch5_tooth,
  FUN = function(y) c(n = length(y), mean = mean(y)))
ch5_summary_table <- data.frame(ch5_cell_summary[1:2],
                                ch5_cell_summary$len)
knitr::kable(ch5_summary_table, digits = 2,
  col.names = c("Method", "Dose (mg/day)", "n", "Mean length"))
```

| Method | Dose (mg/day) | n | Mean length |
|---|---|---|---|
| VC | 0.5 | 10 | 7.98 |
| OJ | 0.5 | 10 | 13.23 |
| VC | 1.0 | 10 | 16.77 |
| OJ | 1.0 | 10 | 22.70 |
| VC | 2.0 | 10 | 26.14 |
| OJ | 2.0 | 10 | 26.06 |

Both delivery methods are observed at each of the three doses, with 10 animals per combination. This gives overlap for comparing methods at those doses. It does not supply observations at every dose between them or outside the observed range.

**Definition 5.3** (Additive Continuous-by-Group Model)
Let $x$ be dose, $x_c = x - 1$, and $z = \mathbb{1}\{\text{OJ}\}$. The additive model is

$$\mu(x, z) = \beta_0 + \beta_x x_c + \beta_g z.$$

| Method | Mean function | Slope in dose |
|---|---|---|
| VC ($z = 0$) | $\beta_0 + \beta_x(x - 1)$ | $\beta_x$ |
| OJ ($z = 1$) | $\beta_0 + \beta_g + \beta_x(x - 1)$ | $\beta_x$ |

The two lines are parallel, and $\mu(x, 1) - \mu(x, 0) = \beta_g$ at every dose. The intercept is the VC mean at **1 mg/day**, because that is where $x_c = 0$.

```
ch5_fit_add <- lm(len ~ dose_c + supp, data = ch5_tooth)
knitr::kable(coef(summary(ch5_fit_add))[, 1:2], digits = 3,
             col.names = c("Estimate", "Standard error"))
```

| | Estimate | Standard error |
|---|---|---|
| (Intercept) | 15.336 | 0.787 |
| dose_c | 9.764 | 0.877 |
| suppOJ | 3.700 | 1.094 |

The fitted equation is

$$\hat{\mu}(x, z) = 15.336 + 9.764(x - 1) + 3.700z.$$

At a fixed delivery method, a one-mg/day dose increase is associated with 9.764 more units of fitted mean length under this linear model. At a fixed dose, the fitted OJ mean exceeds the VC mean by 3.700 units. Neither statement says that every individual response follows its fitted mean exactly.

**Read the design matrix.** There are three columns: intercept, centered dose, and the OJ indicator. The table below shows one row for each predictor combination; the full matrix repeats each combination for 10 animals.

```
ch5_profiles <- unique(ch5_tooth[, c("dose_c", "supp")])
ch5_profile_order <- order(ch5_profiles$supp, ch5_profiles$dose_c)
ch5_profiles <- ch5_profiles[ch5_profile_order, ]
knitr::kable(model.matrix(~ dose_c + supp, data = ch5_profiles),
             row.names = FALSE)
```

| (Intercept) | dose_c | suppOJ |
|---|---|---|
| 1 | -0.5 | 0 |
| 1 | 0.0 | 0 |

| (Intercept) | dose_c | suppOJ |
|---|---|---|
| 1 | 1.0 | 0 |
| 1 | -0.5 | 1 |
| 1 | 0.0 | 1 |
| 1 | 1.0 | 1 |

With full rank, this model has $p = 3$ and $60 - 3 = 57$ residual degrees of freedom. Using numeric dose imposes a straight-line mean within each method; treating the same three doses as a factor permits a separate mean at each dose. We will compare those assumptions below.

## Section 5.4 Interactions and conditional comparisons

**Definition 5.4** (Continuous-by-Group Interaction)
Add the product column $x_c z$:

$$\mu(x, z) = \beta_0 + \beta_x x_c + \beta_g z + \beta_{xg} x_c z.$$

The design row is $(1\ x_c\ z\ x_c z)$, so this two-group model has four parameters when full rank.

| Method | Mean function | Slope in dose |
|---|---|---|
| VC | $\beta_0 + \beta_x(x - 1)$ | $\beta_x$ |
| OJ | $(\beta_0 + \beta_g) + (\beta_x + \beta_{xg})(x - 1)$ | $\beta_x + \beta_{xg}$ |

The group difference at dose $x$ is

$$\boxed{\Delta(x) = \mu(x, 1) - \mu(x, 0) = \beta_g + \beta_{xg}(x - 1).}$$

Thus $\beta_g$ is the group difference at 1 mg/day, and $\beta_{xg}$ is the OJ-minus-VC difference in slopes. An interaction means that one comparison depends on another predictor.

```
ch5_fit_int <- lm(len ~ dose_c * supp, data = ch5_tooth)
knitr::kable(coef(summary(ch5_fit_int))[, 1:2], digits = 3,
             col.names = c("Estimate", "Standard error"))
```

| | Estimate | Standard error |
|---|---|---|
| (Intercept) | 15.011 | 0.772 |
| dose_c | 11.716 | 1.195 |
| suppOJ | 4.351 | 1.091 |
| dose_c:suppOJ | -3.904 | 1.691 |

R expands `dose_c * supp` into `dose_c + supp + dose_c:supp`; `:` denotes the interaction term. See the R formula documentation. The fitted equation is

$$\hat{\mu}(x, z) = 15.011 + 11.716(x - 1) + 4.351z - 3.904(x - 1)z.$$

The VC slope is about 11.716, and the OJ slope is $\hat{\beta}_x + \hat{\beta}_{xg} \approx 7.811$, using the unrounded estimates. The OJ-minus-VC difference is about 4.351 at 1 mg/day and 0.446 at 2 mg/day. A negative interaction coefficient means that the group difference decreases as dose increases; it does not mean that the OJ dose slope is negative.

**The same coefficient label can describe a different comparison**

Read each coefficient within its own model. OLS fits the additive and interaction equations separately, so their coefficient estimates need not agree.

| Coefficient | Additive model | Interaction model |
|---|---|---|
| $\beta_0$ | VC mean at 1 mg/day. | VC mean at 1 mg/day. |
| $\beta_x$ | Common dose slope for both methods. | Dose slope for VC only. |
| $\beta_g$ | OJ-minus-VC difference at every dose. | OJ-minus-VC difference at 1 mg/day. |
| $\beta_{xg}$ | Fixed at zero. | OJ-minus-VC difference in slopes. |

Thus 9.764 is a common slope for both methods, whereas 11.716 is the VC slope. Similarly, 3.700 is a constant group difference, whereas 4.351 is the group difference specifically at 1 mg/day. The intercept retains the same baseline interpretation, but its estimate can still change because the fitted lines are re-estimated under different constraints. Setting the population interaction coefficient to zero recovers the additive model; it does not force the two separate sample fits to have identical remaining estimates.

**Centering changes the baseline, not the fitted model.** Start with the uncentered expression $b_0 + b_x x + b_g z + b_{xg} xz$. Substitute $x = x_c + c$ to obtain

$$(b_0 + cb_x) + b_x x_c + (b_g + cb_{xg})z + b_{xg} x_c z.$$

Centering at $c$ changes the intercept to the reference-group mean at $c$ and the group coefficient to the difference at $c$. The group-specific slopes, fitted means, residuals, and SSE stay the same when all the corresponding terms are retained.

```
ch5_fit_uncentered <- lm(len ~ dose * supp, data = ch5_tooth)
knitr::kable(data.frame(
  Parameterization = c("Raw dose (baseline: 0)",
                       "Centered dose (baseline: 1)"),
  Group_coefficient = c(coef(ch5_fit_uncentered)["suppOJ"],
                        coef(ch5_fit_int)["suppOJ"])), digits = 3,
  col.names = c("Parameterization", "OJ coefficient"),
  row.names = FALSE)
```

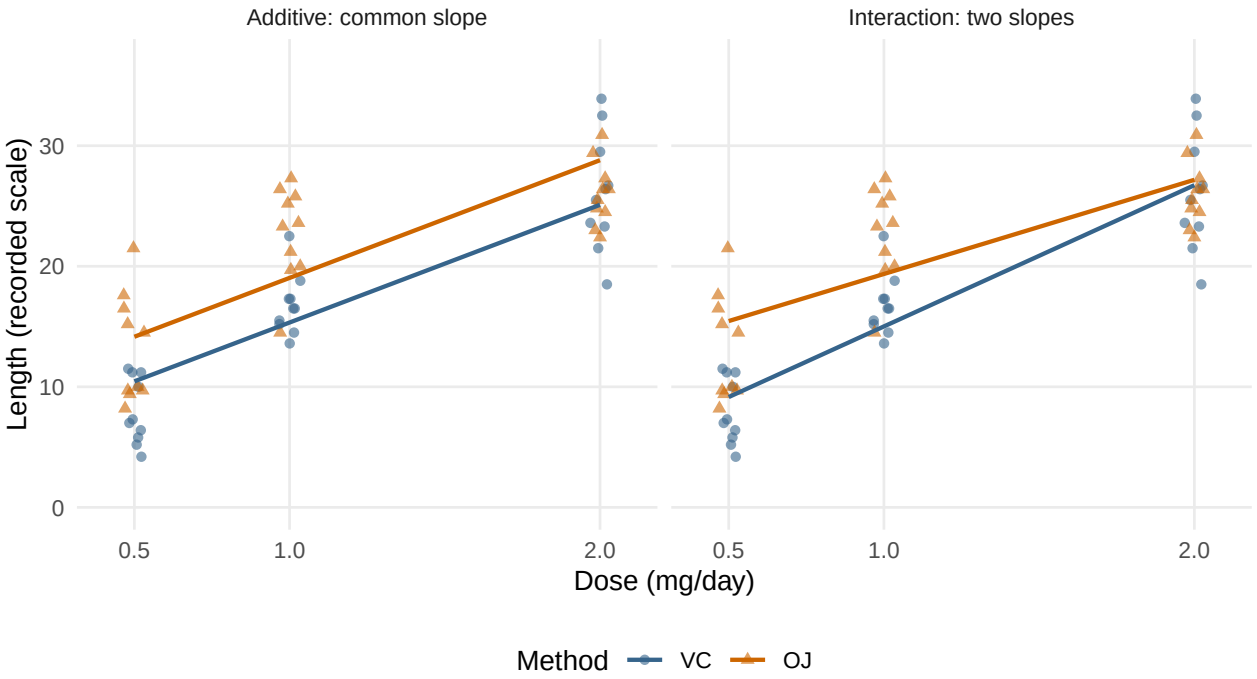



Figure 5.1: Additive and interaction fits to the same observations. Horizontal jitter separates overlapping points for display only; both models use the original doses. The additive model imposes parallel lines. The interaction model permits different slopes, while still requiring each method's mean to be linear in dose.

| Parameterization | OJ coefficient |
|---|---|
| Raw dose (baseline: 0) | 8.255 |
| Centered dose (baseline: 1) | 4.351 |

```
stopifnot(isTRUE(all.equal(unname(fitted(ch5_fit_uncentered)),
                           unname(fitted(ch5_fit_int)))))
```

Zero dose was not observed. Centering at an observed dose gives the group coefficient a more directly supported meaning. It does not establish linearity or create additional observations.

> **!** Keep the lower-order terms when interpreting an interaction
>
> A model such as `dose_c * supp` includes both main effects and their interaction. Do not remove a main effect merely because its coefficient has a large p-value in the interaction model. That coefficient concerns a particular baseline, and dropping it imposes a constraint that can depend on the choice of center or reference group. Start with the scientific restriction you intend to test.

**Other interactions.** For two continuous predictors, $\mu = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$ has slope $\beta_1 + \beta_{12} x_2$ in $x_1$. For two factors with $a$ and $b$ levels and all cells observed, reference coding uses $(a-1)(b-1)$ interaction columns. With all main effects, there are $1+(a-1)+(b-1)+(a-1)(b-1) = ab$

parameters, one mean per cell. The same principle applies: a comparison varies with another predictor.

**Exercise 5.3** (In Class: Which Difference Is Six?)
Suppose $x_c = x - 1$ and $\hat{\mu} = 10 + 2x_c + 6z - x_c z$.

1. What is the slope for $z = 1$?
2. At which original $x$ value is the group difference 6?
3. What is the group difference at $x = 3$?

**Solution 5.3**
The slope for $z = 1$ is $2 - 1 = 1$. The group difference is 6 at $x = 1$, where $x_c = 0$. At $x = 3$, $x_c = 2$ and the difference is $6 - 2 = 4$.

## Section 5.5 Match the hypothesis to the model comparison

The partial F test from Chapter 4 compares nested mean spaces on the **same observations**:

$$F = \frac{(\text{SSE}_R - \text{SSE}_F)/(p_F - p_R)}{\text{SSE}_F/(n - p_F)}.$$

Under the reduced model and the stated normal, constant-variance error assumptions, the reference distribution is $F_{p_F - p_R, n - p_F}$. These comparisons answer different questions:

| Question | Reduced model | Full model | Numerator df |
|---|---|---|---|
| Are the dose slopes equal? | `dose_c + supp` | `dose_c * supp` | 1 |
| Are the two entire dose lines identical? | `dose_c` | `dose_c * supp` | 2 |
| Is a straight line within each method adequate at the observed doses? | `dose_c * supp` | `dose_f * supp` | 2 |

**Equal slopes.** The additive model sets $\beta_{xg} = 0$. Comparing it with the interaction model tests whether the OJ-minus-VC difference is constant across dose under the two-line framework.

```
anova(ch5_fit_add, ch5_fit_int)
```

```
Analysis of Variance Table

Model 1: len ~ dose_c + supp
Model 2: len ~ dose_c * supp
  Res.Df     RSS Df Sum of Sq      F  Pr(>F)
```

```
1     57 1022.56
2     56  933.63  1     88.92 5.3335 0.02463 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The comparison has 1 numerator and 56 denominator degrees of freedom, with $F \approx 5.334$ and $p \approx 0.0246$. Under the model assumptions, this is evidence against equal slopes. It is not a test that the two groups have identical means at every dose.

**Identical lines.** For the two lines to coincide, both $\beta_g = 0$ and $\beta_{xg} = 0$ must hold. A single p-value for `suppOJ` tests only the difference at the centering dose. Test the two restrictions together instead:

```
ch5_fit_dose <- lm(len ~ dose_c, data = ch5_tooth)
anova(ch5_fit_dose, ch5_fit_int)
```

```
Analysis of Variance Table

Model 1: len ~ dose_c
Model 2: len ~ dose_c * supp
  Res.Df     RSS Df Sum of Sq      F   Pr(>F)
1     58 1227.91
2     56  933.63  2    294.27 8.8253 0.000466 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**Check the straight-line restriction.** Dose takes three numerical values; the two-line model constrains each group's three fitted means to lie on a line. A factor-dose-by-method model estimates all six cell means:

```
ch5_fit_cells <- lm(len ~ dose_f * supp, data = ch5_tooth)
ch5_fits <- list(Additive = ch5_fit_add, Interaction = ch5_fit_int,
                 `Six cell means` = ch5_fit_cells)
knitr::kable(data.frame(
  Model = names(ch5_fits),
  p = vapply(ch5_fits, function(fit) fit$rank, numeric(1)),
  Residual_df = vapply(ch5_fits, df.residual, numeric(1)),
  SSE = vapply(ch5_fits, deviance, numeric(1))),
  digits = 2, row.names = FALSE,
  col.names = c("Model", "p", "Residual df", "SSE"))
```

| Model | p | Residual df | SSE |
|---|---|---|---|
| Additive | 3 | 57 | 1022.56 |
| Interaction | 4 | 56 | 933.63 |
| Six cell means | 6 | 54 | 712.11 |

```
anova(ch5_fit_int, ch5_fit_cells)
```

```
Analysis of Variance Table

Model 1: len ~ dose_c * supp
Model 2: len ~ dose_f * supp
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1     56 933.63
2     54 712.11  2    221.53 8.3994 0.0006667 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The six-cell model has $p = 6$ and 54 residual degrees of freedom. The comparison adds two independent directions, one departure from a straight line for each method's three means. Here $F \approx 8.399$ and $p \approx 0.000667$ provide evidence against the straight-line restriction under the common-variance normal model. The fitted lines were useful for learning coefficient interpretation; this check shows why their shape should not be accepted automatically for these data.

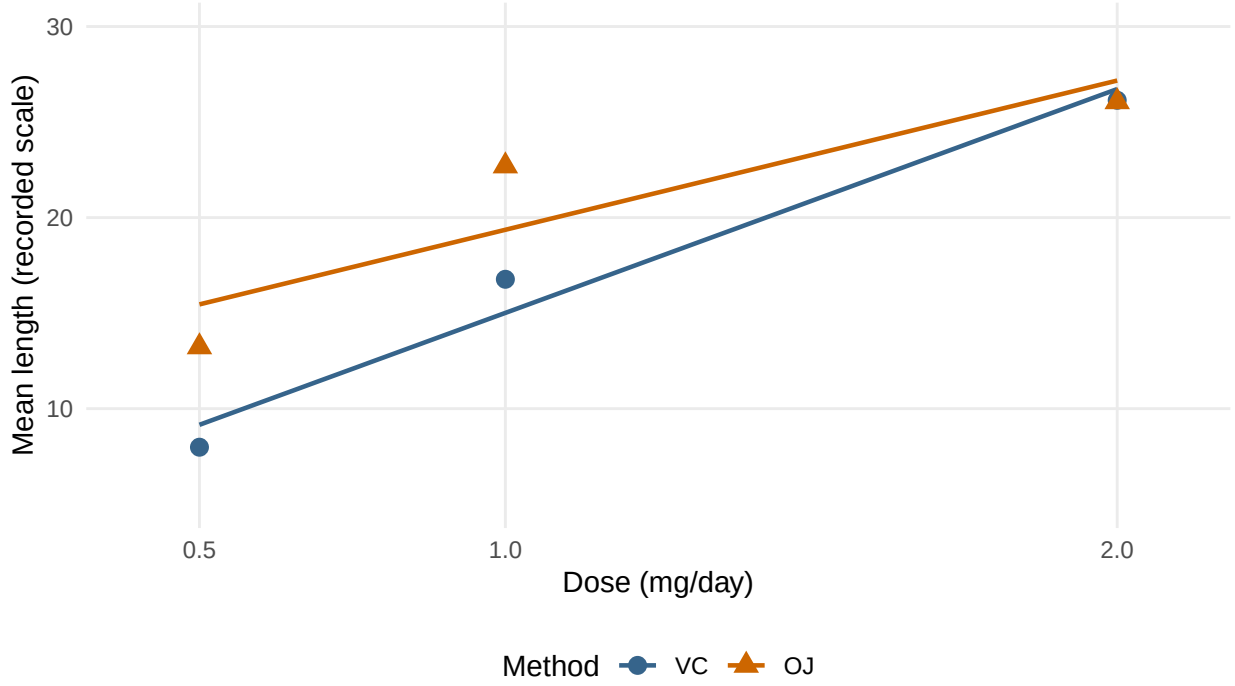



Figure 5.2: Cell means and the two-line fit. Points are the observed mean responses at each method-dose combination. The six-cell model reproduces these means; the straight lines impose additional restrictions. No observations outside 0.5–2 mg/day are shown.

> **A factor is a block of columns**
>
> The three-level `dose_f` main effect uses two columns. The `dose_f:supp` interaction uses two more. A scientific test concerning an entire factor or interaction may therefore require a joint test, rather than selecting one printed coefficient p-value. Always count independent

restrictions and use the full model's residual degrees of freedom.

## Section 5.6 Coding checks and the bridge to contrasts

R's `model.matrix()` reveals the columns created by a formula. Use it to connect the software coefficients to your written mean model.

```
knitr::kable(model.matrix(~ dose_c * supp, data = ch5_profiles),
             row.names = FALSE)
```

| (Intercept) | dose_c | suppOJ | dose_c:suppOJ |
|---|---|---|---|
| 1 | -0.5 | 0 | 0.0 |
| 1 | 0.0 | 0 | 0.0 |
| 1 | 1.0 | 0 | 0.0 |
| 1 | -0.5 | 1 | -0.5 |
| 1 | 0.0 | 1 | 0.0 |
| 1 | 1.0 | 1 | 1.0 |

The last column is the product of centered dose and the OJ indicator. It is zero for every VC observation; it changes the OJ slope relative to VC.

**Verify a reference change.** Refit with OJ as the reference while retaining the same model terms and observations:

```
ch5_relevel <- ch5_tooth
ch5_relevel$supp <- relevel(ch5_relevel$supp, ref = "OJ")
contrasts(ch5_relevel$supp) <-
  contr.treatment(levels(ch5_relevel$supp), base = 1)
ch5_fit_ojref <- lm(len ~ dose_c * supp, data = ch5_relevel)
knitr::kable(data.frame(
  Reference = c("VC", "OJ"),
  Intercept = c(coef(ch5_fit_int)[1], coef(ch5_fit_ojref)[1]),
  Dose_slope = c(coef(ch5_fit_int)[2], coef(ch5_fit_ojref)[2]),
  SSE = c(deviance(ch5_fit_int), deviance(ch5_fit_ojref))),
  digits = 3, row.names = FALSE)
```

| Reference | Intercept | Dose_slope | SSE |
|---|---|---|---|
| VC | 15.011 | 11.716 | 933.635 |
| OJ | 19.361 | 7.811 | 933.635 |

```
stopifnot(isTRUE(all.equal(unname(fitted(ch5_fit_int)),
                           unname(fitted(ch5_fit_ojref)))))
```

The intercept now describes OJ at 1 mg/day, and the dose coefficient is the OJ slope. The fitted values and SSE agree with the VC-reference fit because the two design matrices span the same column space.

**Express a comparison before calculating it.** In the VC-reference interaction model, order the coefficient vector as $(\beta_0, \beta_x, \beta_g, \beta_{xg})$. The OJ-minus-VC comparison at dose $x_0$ is

$$\Delta(x_0) = \mathbf{l}^\top \boldsymbol{\beta}, \qquad \mathbf{l} = (0, 0, 1, x_0 - 1).$$

Here the comma-separated tuple denotes a column vector, as in Chapter 4. At $x_0 = 1$, the comparison is $\beta_g$; at $x_0 = 2$, it is $\beta_g + \beta_{xg}$. The linear-combination machinery from Chapter 3 therefore applies. Its standard error uses the full coefficient covariance matrix, including the covariance between the group and interaction estimates. Chapter 6 develops these comparisons, coding invariance, and the distinction between one comparison and a family of comparisons.

## Section 5.7 Concept checks and practice

**Exercise 5.4** (Concept Check: Choose or Mark True/False)

1. **T/F.** A multiple-regression coefficient always has the same interpretation as the slope in a simple regression using that predictor alone.
2. **Multiple choice.** With an intercept, a three-level factor normally contributes **(A)** one, **(B)** two, or **(C)** three independent indicator columns.
3. **T/F.** Reversing the reference category, while preserving the same model space, changes fitted values and SSE.
4. **Multiple choice.** In a binary-group interaction model centered at $x = 1$, the group coefficient compares the groups **(A)** at every $x$, **(B)** at $x = 0$, or **(C)** at $x = 1$.
5. **T/F.** A negative interaction coefficient implies a negative slope in the non-reference group.
6. **Multiple choice.** Testing identical lines requires **(A)** the interaction coefficient alone, **(B)** the group and interaction coefficients jointly, or **(C)** the intercept alone to be zero.
7. **T/F.** Replacing numeric dose by a three-level factor merely renames the coefficients and must preserve the fitted means.
8. **T/F.** Failing to reject equal slopes proves that the two population slopes are exactly equal.
9. **T/F.** If the added predictor's population coefficient is zero, fitting the unrestricted larger model must leave the other estimated coefficients unchanged in every sample.
10. **Multiple choice.** In the same-data OLS identity above, suppose $\hat{\beta}_1 = -1$, $\hat{\beta}_2 = 2$, and $\hat{\delta}_1 = 3$. The reduced-model slope is **(A)** $-1$, **(B)** 5, or **(C)** $-7$.
11. **T/F.** Two models whose intercepts describe the same reference group at the same predictor value must give the same estimated intercept.

### Solution 5.4

| Item | Answer | Reason |
|---|---|---|
| 1 | F | Adjustment changes the comparison. |
| 2 | B | One level serves as the reference. |
| 3 | F | The fitted mean space is unchanged. |
| 4 | C | The group coefficient is evaluated at centered dose zero. |
| 5 | F | Add the interaction coefficient to the reference-group slope. |
| 6 | B | Both intercept difference and slope difference must vanish. |
| 7 | F | A factor can relax the straight-line restriction. |
| 8 | F | Non-rejection is not proof of equality. |
| 9 | F | The restriction concerns population parameters; the unrestricted fit can estimate a nonzero added coefficient and re-estimate the others. |
| 10 | B | $-1 + 2(3) = 5$. An unadjusted and an adjusted slope can even have opposite signs. |
| 11 | F | The same baseline comparison can receive different estimates under different model constraints. |

**Exercise 5.5** (Practice: State the Restrictions and Degrees of Freedom)
Consider $n = 48$ observations, a binary group $z$, and a continuous predictor $x$. Assume the relevant design matrices have full rank and the normal-model assumptions hold. The full model is $\mu = \beta_0 + \beta_x x + \beta_g z + \beta_{xg} xz$.

1. Write the null hypothesis for parallel lines and give both F degrees of freedom.
2. Write the null hypothesis for identical lines and give both F degrees of freedom.
3. Express the group difference at $x = 2$ as a linear combination of coefficients.
4. If the full model has SSE 440 and the additive model has SSE 500, compute the partial F statistic for equal slopes.

### Solution 5.5

1. $H_0 : \beta_{xg} = 0$; degrees of freedom $(1, 44)$.
2. $H_0 : \beta_g = \beta_{xg} = 0$; degrees of freedom $(2, 44)$.
3. $\Delta(2) = \beta_g + 2\beta_{xg}$, with column vector $\mathbf{l} = (0, 0, 1, 2)$ in the displayed coefficient order.
4. $F = \{(500 - 440)/1\}/(440/44) = 6$.

**Summary.** The model's columns determine what its coefficients mean. An additive slope is an adjusted mean comparison; factor coefficients depend on a reference group; an interaction makes a comparison depend on another predictor. Centering and releveling preserve fitted values when they only reparameterize the same mean space. Changing the mean space, such as relaxing a straight-line

restriction, is a substantive modelling decision that can be assessed with the nested-model tools from Chapter 4.

# Part I

# Optional topic

# Quantile Regression

> **Self-study topic**
>
> This reading extends the regression framework to conditional quantiles. It is supplementary to the main lecture sequence. Read Chapter 2 — Least Squares Estimation, Chapter 5 — Multiple Regression and Categorical Predictors, and Chapter 8 — Diagnostics and Model Adequacy (not included in this edition) first: least squares, coefficient interpretation, and diagnostics are the useful prerequisites. The R examples use the `quantreg` package; install it once with `install.packages("quantreg")` if needed.

## When the mean is not the whole question

OLS describes a conditional mean. In some studies, the scientific question concerns a typical response, a low outcome, or an unusually high outcome. For example, average delivery time and the 90th percentile of delivery time answer different service-planning questions. A covariate may change the spread of the response even when its relationship with the mean is simple.

**Definition 5.5** (Conditional Quantile)
For $0 < \tau < 1$, define

$$Q_\tau(Y \mid \mathbf{X} = \mathbf{x}) = \inf\{q : P(Y \le q \mid \mathbf{X} = \mathbf{x}) \ge \tau\}.$$

Here $Y$ is a scalar response, $\mathbf{X}$ is a random predictor vector, and $\mathbf{x}$ is its observed value. A linear quantile model specifies $Q_\tau(Y \mid \mathbf{X} = \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_\tau$. The coefficients may vary with $\tau$. At $\tau = 0.5$, the target is a conditional median.

**Interpretation.** Without an interaction involving $x_j$, $\beta_{\tau,j}$ is the change in the modeled conditional $\tau$-quantile for a one-unit change in $x_j$, holding the other predictors fixed. This compares conditional distributions. It does not track a particular person who remains at the same percentile, and it is not automatically a causal effect.

> **Use all observations to estimate a conditional quantile**
>
> Fitting the 90th percentile does not mean selecting the largest 10% of responses and fitting OLS to them. Quantile regression uses the full dataset with a loss function that gives different penalties to positive and negative residuals. A marginal percentile of all responses is also different from a percentile conditional on the predictors.

**Why a different loss?** Least squares penalizes a residual $u = y - \mathbf{x}^\top \mathbf{b}$ by $u^2$. Quantile regression instead uses the check loss, also called pinball loss:

$$\rho_\tau(u) = u\{\tau - \mathbb{1}\{u < 0\}\} = \begin{cases} \tau u, & u \geq 0, \\ (1-\tau)(-u), & u < 0, \end{cases}$$

$$\hat{\boldsymbol{\beta}}_\tau \in \arg\min_{\mathbf{b}} \sum_{i=1}^{n} \rho_\tau(y_i - \mathbf{x}_i^\top \mathbf{b}).$$

At $\tau = 0.5$, minimizing check loss is equivalent to minimizing the sum of absolute residuals. At $\tau = 0.9$, underpredicting the response costs nine times as much per unit as overpredicting it. The fitted target consequently moves toward the upper tail.

**Proposition 5.1** (Why Check Loss Targets a Quantile)
If $\mathbb{E}(|Y|) < \infty$, the minimizers of $\mathbb{E}\{\rho_\tau(Y - a)\}$ are characterized by $F(a-) \leq \tau \leq F(a)$; the conditional-quantile definition above selects the lower endpoint when the minimizer is not unique. Where $F$ is continuous, the derivative with respect to $a$ is $F(a) - \tau$. It is nonpositive below a minimizing quantile and nonnegative above it. The same argument applies to the conditional distribution at a fixed $\mathbf{x}$.

**Scope of the linear model.** The population argument allows a different quantile at every $\mathbf{x}$. A linear quantile regression restricts these values to one linear function. If the true conditional quantile is nonlinear, the fitted line is a best approximation under the chosen loss, rather than an exact description at every predictor value.

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

check_loss <- function(u, tau) u * (tau - (u < 0))
taus <- c(0.1, 0.5, 0.9)
loss <- expand.grid(u = seq(-3, 3, length.out = 301), tau = taus)
loss$value <- check_loss(loss$u, loss$tau)
ggplot(loss, aes(u, value, colour = factor(tau))) +
  geom_vline(xintercept = 0, linetype = "dotted", colour = "grey70") +
  geom_line(linewidth = 0.9) +
  scale_colour_manual(values = c("steelblue4", "grey35", "darkorange3")) +
  labs(x = "Residual: observed - predicted", y = "Check loss",
    colour = "Quantile")
```

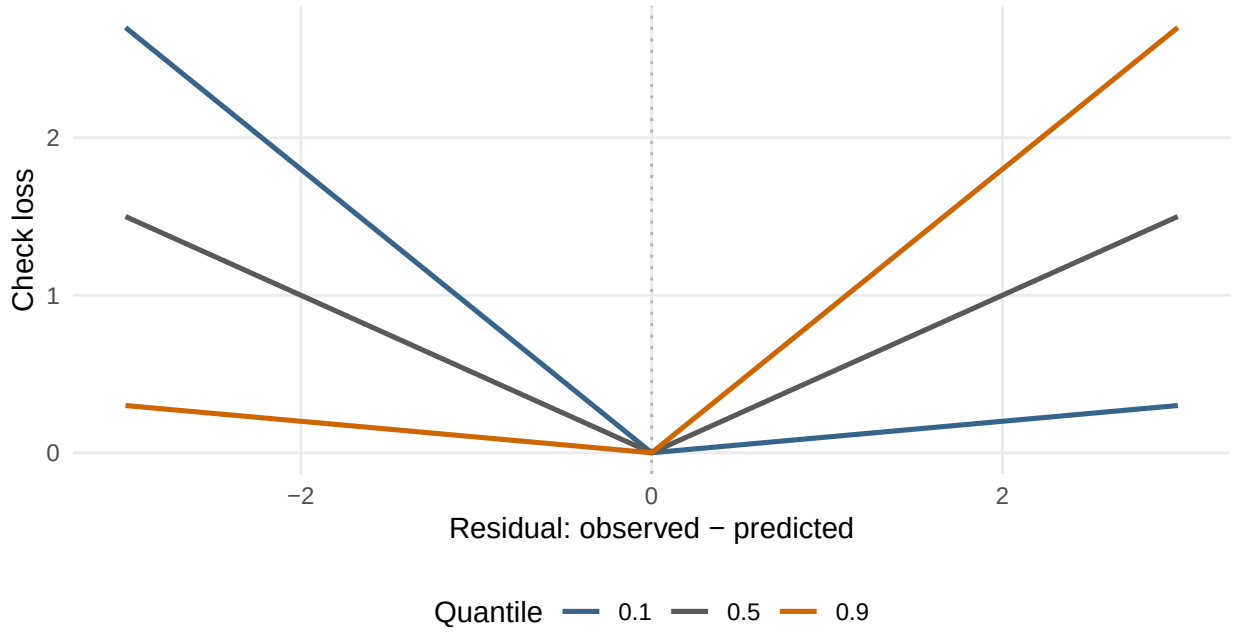



Figure 5.3: Check loss for three quantiles. A positive residual means the prediction is too low. The upper-quantile loss penalizes that side more strongly.

## A mean model can miss changing spread

Consider the simulated model $Y = 5 + 2x + (0.5 + 0.8x)Z$, with $0 \le x \le 4$ and $Z \sim N(0, 1)$ independent of $x$. Its conditional mean and median are both $5 + 2x$, while

$$Q_\tau(Y \mid x) = 5 + 2x + (0.5 + 0.8x)\Phi^{-1}(\tau).$$

The conditional-quantile slopes are $2 + 0.8\Phi^{-1}(\tau)$. Different slopes here reflect changing spread, even though every observation follows the same data-generating model. Quantile regression need not be reserved for skewed distributions.

```
set.seed(8561)
n <- 350
x <- runif(n, 0, 4)
dat_qr <- data.frame(x, y = 5 + 2 * x +
                       (0.5 + 0.8 * x) * rnorm(n))
fit_mean <- lm(y ~ x, data = dat_qr)
fit_quantiles <- quantreg::rq(y ~ x, tau = taus, data = dat_qr)
round(coef(fit_quantiles), 3)
```

```
            tau= 0.1 tau= 0.5 tau= 0.9
(Intercept)    4.509    5.078    5.850
x              0.768    1.816    2.965
```

```
data.frame(tau = taus, true_slope = 2 + 0.8 * qnorm(taus),
           estimated_slope = unname(coef(fit_quantiles)["x", ]))
```

```
  tau true_slope estimated_slope
1 0.1  0.9747587       0.7676385
2 0.5  2.0000000       1.8156789
3 0.9  3.0252413       2.9651924
```

The matrix returned by `coef()` contains one column per requested quantile. R's `rq()` documentation describes its formula interface and the `tau` argument.

```
new_x <- data.frame(x = seq(0, 4, length.out = 200))
qhat <- predict(fit_quantiles, newdata = new_x)
quantile_lines <- data.frame(x = rep(new_x$x, times = length(taus)),
                             prediction = as.vector(qhat),
                             target = rep(paste("Quantile", taus),
                               each = nrow(new_x)))
prediction_lines <- rbind(quantile_lines,
  data.frame(x = new_x$x, prediction = predict(fit_mean, newdata = new_x),
             target = "OLS mean"))
prediction_lines$target <- factor(prediction_lines$target,
                                  levels = c(paste("Quantile", taus),
                                    "OLS mean"))
ggplot(dat_qr, aes(x, y)) +
  geom_point(alpha = 0.35, colour = "grey40", size = 1.5) +
  geom_line(data = prediction_lines,
            aes(y = prediction, colour = target, linetype = target),
              linewidth = 0.8) +
  scale_colour_manual(values = c("steelblue4", "grey35", "darkorange3",
    "black")) +
  scale_linetype_manual(values = c("solid", "solid", "solid", "dashed")) +
  labs(x = "x", y = "Response", colour = NULL, linetype = NULL)
```

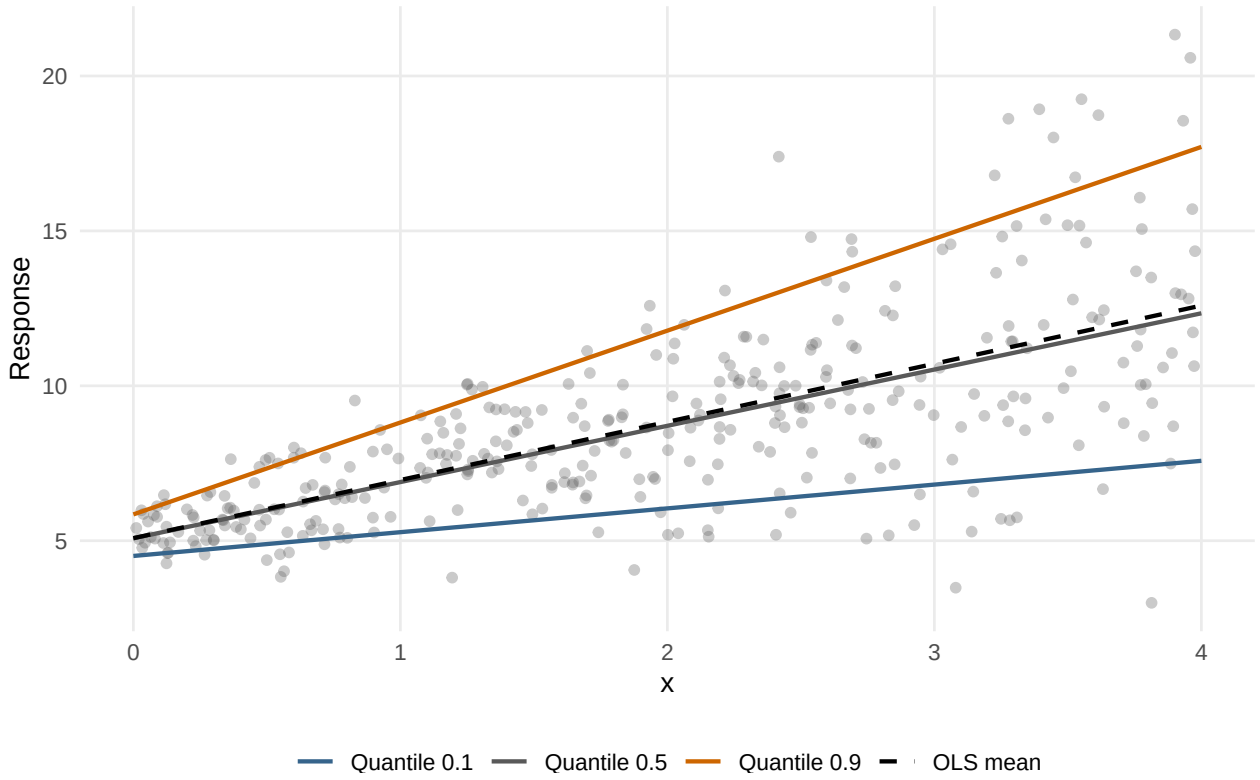



Figure 5.4: Fitted conditional quantiles and the OLS mean in simulated data with increasing spread. The growing distance between the lower and upper quantiles reveals variation that a mean line alone does not describe.

**Read the graph.** Compare the predicted 10th, 50th, and 90th percentiles at a fixed $x$. The gap between the 10th and 90th percentiles describes an interval containing the central 80% of a continuous conditional distribution when the quantile model is correct. The fitted gap is not a confidence interval for the mean or for a regression coefficient, and its predictive coverage is not guaranteed in a finite sample.

## Uncertainty and practical checks

Quantile regression avoids a normal-error or constant-variance requirement for defining its target, but it is not assumption-free. Reliable estimation and inference still depend on the quantile specification, the design, the sampling structure, and sufficient information near the quantile of interest. Very extreme quantiles are difficult to estimate from small datasets.

**A pairs-bootstrap illustration.** These simulated observations are independent draws, so resampling whole $(x_i, y_i)$ pairs is appropriate. The code requests a bootstrap standard error for the median-regression slope and forms an approximate normal interval. The replicate count is kept moderate for the notes; increase it and check Monte Carlo stability in a substantive analysis.

```
set.seed(8561)
fit_median <- quantreg::rq(y ~ x, tau = 0.5, data = dat_qr)
```

```
median_summary <- summary(fit_median, se = "boot", R = 499,
                          bsmethod = "xy")
b <- coef(fit_median)["x"]
se_b <- median_summary$coefficients["x", "Std. Error"]
c(slope = unname(b), bootstrap_SE = se_b,
  lower = unname(b - qnorm(0.975) * se_b),
  upper = unname(b + qnorm(0.975) * se_b))
```

```
      slope bootstrap_SE       lower       upper
   1.815679     0.136845    1.547468    2.083890
```

This is not the exact finite-sample OLS t interval from Chapter 3 — Distribution Theory of OLS and Inference. For dependent or clustered observations, an independent-pairs bootstrap generally fails to preserve the sampling structure. The package's `summary.rq()` documentation describes alternative inference methods.

**!** Two fitted quantiles need a joint comparison

If one slope is significant at $\tau = 0.1$ and another is not significant at $\tau = 0.9$, that does not establish that the slopes differ. They are estimated from the same observations and are dependent. A difference such as $\beta_{0.9,j} - \beta_{0.1,j}$ requires joint inference that retains this dependence, for example refitting both quantiles in each common bootstrap sample.

**Before interpreting a fit:** check the functional form, influential predictor values, the amount of information in the tails, and whether separately fitted quantiles cross. Linear loss is less sensitive to very large response residuals than squared loss, but it does not eliminate leverage problems. Crossing fitted quantiles cannot describe a valid ordered distribution at that $x$; investigate misspecification or use methods that enforce ordering rather than silently relabeling the curves.

**Evaluate the stated target.** For held-out data, compare candidate predictions at the same $\tau$ using mean check loss. A coverage diagnostic compares the fraction of responses below the predicted quantile with $\tau$, including checks across meaningful predictor ranges. Good overall coverage can hide poor conditional coverage. Use splits that match future prediction, as in Chapter 10 — Model Assessment and Prediction (not included in this edition).

## Application and self-study

**i** Application: household expenditure

The `engel` dataset bundled with `quantreg` contains income and food expenditure. Use `data("engel", package = "quantreg")`, plot the variables, and fit the 0.1, 0.5, and 0.9 conditional quantiles of `foodexp` given `income`. Explain how a household-expenditure quantile differs from mean expenditure. Treat the fitted relationships as descriptive associations, and avoid extrapolating beyond the observed incomes.

1. Use the loss definition to explain why the 90th percentile is penalized more for underprediction.
2. Derive the true intercept and slope at each quantile in the simulated example. Explain why the median and mean agree here.
3. Predict the 10th and 90th percentiles at $x = 1$ and $x = 3$. Explain what their widening gap means.
4. Design a bootstrap for comparing two quantile slopes. State which observations must stay together if the data are clustered.

**Suggested reading.** Start with the `quantreg` package documentation, then the vignette available through `vignette("rq", package = "quantreg")`. Roger Koenker's *Quantile Regression* develops the estimation and inference theory in greater depth.

# Mixed-Effects Models

> **i** Self-study topic
>
> This reading extends linear models to grouped and repeated observations. It is supplementary to the main lecture sequence. Chapter 1 — Matrix Representation and Notation, Chapter 3 — Distribution Theory of OLS and Inference, and Chapter 7 — Experimental Design Models: ANOVA and ANCOVA (not included in this edition) supply the useful prerequisites: covariance, inference, and the experimental unit. The R examples use `lme4` and `patchwork`; install them once with `install.packages(c("lme4", "patchwork"))` if needed. The covariance connection to GLS is developed here, so the Generalized Least Squares and Correlated Errors topic (not included in this edition) is not a prerequisite.

## Why group membership changes the analysis

Measurements from the same person, classroom, clinic, or production batch often share influences. Treating all rows as independent can give misleading standard errors. Repeated measurements provide information about within-unit change, but they do not create additional independently sampled or randomized units.

A linear mixed-effects model combines shared regression coefficients with group-specific deviations. **Fixed effects** describe common coefficients, possibly including measured group characteristics. **Random effects** represent latent group deviations through a distribution. These terms describe components of the model; they do not mean that a predictor must be nonrandom or that treatment assignment was randomized.

**Definition 5.6** (A Random-Intercept Model)
For observation $i$ in group $j$, let

$$Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_j + \varepsilon_{ij},$$
$$b_j \overset{\text{iid}}{\sim} N(0, \tau^2), \qquad \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

Condition on the observed predictors. Assume that the group effects and residual errors are mutually independent, with the displayed distributions unchanged after conditioning on those predictors. Conditional on $b_j$, the group has intercept $\beta_0 + b_j$. After integrating over $b_j$, its mean is $\beta_0 + \beta_1 x_{ij}$.

For two distinct observations in the same group, the shared $b_j$ gives covariance $\tau^2$; their marginal variances are $\tau^2 + \sigma^2$. Therefore the within-group correlation in this simple model is

$$\text{ICC} = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

This intraclass correlation describes the fraction of conditional-on-predictors variation attributable to a shared group intercept. It is not a universal correlation formula: random slopes or additional residual correlation change it.

**Connection to the course.** In matrix notation,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\varepsilon}, \qquad \mathbf{b} \sim N(\mathbf{0}, \mathbf{G}), \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{R}),$$
$$\text{Var}(\mathbf{Y} \mid \mathbf{X}, \mathbf{Z}) = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R},$$

with $\mathbf{b}$ and $\boldsymbol{\varepsilon}$ independent and the design conditioned on. For a known covariance $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}$, estimation of $\boldsymbol{\beta}$ has the GLS form. In practice, the covariance parameters are estimated too, so the exact OLS t/F theory does not transfer unchanged.

**!** A random intercept does not repair every dependence problem

The basic model assumes independent groups and independent residual errors conditional on the random effects. Serial correlation, household relationships across groups, or an omitted random slope may require a different covariance structure. A random-effects model also does not remove confounding: correlation between group effects and predictors can invalidate its interpretation unless modeled appropriately.

## Partial pooling and shrinkage

Three possible analyses of group means illustrate what the random-effects assumption adds:

| Approach | What is shared? | Main tradeoff |
|---|---|---|
| Complete pooling | One mean for every group | Ignores genuine group differences. |
| Separate group means | Each group has its own unrestricted mean | Small groups can have very noisy estimates. |
| Partial pooling | Groups share a distribution of deviations | Stabilizes estimates using an explicit exchangeability assumption. |

**Proposition 5.2** (Shrinkage in the Intercept-Only Model)
For $Y_{ij} = \mu + b_j + \varepsilon_{ij}$ under the Gaussian assumptions above, suppose $\mu$, $\tau^2 > 0$, and $\sigma^2 > 0$ are known. If group $j$ has $n_j$ observations, then

$$\mathbb{E}(\mu + b_j \mid \bar{Y}_j) = \mu + w_j(\bar{Y}_j - \mu), \qquad w_j = \frac{\tau^2}{\tau^2 + \sigma^2/n_j}.$$

The conditional group mean is a weighted average of the observed group mean and the common

mean. Small groups have less precise sample means and receive more shrinkage toward the common mean. Larger between-group variation reduces shrinkage.

**Why the weight appears.** The sample mean contains a group signal $b_j$ with variance $\tau^2$ and averaged residual noise with variance $\sigma^2/n_j$. Conditioning a jointly normal pair yields the signal-to-total-variance weight. Plugging estimated parameters into this expression gives an empirical best linear unbiased prediction (empirical BLUP) in this simple model. More general random-slope models have a matrix version rather than this scalar weight.

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

set.seed(8561)
n_group <- c(3, 4, 5, 6, 8, 10, 12, 16, 20, 24, 30, 40)
group <- factor(rep(LETTERS[seq_along(n_group)], n_group))
b_true <- rnorm(length(n_group), sd = 2)
dat_group <- data.frame(group,
  y = 12 + b_true[as.integer(group)] + rnorm(length(group), sd = 3))
fit_pool <- lme4::lmer(y ~ 1 + (1 | group), data = dat_group,
                       REML = TRUE)
mu_hat <- unname(lme4::fixef(fit_pool)[1])
tau2_hat <- as.numeric(lme4::VarCorr(fit_pool)$group[1, 1])
sigma2_hat <- sigma(fit_pool)^2
raw_mean <- tapply(dat_group$y, dat_group$group, mean)
shrunk_mean <- mu_hat + lme4::ranef(fit_pool)$group[, "(Intercept)"]
w_hat <- tau2_hat / (tau2_hat + sigma2_hat / n_group)
round(data.frame(n = n_group, raw_mean, mixed_mean = shrunk_mean,
                 weight_on_group_mean = w_hat), 3)
```

```
   n raw_mean mixed_mean weight_on_group_mean
A  3   16.277     14.780                0.647
B  4    9.330     10.117                0.709
C  5   15.146     14.380                0.753
D  6   12.371     12.300                0.786
E  8   12.053     12.051                0.830
F 10   11.601     11.663                0.859
G 12    7.901      8.399                0.880
H 16   11.816     11.837                0.907
I 20   12.033     12.033                0.924
J 24   16.297     16.025                0.936
K 30    8.738      8.909                0.948
L 40   11.989     11.991                0.961
```

```
c(ICC = tau2_hat / (tau2_hat + sigma2_hat))
```

```
      ICC
0.3790268
```

The syntax `(1 | group)` requests a random intercept for each group. The shared intercept is returned by `fixef()`, while `ranef()` returns estimated group deviations. These are different quantities; adding them gives the fitted group mean in this intercept-only example. See the `lmer()` reference for model syntax.

```
pooling <- data.frame(label = paste0(levels(group), " (", n_group, ")"),
                      raw = as.numeric(raw_mean),
                        pooled = as.numeric(shrunk_mean))
pooling$label <- factor(pooling$label, levels = pooling$label)
ggplot(pooling, aes(y = label)) +
  geom_vline(xintercept = mu_hat, linetype = "dashed",
    colour = "darkorange3") +
  geom_segment(aes(x = raw, xend = pooled, yend = label), colour = "grey60") +
  geom_point(aes(x = raw, colour = "Separate means",
    shape = "Separate means"), size = 2.3) +
  geom_point(aes(x = pooled, colour = "Partial pooling",
    shape = "Partial pooling"), size = 2.3) +
  scale_colour_manual(values = c("Separate means" = "black",
    "Partial pooling" = "steelblue4")) +
  scale_shape_manual(values = c("Separate means" = 1,
    "Partial pooling" = 16)) +
  labs(x = "Group mean", y = "Group (sample size)", colour = NULL,
    shape = NULL)
```

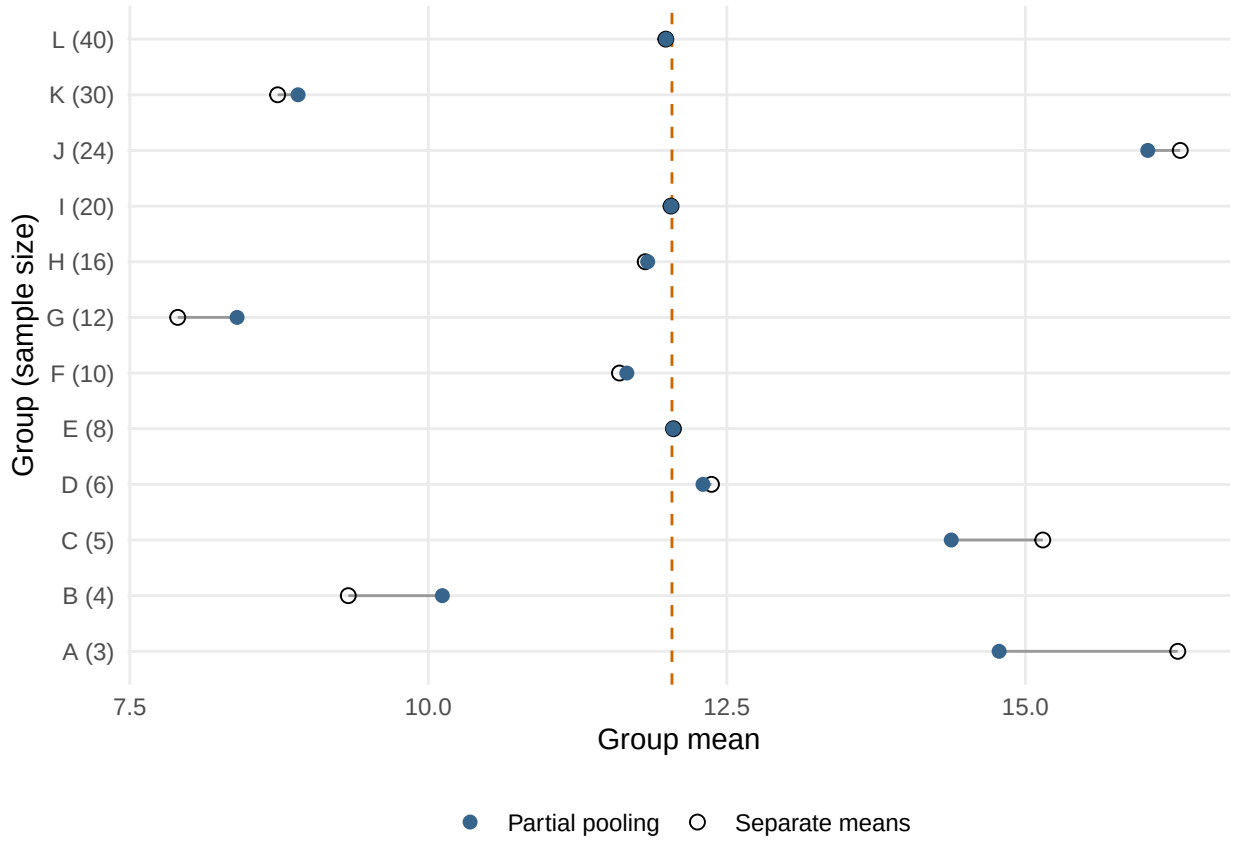



Figure 5.5: Unpooled sample means and partially pooled group means in simulated data. The dashed line is the fitted common mean. The amount of shrinkage reflects group sample size and the estimated variance components.

**What shrinkage does not establish.** It is not evidence that a small group's true mean equals the common mean. It reflects uncertainty and the assumed distribution of group effects. Point predictions of group deviations are also not confidence intervals or a justified ranking of all groups.

## Application: repeated measurements and random slopes

> **i** Application: reaction times during sleep restriction
>
> The public `sleepstudy` data bundled with `lme4` contain repeated reaction-time measurements for 18 subjects. `Reaction` is the average reaction time for a subject on a recorded day, measured in milliseconds. Days 0–1 are adaptation/training and day 2 is baseline. We use baseline and later measurements, with `day = Days - 2`. The independent grouping unit is the subject. This linear-trend example is not an estimate of a randomized treatment contrast. See the dataset documentation for the study and variables.

A random intercept shifts a subject's line vertically. A **random slope** also allows the subject's rate of change to vary:

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + b_{0j} + b_{1j} t_{ij} + \varepsilon_{ij}, \qquad \begin{pmatrix} b_{0j} \\ b_{1j} \end{pmatrix} \sim N_2(\mathbf{0}, \mathbf{G}).$$

Here $\beta_1$ describes the average linear trend across the modeled subject population, while $\beta_1 + b_{1j}$ is the subject-specific slope. Allowing covariance between $b_{0j}$ and $b_{1j}$ recognizes that baseline level and rate of change may be related.

**A covariance model as well as a mean model.** For two measurements on the same subject, at distinct occasions $t$ and $s$, the covariance contributed by the random effects is $g_{00} + (t+s)g_{01} + tsg_{11}$. Thus correlations can depend on the observation times. One constant random-intercept ICC no longer summarizes the whole dependence structure.

```
data("sleepstudy", package = "lme4")
study <- subset(sleepstudy, Days >= 2)
study$day <- study$Days - 2
fit_sleep <- lme4::lmer(Reaction ~ day + (day | Subject),
                        data = study, REML = TRUE)
lme4::fixef(fit_sleep)
```

```
(Intercept)         day
  267.96742    11.43543
```

```
lme4::VarCorr(fit_sleep)
```

```
 Groups   Name        Std.Dev. Corr
 Subject  (Intercept) 30.9573
          day          6.7659  0.178
 Residual             25.5264
```

```
lme4::isSingular(fit_sleep)
```

```
[1] FALSE
```

In `(day | Subject)`, the random intercept is included along with the random slope, and their covariance is estimated. This is a model for continuous Gaussian responses. Binary or count outcomes with random effects lead to **generalized linear mixed models**, which add a link and response family; they require further study beyond this topic.

```
subjects <- levels(study$Subject)[1:4]
plot_grid <- expand.grid(day = seq(0, 7, length.out = 40), Subject = subjects)
plot_grid$conditional <- predict(fit_sleep, newdata = plot_grid)
plot_grid$population <- predict(fit_sleep, newdata = plot_grid, re.form = NA)
subject_lines <- rbind(
  transform(plot_grid, prediction = conditional, fit = "Subject-specific"),
```

```
  transform(plot_grid, prediction = population, fit = "Population"))
ggplot(subset(study, Subject %in% subjects), aes(day, Reaction)) +
  geom_point(size = 1.7) +
  geom_line(data = subject_lines,
            aes(y = prediction, colour = fit, linetype = fit),
              linewidth = 0.8) +
  scale_colour_manual(values = c("Subject-specific" = "steelblue4",
    Population = "black")) +
  scale_linetype_manual(values = c("Subject-specific" = "solid",
    Population = "dashed")) +
  facet_wrap(~ Subject, ncol = 2, labeller = label_both) +
  coord_cartesian(ylim = range(study$Reaction)) +
  labs(x = "Days since baseline", y = "Reaction time (ms)", colour = NULL,
    linetype = NULL)
```

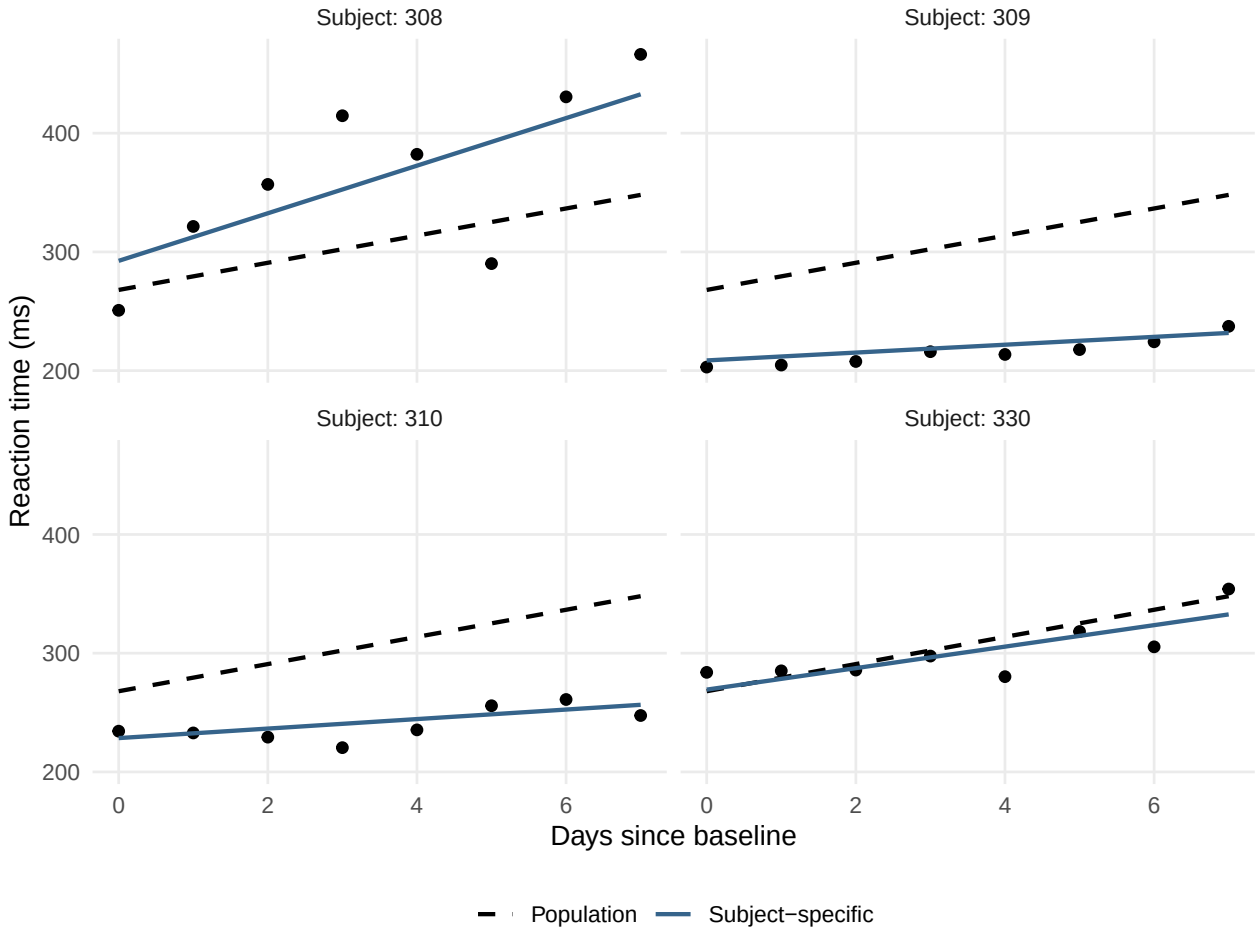



Figure 5.6: Four subjects from the sleepstudy data. Blue lines include each subject's fitted random effects; dashed lines use only the common fixed effects. A common linear trend does not require identical subject trajectories.

**Prediction target.** For a previously observed subject, predictions can use that subject's estimated

random effects. For a new subject with no response data yet, those deviations are unknown; setting them to zero gives the population mean in this Gaussian identity-link model. A prediction interval for the new subject must still include between-subject and residual variation, together with parameter uncertainty. A fixed-effect line alone is not such an interval. With a nonlinear GLMM link, setting random effects to zero generally does not equal averaging the response over their distribution.

## Inference, model checks, and further reading

**ML and REML.** Maximum likelihood estimates regression and covariance parameters through the marginal model. Restricted maximum likelihood (REML) accounts for the estimation of fixed effects when fitting covariance parameters. When comparing different fixed-effect structures by likelihood, use ML fits on the same observations and keep the specified random-effects structure comparable; the REML criterion is not directly comparable across different fixed-effect designs.

The following comparison asks whether the average linear trend is zero while allowing the same subject-specific random intercept and slope structure in both models. Its likelihood-ratio reference is asymptotic, not the exact F distribution from Chapter 4 — Model Comparison, Partial F Tests, and ANOVA.

```
fit_sleep_ml <- update(fit_sleep, REML = FALSE)
fit_no_trend_ml <- lme4::lmer(Reaction ~ 1 + (day | Subject),
                              data = study, REML = FALSE)
anova(fit_no_trend_ml, fit_sleep_ml, refit = FALSE)
```

```
Data: study
Models:
fit_no_trend_ml: Reaction ~ 1 + (day | Subject)
fit_sleep_ml: Reaction ~ day + (day | Subject)
                npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
fit_no_trend_ml    5 1444.4 1459.3 -717.21    1434.4
fit_sleep_ml       6 1425.2 1443.0 -706.58    1413.2 21.266  1  3.997e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **!** Variance components require special care
>
> Testing whether a random-effect variance is zero places the null parameter on a boundary. The usual chi-square rule based only on the parameter-count difference need not apply. Small numbers of independent groups also limit large-sample approximations. Methods such as parametric bootstrap or appropriate small-sample corrections need additional justification. Do not use the ordinary OLS F test simply because the model contains a linear predictor.

**Check what the design can support.** Examine conditional residuals versus fitted values, residual patterns across time and groups, convergence messages, and near-singularity. A singular fit means the fitted random-effects covariance is on or near a lower-dimensional boundary, such as an estimated variance near zero or a redundant intercept/slope combination. It is not synonymous with an

optimizer failing to converge. Review the design and model complexity before changing optimizer settings. See `isSingular()`.

```
conditional_diagnostics <- data.frame(fitted = fitted(fit_sleep),
                                      residual = resid(fit_sleep))
residual_plot <- ggplot(conditional_diagnostics, aes(fitted, residual)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_point(alpha = 0.5, colour = "steelblue4", size = 1.5) +
  labs(x = "Conditional fitted value", y = "Conditional residual")
qq_plot <- ggplot(conditional_diagnostics, aes(sample = residual)) +
  stat_qq_line(colour = "darkorange3") +
  stat_qq(size = 1.5, colour = "steelblue4") +
  labs(x = "Theoretical normal quantile", y = "Conditional residual")
patchwork::wrap_plots(residual_plot, qq_plot, nrow = 1)
```

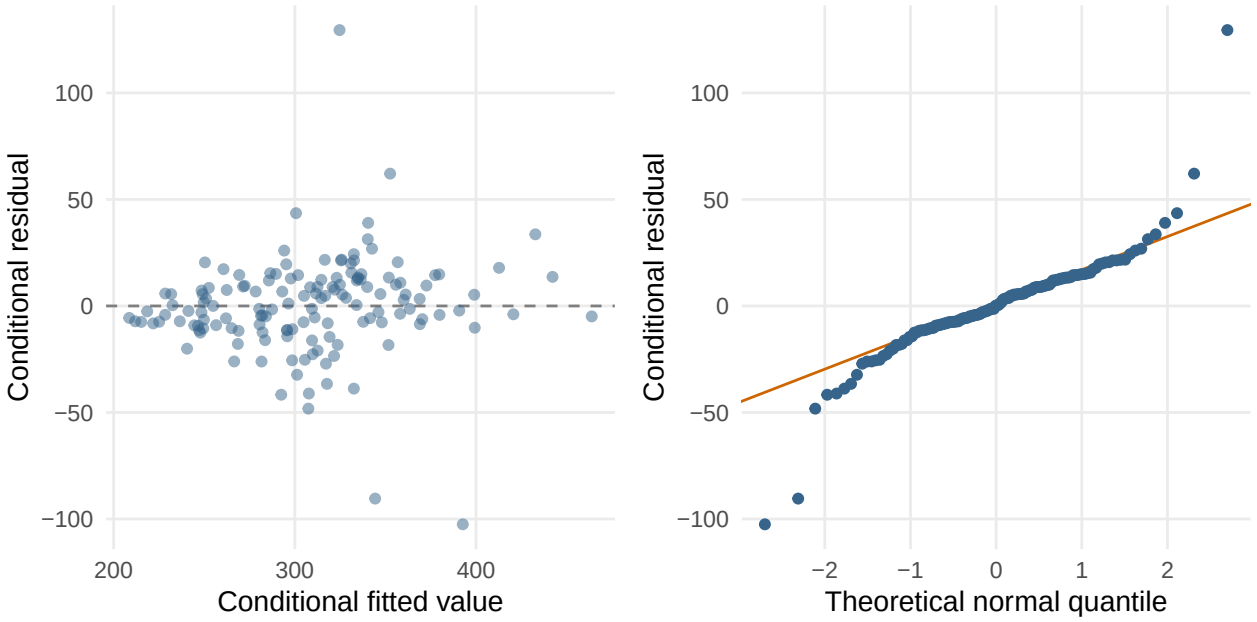



Figure 5.7: Basic conditional-residual checks for the fitted sleepstudy model. These plots can reveal mean or residual-distribution problems, but they do not prove that the full dependence structure is correct.

Plots of estimated random effects can be informative, but they display shrunken estimates with unequal uncertainty, not directly observed independent normal draws. With few groups, distributional checks are particularly limited. For prediction to new subjects, evaluate with subject-level holdouts; random row-wise splitting can let the same subject appear in both training and test data.

**Self-study questions.**

1. Derive the covariance and ICC for the random-intercept model. Explain why increasing observations within a group is different from adding new independent groups.
2. In the simulation, verify that `mu_hat + w_hat * (raw_mean - mu_hat)` matches the mixed-model group predictions. Explain the role of $n_j$.

3. Write down the fixed and random portions of the `sleepstudy` model. Explain what changes if `(day | Subject)` is replaced with `(1 | Subject)`.
4. Explain why a new subject's prediction interval is wider than an interval that conditions on known subject effects, and why a row-wise validation split may answer the wrong question.
5. State the null hypothesis in the ML comparison and explain why testing a random-effect variance of zero is a different inferential problem.

**Suggested reading.** The authors' Fitting Linear Mixed-Effects Models Using lme4 develops the model and computation. The package's inference guide discusses likelihood, bootstrap, and small-sample approaches. Continue to generalized mixed models only after distinguishing conditional and marginal prediction targets in the Gaussian case.

# References

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. John wiley & Sons.

Montgomery, Douglas C, Elizabeth A Peck, and G Geoffrey Vining. 2021. *Introduction to Linear Regression Analysis*. John Wiley & Sons.

Petersen, Kaare Brandt, and Michael Syskind Pedersen. 2008. *The Matrix Cookbook*. Technical University of Denmark.

Seber, George AF, and Alan J Lee. 2003. *Linear Regression Analysis*. John Wiley & Sons.

# Part II

# Appendix

# Matrix Algebra Review

### Matrix dimensions

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ is defined and is $m \times p$.

### Transpose rules

For conformable matrices,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

### Inverse rule

If $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}.$$

### Symmetric matrices

A matrix $\mathbf{A}$ is symmetric if

$$\mathbf{A}^\top = \mathbf{A}.$$

Covariance matrices are always symmetric.

---

A good reference of the linear algebra can be found from Petersen and Pedersen (2008).

- Petersen, Kaare Brandt, and Michael Syskind Pedersen. (2008). *The Matrix Cookbook*. Technical University of Denmark.

# Basic Mathematical Statistics

The chi-square, Student's $t$, and $F$ distributions all arise from normal random variables. This appendix explains the constructions, proves the distributional results, and connects them to the general linear hypothesis in Chapter 3 — Distribution Theory of OLS and Inference, Section 3.8: General Linear Hypotheses.

The logical order is **normal variables** $\longrightarrow$ **sums of squares** $\longrightarrow$ **standardized ratios**. Degrees of freedom count independent normal components. Independence is essential when forming the ratios.

## Chi-square: a sum of squared standard normals

**Definition 5.7** (Chi-square distribution)
If $Z_1, \ldots, Z_r$ are independent $N(0,1)$ variables, then

$$U = \sum_{j=1}^{r} Z_j^2 \sim \chi_r^2.$$

Each squared standardized component contributes one degree of freedom. Thus $U$ measures a total squared distance from zero. Its density is

$$\boxed{f_U(u) = \frac{u^{r/2-1} e^{-u/2}}{2^{r/2}\Gamma(r/2)}, \qquad u > 0.}$$

This is a gamma density with shape $r/2$ and **scale** 2. Here $\Gamma(a) = \int_0^\infty x^{a-1}e^{-x}\,dx$ for $a > 0$. For the sum-of-squares construction, $r$ is a positive integer; the density also defines $\chi_r^2$ for any real $r > 0$.

Proof: chi-square density and moments

For $s < 1/2$,

$$\begin{aligned}\mathbb{E}(e^{sZ_j^2}) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{1-2s}{2} z^2\right\} dz \\ &= (1-2s)^{-1/2}.\end{aligned}$$

The substitution $w = \sqrt{1-2s}\, z$ gives the second equality. Independence therefore gives

$$M_U(s) = \prod_{j=1}^{r} \mathbb{E}(e^{sZ_j^2}) = (1-2s)^{-r/2}.$$

To identify the law, integrate the displayed gamma density against $e^{su}$:

$$\begin{aligned}\int_0^\infty e^{su} \frac{u^{r/2-1}e^{-u/2}}{2^{r/2}\Gamma(r/2)}\, du &= \frac{\Gamma(r/2)(1/2-s)^{-r/2}}{2^{r/2}\Gamma(r/2)} \\ &= (1-2s)^{-r/2}.\end{aligned}$$

Moment-generating functions finite in a neighborhood of zero uniquely determine a probability law, so $U$ has the claimed density. Differentiating at zero gives

$$\mathbb{E}(U) = r, \qquad \mathrm{Var}(U) = 2r.$$

In particular, $U/r$ has mean 1 and variance $2/r$: it estimates a unit variance more stably as the degrees of freedom increase.

## Why normal quadratic forms are chi-square

**Theorem 5.1** (Normal quadratic forms)
Let $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I}_n)$, and let $\mathbf{A}$ be symmetric and idempotent, with rank $r \geq 1$. Then

$$\boxed{\mathbf{Z}^\top \mathbf{A} \mathbf{Z} \sim \chi^2_r.}$$

Proof: normal quadratic forms

The spectral theorem gives an orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{A} = \mathbf{Q}\,\mathrm{diag}(\mathbf{I}_r, \mathbf{0})\mathbf{Q}^\top.$$

Indeed, idempotence implies $\lambda^2 = \lambda$, so every eigenvalue is 0 or 1; exactly $r$ eigenvalues are 1. Set $\mathbf{W} = \mathbf{Q}^\top \mathbf{Z}$. An orthogonal transformation preserves the standard multivariate normal law, because its mean is zero and its covariance is $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n$. Hence the $W_j$ are independent $N(0,1)$ variables, and

$$\mathbf{Z}^\top \mathbf{A} \mathbf{Z} = \mathbf{W}^\top \mathrm{diag}(\mathbf{I}_r, \mathbf{0})\mathbf{W} = \sum_{j=1}^{r} W_j^2 \sim \chi^2_r.$$

Geometrically, $\mathbf{A}$ projects onto an $r$-dimensional subspace. The quadratic form is the squared length of the projected normal vector. If $r = 0$, it is identically zero rather than having a positive-degree chi-square density.

**Corollary 5.1** (Residual variation in the normal linear model)
Throughout the regression applications, assume

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I}_n), \quad \sigma^2 > 0, \quad \operatorname{rank}(\mathbf{X}) = p < n,$$

with fixed $\mathbf{X}$. Write

$$\mathbf{V} = (\mathbf{X}^\top\mathbf{X})^{-1}, \quad \mathbf{H} = \mathbf{X}\mathbf{V}\mathbf{X}^\top, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{H}, \quad \nu = n - p.$$

Since $\mathbf{M}$ is symmetric and idempotent with rank $\nu$, and $\mathbf{M}\mathbf{X} = \mathbf{0}$,

$$\frac{\text{SSE}}{\sigma^2} = \left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right)^\top \mathbf{M} \left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right) \sim \chi^2_\nu.$$

Consequently, $\hat{\sigma}^2 = \text{SSE}/\nu$ is unbiased for $\sigma^2$.

**Proposition 5.3** (Independence of OLS estimates and residual variation)
Both $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{V}\mathbf{X}^\top\boldsymbol{\varepsilon}$ and $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ are jointly normal, and

$$\operatorname{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = \sigma^2\mathbf{V}\mathbf{X}^\top\mathbf{M} = \mathbf{0}.$$

A jointly normal vector with zero cross-covariance has a characteristic function that factors into the two marginal characteristic functions. Thus these vectors are independent, even though the residual covariance is singular. Since $\text{SSE} = \mathbf{e}^\top\mathbf{e}$, $\hat{\boldsymbol{\beta}}$ and SSE are independent. Zero covariance alone would not suffice without joint normality.

## Student's t: estimating an unknown scale

**Definition 5.8** (Student's t distribution)
Let $Z \sim N(0,1)$ and $U \sim \chi^2_\nu$ be independent, with $\nu > 0$. Then

$$\boxed{T = \frac{Z}{\sqrt{U/\nu}} \sim t_\nu.}$$

The denominator is a random estimate of a standard deviation. It can be small, producing larger standardized values than a normal denominator fixed at 1 would produce. The result is a symmetric density with heavier tails than the standard normal density.

Proof: Student t density

Use $t = z/\sqrt{u/\nu}$ and retain $u$. The inverse map and absolute Jacobian determinant are

$$z = t\sqrt{u/\nu}, \qquad \left|\frac{\partial(z,u)}{\partial(t,u)}\right| = \sqrt{u/\nu}.$$

Independence gives a product joint density. Integrating out $u$ yields

$$f_T(t) = \int_0^\infty \frac{e^{-t^2u/(2\nu)}}{\sqrt{2\pi}} \frac{u^{\nu/2-1}e^{-u/2}}{2^{\nu/2}\Gamma(\nu/2)} \sqrt{\frac{u}{\nu}}\, du$$
$$= \frac{1}{\sqrt{2\pi\nu}\, 2^{\nu/2}\Gamma(\nu/2)} \int_0^\infty u^{(\nu+1)/2-1} e^{-(1+t^2/\nu)u/2}\, du.$$

Using $\int_0^\infty u^{a-1}e^{-bu}\, du = \Gamma(a)b^{-a}$ for $a, b > 0$, we obtain

$$\boxed{f_T(t) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}, \qquad t \in \mathbb{R}.}$$

This is the Student $t$ density. Its dependence on $t^2$ proves symmetry. Its tails decay polynomially, as $|t|^{-(\nu+1)}$, rather than exponentially as for a normal density. Also, $\mathrm{Var}(U/\nu) = 2/\nu \to 0$, so $U/\nu \to 1$ in probability and $t_\nu$ approaches $N(0,1)$ as $\nu \to \infty$.

**Corollary 5.2** (Student t inference for a regression contrast)
For any fixed nonzero $\mathbf{a} \in \mathbb{R}^p$, let $v_a = \mathbf{a}^\top \mathbf{V}\mathbf{a} > 0$. Then

$$Z = \frac{\mathbf{a}^\top\hat{\boldsymbol{\beta}} - \mathbf{a}^\top\boldsymbol{\beta}}{\sigma\sqrt{v_a}} \sim N(0,1), \qquad U = \frac{\mathrm{SSE}}{\sigma^2} \sim \chi^2_\nu.$$

The normal quadratic-form section proves their independence. Therefore

$$\frac{\mathbf{a}^\top\hat{\boldsymbol{\beta}} - \mathbf{a}^\top\boldsymbol{\beta}}{\hat{\sigma}\sqrt{v_a}} \sim t_\nu.$$

Under $H_0 : \mathbf{a}^\top\boldsymbol{\beta} = d$, replace the true contrast in the numerator by $d$. Choosing a coordinate vector for $\mathbf{a}$ gives the usual coefficient $t$ test.

## F: comparing two independent sums of squares

**Definition 5.9** (F distribution)
If $U_1 \sim \chi^2_r$ and $U_2 \sim \chi^2_\nu$ are independent, with $r, \nu > 0$, then

$$\boxed{F = \frac{U_1/r}{U_2/\nu} \sim F_{r,\nu}.}$$

The two degrees of freedom have different roles: $r$ belongs to the numerator and $\nu$ to the denominator. Both normalized sums of squares have expectation 1. Their ratio is positive and can be large when the numerator is unusually large or the denominator is small. The ratio itself does not have mean 1: its mean is $\nu/(\nu-2)$ when $\nu > 2$.

Proof: F density

Put $a = r/2$, $b = \nu/2$, and $c = r/\nu$. Transform $(U_1, U_2)$ to $(F, V)$, where $V = U_2$. The inverse map is

$$u_1 = cfv, \qquad u_2 = v, \qquad f > 0,\ v > 0,$$

and its absolute Jacobian determinant is

$$\left|\det \begin{pmatrix} cv & cf \\ 0 & 1 \end{pmatrix}\right| = cv.$$

Using independence and the two chi-square densities,

$$\begin{aligned} f_F(f) &= \int_0^\infty \frac{(cfv)^{a-1} e^{-cfv/2}}{2^a \Gamma(a)} \frac{v^{b-1} e^{-v/2}}{2^b \Gamma(b)} cv\, dv \\ &= \frac{c^a f^{a-1}}{2^{a+b}\Gamma(a)\Gamma(b)} \int_0^\infty v^{a+b-1} e^{-(1+cf)v/2}\, dv \\ &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} c^a f^{a-1} (1+cf)^{-(a+b)}. \end{aligned}$$

Substituting the definitions of $a, b, c$ gives the $F$ density:

$$\boxed{f_F(f) = \frac{\Gamma((r+\nu)/2)}{\Gamma(r/2)\Gamma(\nu/2)} \left(\frac{r}{\nu}\right)^{r/2} f^{r/2-1} \left(1 + \frac{r}{\nu} f\right)^{-(r+\nu)/2}, \quad f > 0.}$$

Remark: normalization and order of the degrees of freedom

The raw ratio $U_1/U_2$ equals $(r/\nu)F$, and hence is not generally $F_{r,\nu}$. Reversing the numerator and denominator gives $1/F \sim F_{\nu,r}$.

## Proof of the general linear hypothesis F test

**Theorem 5.2** (F test for the general linear hypothesis)
Consider $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$, where $\mathbf{C}$ is $r \times p$ with row rank $r$, $1 \le r \le p$, and $\mathbf{d} \in \mathbb{R}^r$. Keep the normal-model assumptions from the normal quadratic-form section, and write

$$\mathbf{D} = \mathbf{C}\mathbf{V}\mathbf{C}^\top, \qquad \mathbf{L} = \mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}.$$

Under $H_0$, the statistic

$$F = \frac{\mathbf{L}^\top \mathbf{D}^{-1} \mathbf{L}}{r\hat{\sigma}^2} \sim F_{r,n-p}.$$

Proof: general linear hypothesis F test

**Step 1: standardize the restrictions.** The matrix $\mathbf{D}$ is positive definite: for nonzero $\mathbf{b}$, full row rank implies $\mathbf{C}^\top\mathbf{b} \neq \mathbf{0}$, so $\mathbf{b}^\top\mathbf{D}\mathbf{b} = (\mathbf{C}^\top\mathbf{b})^\top\mathbf{V}(\mathbf{C}^\top\mathbf{b}) > 0$. Under $H_0$, $\mathbf{L} \sim N_r(\mathbf{0}, \sigma^2\mathbf{D})$. Consequently,

$$\mathbf{W} = \sigma^{-1}\mathbf{D}^{-1/2}\mathbf{L} \sim N_r(\mathbf{0}, \mathbf{I}_r),$$

where $\mathbf{D}^{-1/2}$ is the symmetric positive definite inverse square root. The numerator quadratic form therefore satisfies

$$Q = \frac{\mathbf{L}^\top\mathbf{D}^{-1}\mathbf{L}}{\sigma^2} = \mathbf{W}^\top\mathbf{W} \sim \chi^2_r.$$

**Step 2: identify the independent denominator.** We already proved $U = \mathrm{SSE}/\sigma^2 \sim \chi^2_\nu$, with $\nu = n - p$. The quantity $Q$ is a function of $\hat{\boldsymbol{\beta}}$, whereas $U$ is a function of SSE. Their independence follows from the normal-model independence proved in the normal quadratic-form section.

**Step 3: form the ratio.** Apply the F ratio result and cancel $\sigma^2$:

$$\boxed{F = \frac{Q/r}{U/\nu} = \frac{\mathbf{L}^\top\mathbf{D}^{-1}\mathbf{L}}{r\hat{\sigma}^2} \sim F_{r,n-p} \quad \text{under } H_0.}$$

Substituting $\mathbf{L} = \mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}$ and $\mathbf{D} = \mathbf{C}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{C}^\top$ recovers the statistic in Chapter 3 — Distribution Theory of OLS and Inference, Section 3.8: General Linear Hypotheses. Thus $r$ counts the independent restrictions and $n - p$ counts residual degrees of freedom. Large values indicate a departure from $H_0$; a level-$\alpha$ test rejects when $F > F_{1-\alpha;r,n-p}$, where the right-hand side is an upper quantile. The central $F$ law above is a null distribution; under an alternative the numerator is generally noncentral chi-square.

**Corollary 5.3** (The identity $T^2 = F$ for one restriction)
When $r = 1$ and $\mathbf{C} = \mathbf{a}^\top$,

$$F = \frac{(\mathbf{a}^\top\hat{\boldsymbol{\beta}} - d)^2}{\hat{\sigma}^2\mathbf{a}^\top\mathbf{V}\mathbf{a}} = T^2.$$

Equivalently, squaring the defining ratio for $T$ gives $T^2 = (Z^2/1)/(U/\nu)$, and $Z^2 \sim \chi^2_1$ independently of $U$. Hence $T^2 \sim F_{1,\nu}$. The upper-tail $F$ test matches the **two-sided** $t$ test, not a one-sided test.

## A compact regression example and reading guide

**Example 5.3** (Regression with 20 observations and 3 coefficients)
Suppose the normal linear model has $n = 20$ observations and $p = 3$ coefficients, including an intercept. Then $\nu = 17$. If $\mathrm{SSE} = 34$, the observed variance estimate is $\hat{\sigma}^2 = 34/17 = 2$.

- The sampling law of $\mathrm{SSE}/\sigma^2$ is $\chi^2_{17}$; the observed value 34 does not identify the unknown

$\sigma^2$.

- For $H_0 : \beta_1 = 0$, suppose $\hat{\beta}_1 = 1.5$ and $V_{22} = 0.05$ (the intercept is the first coordinate). Then

$$T = \frac{1.5}{\sqrt{2(0.05)}} \approx 4.743, \qquad F = T^2 = 22.5.$$

The null reference distributions are $t_{17}$ and $F_{1,17}$, respectively.

- For a joint test of two independent restrictions, $r = 2$. If the observed unscaled numerator quadratic form $\mathbf{L}^\top \mathbf{D}^{-1} \mathbf{L}$ is 12, then $F = 12/(2 \times 2) = 3$, compared with $F_{2,17}$.

What to check before using these results

Confirm the normal-model assumptions, the rank of the projection or restriction matrix, the residual degrees of freedom, and the required independence. A symmetric matrix alone is not enough for the quadratic-form chi-square result: idempotence is also needed. Likewise, correct marginal distributions alone do not establish a $t$ or $F$ ratio law.

**Reference books.**

1. **Seber, G. A. F., and Lee, A. J. (2003). *Linear Regression Analysis*, 2nd ed. Wiley.** Read Chapter 2, *Multivariate Normal Distribution*, for the normal-vector background; Chapter 3, *Linear Regression: Estimation and Distribution Theory*, for regression sampling distributions; and Chapter 4, *Hypothesis Testing*, for linear restrictions. This is the main reference for the regression applications above. Publisher and contents.
2. **Casella, G., and Berger, R. (2024 reprint). *Statistical Inference*, 2nd ed. Chapman & Hall/CRC.** Chapters 3, *Common Families of Distributions*, and 5, *Properties of a Random Sample*, provide the probability background for chi-square, $t$, and $F$ distributions. The publisher identifies this as a reprint of the earlier second edition. Publisher and contents.

# Part III

# Assignment

# Assignment 1

**STAT 8561 | Fall 2026**
**Total:** 40 points
**Coverage:** Chapter 1 — Matrix Representation and Notation and Chapter 2 — Least Squares Estimation.

This assignment focuses on matrix representation, least squares estimation, projection geometry, and linear regression in R.

| Question | Topic | Points |
|---|---|---|
| 1 | Matrix representation | 8 |
| 2 | Least squares and projection | 12 |
| 3 | Consequences and geometry of least squares | 12 |
| 4 | Regression in R | 8 |
| **Total** | | **40** |

**Instructions**

Use **Quarto** or **R Markdown**. Upload **both** files to the **Assignment 1** submission folder on **iCollege**:

1. Your rendered **PDF**, named `A1_LastName_FirstName.pdf`.
2. Its matching source file, named `A1_LastName_FirstName.qmd` or `A1_LastName_FirstName.Rmd`.

The PDF must be generated from the submitted source file. **HTML alone does not satisfy the submission requirement.**

- Put your name and course number at the beginning of the PDF.
- Clearly label every question and subpart.
- For each multiple-choice item, select **exactly one** answer and report its letter. Include an explanation only when requested.
- For True/False items, enter only `T` or `F`.
- For short-answer questions, one or two clear sentences are sufficient unless a derivation is requested.
- Round numerical answers to **three decimal places**, retaining full precision during calculations.
- Include all R code used to obtain your answers in executable chunks.
- Your source must render from beginning to end without errors. Type mathematical expressions using LaTeX; do not submit screenshots of answers or code.

# Question 1: Matrix Representation of the Linear Model [8]

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, 5,$$

with the following observed data:

| $i$ | $x_i$ | $Y_i$ |
|---|---|---|
| 1 | 0 | 2 |
| 2 | 1 | 4 |
| 3 | 2 | 5 |
| 4 | 3 | 8 |
| 5 | 4 | 9 |

## (a) [2]

Which of the following is the correct design matrix $\mathbf{X}$?

A.
$$\begin{pmatrix} 0 & 2 \\ 1 & 4 \\ 2 & 5 \\ 3 & 8 \\ 4 & 9 \end{pmatrix}$$

B.
$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$$

C.
$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$$

D.
$$\begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}$$

Solution 1(a)

**B.** Each row is $(1, x_i)$: the first column represents the intercept, and the second contains the predictor values.

## (b) [2]

Fill in the following table.

| Quantity | Dimension / Rank |
|---|---|
| $\mathbf{Y}$ | ___ |
| $\mathbf{X}$ | ___ |
| $\boldsymbol{\beta}$ | ___ |
| $\text{rank}(\mathbf{X})$ | ___ |

Solution 1(b)

| Quantity | Dimension / Rank |
|---|---|
| $\mathbf{Y}$ | $5 \times 1$ |
| $\mathbf{X}$ | $5 \times 2$ |
| $\boldsymbol{\beta}$ | $2 \times 1$ |
| $\text{rank}(\mathbf{X})$ | 2 |

The two columns of $\mathbf{X}$ are linearly independent because the predictor is not constant.

## (c) [2]

Which pair is correct?

A.
$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 30 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}$$

B.
$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 5 & 5 \\ 5 & 30 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}$$

C.
$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 30 & 10 \\ 10 & 5 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 77 \\ 28 \end{pmatrix}$$

D.
$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 28 \\ 77 \end{pmatrix}$$

Solution 1(c)

**A.** Here $\sum_i x_i = 10$, $\sum_i x_i^2 = 30$, $\sum_i Y_i = 28$, and $\sum_i x_i Y_i = 74$. Thus

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 30 \end{pmatrix}, \qquad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}.$$

### (d) [2]

Assume

$$\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}, \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_5.$$

For each statement, enter `T` or `F`.

| Statement | T/F |
|---|---|
| $\mathbb{E}[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}$ | --- |
| $\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_5$ | --- |
| $\mathbb{E}[\mathbf{Y}] = \mathbf{0}$ | --- |
| The five errors all have variance $\sigma^2$ | --- |

Solution 1(d)

| Statement | T/F |
|---|---|
| $\mathbb{E}[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}$ | T |
| $\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_5$ | T |
| $\mathbb{E}[\mathbf{Y}] = \mathbf{0}$ | F |
| The five errors all have variance $\sigma^2$ | T |

## Question 2: Least Squares and Projection [12]

Continue using the data and design matrix from Question 1.

### (a) [4]

Starting from

$$S(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}),$$

derive the normal equations

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y},$$

and then state the OLS estimator when $\mathbf{X}$ has full column rank.

Keep your derivation concise.

Solution 2(a)

Expanding,

$$S(\boldsymbol{\beta}) = \mathbf{Y}^\top \mathbf{Y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Differentiating,

$$\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Setting the derivative equal to zero gives

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}.$$

Hence,

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}}.$$

## (b) [3]

Using the quantities from Question 1, compute

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}.$$

Fill in the blanks:

$$\hat{\beta}_0 = _____, \qquad \hat{\beta}_1 = _____.$$

Hence,

$$\hat{Y} = _____ + _____ x.$$

Solution 2(b)

Using the data in Question 1,

$$\hat{\boldsymbol{\beta}} = \frac{1}{50}\begin{pmatrix} 30 & -10 \\ -10 & 5 \end{pmatrix}\begin{pmatrix} 28 \\ 74 \end{pmatrix} = \begin{pmatrix} 2.0 \\ 1.8 \end{pmatrix}.$$

Therefore $\hat{\beta}_0 = 2.000$, $\hat{\beta}_1 = 1.800$, and

$$\boxed{\hat{Y} = 2.0 + 1.8x}.$$

These are the coefficients used in Question 3.

## (c) [2]

For the least squares residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}},$$

which statement must be true?

A. $\mathbf{Xe} = \mathbf{0}$

B. $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$

C. $\mathbf{e}^\top \mathbf{e} = 0$

D. $\mathbf{He} = \mathbf{e}$

In one sentence, explain the geometric meaning of your choice.

**Your explanation:**

---

Solution 2(c)

**B.** The residual vector is orthogonal to the column space of $\mathbf{X}$.

## (d) [3]

For the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top,$$

enter `T` or `F`.

| Statement | T/F |
|---|---|
| $\mathbf{H}^\top = \mathbf{H}$ | --- |
| $\mathbf{H}^2 = \mathbf{H}$ | --- |
| $\hat{\mathbf{Y}} = \mathbf{HY}$ | --- |
| $\mathbf{H}$ is an orthogonal projection matrix | --- |

Solution 2(d)

| Statement | T/F |
|---|---|
| $\mathbf{H}^\top = \mathbf{H}$ | T |
| $\mathbf{H}^2 = \mathbf{H}$ | T |
| $\hat{\mathbf{Y}} = \mathbf{HY}$ | T |
| $\mathbf{H}$ is an orthogonal projection matrix | T |

## Question 3: Consequences and Geometry of Least Squares [12]

Continue using the data and fitted regression model from Questions 1 and 2.

Recall that

$$\hat{\beta}_0 = 2.0, \qquad \hat{\beta}_1 = 1.8.$$

### (a) Fitted Values and Residuals [3]

Compute the fitted values

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

and the residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

Fill in the blanks:

$$\hat{\mathbf{Y}} = \begin{pmatrix} _____ \\ _____ \\ _____ \\ _____ \\ _____ \end{pmatrix}, \qquad \mathbf{e} = \begin{pmatrix} _____ \\ _____ \\ _____ \\ _____ \\ _____ \end{pmatrix}.$$

Solution 3(a)

$$\hat{\mathbf{Y}} = \begin{pmatrix} 2.0 \\ 3.8 \\ 5.6 \\ 7.4 \\ 9.2 \end{pmatrix}, \qquad \mathbf{e} = \begin{pmatrix} 0 \\ 0.2 \\ -0.6 \\ 0.6 \\ -0.2 \end{pmatrix}.$$

## (b) Consequences of Including an Intercept [3]

For each statement, enter `T` or `F`.

| Statement | T/F |
|---|---|
| The residuals satisfy $\sum_{i=1}^{n} e_i = 0$. | ___ |
| The fitted values satisfy $\hat{Y} = \bar{Y}$. | ___ |
| The vector $\mathbf{1}_n$ belongs to $\mathcal{C}(\mathbf{X})$. | ___ |
| These properties necessarily hold for every regression model without an intercept. | ___ |

Solution 3(b)

| Statement | T/F |
|---|---|
| The residuals satisfy $\sum_{i=1}^{n} e_i = 0$. | T |
| The fitted values satisfy $\hat{Y} = \bar{Y}$. | T |
| The vector $\mathbf{1}_n$ belongs to $\mathcal{C}(\mathbf{X})$. | T |
| These properties necessarily hold for every regression model without an intercept. | F |

Because the model contains an intercept, $\mathbf{1}_n$ is a column of $\mathbf{X}$. Since $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$,

$$\mathbf{1}_n^\top \mathbf{e} = 0,$$

which implies

$$\sum_{i=1}^{n} e_i = 0$$

and therefore

$$\bar{\hat{Y}} = \bar{Y}.$$

## (c) Residual-Maker Matrix [3]

Define

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

For each statement, enter `T` or `F`.

| Statement | T/F |
|---|---|
| $\mathbf{e} = \mathbf{M}\mathbf{Y}$ | ___ |
| $\mathbf{M}^\top = \mathbf{M}$ | ___ |

| Statement | T/F |
|---|---|
| $\mathbf{M}^2 = \mathbf{M}$ | --- |
| $\mathbf{HM} = \mathbf{0}$ | --- |

Solution 3(c)

| Statement | T/F |
|---|---|
| $\mathbf{e} = \mathbf{MY}$ | T |
| $\mathbf{M}^\top = \mathbf{M}$ | T |
| $\mathbf{M}^2 = \mathbf{M}$ | T |
| $\mathbf{HM} = \mathbf{0}$ | T |

The matrix $\mathbf{H}$ projects onto $\mathcal{C}(\mathbf{X})$, whereas $\mathbf{M}$ projects onto $\mathcal{C}(\mathbf{X})^\perp$.

## (d) Rank and Uniqueness [3]

Suppose now that a design matrix $\mathbf{X}$ does **not** have full column rank.

For each statement, enter `T` or `F`.

| Statement | T/F |
|---|---|
| $(\mathbf{X}^\top\mathbf{X})^{-1}$ necessarily exists. | --- |
| The least squares coefficient vector $\hat{\boldsymbol{\beta}}$ may not be unique. | --- |
| The fitted vector $\hat{\mathbf{Y}}$ is still unique. | --- |
| The residual vector $\mathbf{e}$ is still unique. | --- |

Solution 3(d)

| Statement | T/F |
|---|---|
| $(\mathbf{X}^\top\mathbf{X})^{-1}$ necessarily exists. | F |
| The least squares coefficient vector $\hat{\boldsymbol{\beta}}$ may not be unique. | T |
| The fitted vector $\hat{\mathbf{Y}}$ is still unique. | T |
| The residual vector $\mathbf{e}$ is still unique. | T |

Even when several coefficient vectors produce the same least squares solution, the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ is unique.

# Question 4: Linear Regression in R [8]

For this question, use the built-in R dataset `cars`.

The variables are:

- `speed`: speed of the car in miles per hour;
- `dist`: stopping distance in feet.

Consider the model

$$\text{dist}_i = \beta_0 + \beta_1 \text{speed}_i + \varepsilon_i.$$

Load the data using

```
data(cars)
```

## (a) [4]

Fit the model using `lm()` and complete the table.

```
fit <- lm(dist ~ speed, data = cars)
summary(fit)
```

| Quantity | Your answer |
|---|---|
| $\hat{\beta}_0$ | --- |
| $\hat{\beta}_1$ | --- |

Which interpretation of $\hat{\beta}_1$ is correct?

A. For each 1 mph increase in speed, the stopping distance increases by exactly 3.9324 feet for every car.

B. For each 1 mph increase in speed, the fitted mean stopping distance increases by approximately 3.9324 feet.

C. The average speed increases by 3.9324 mph for every extra foot of stopping distance.

D. A car travelling at 0 mph has a stopping distance of 3.9324 feet.

Solution 4(a)

```
data(cars)
fit <- lm(dist ~ speed, data = cars)
coef(fit)
```

```
(Intercept)       speed
```

```
-17.579095     3.932409
```

$$\hat{\beta}_0 = -17.5791, \qquad \hat{\beta}_1 = 3.9324.$$

The correct interpretation is **B**.

## (b) [4]

Construct $\mathbf{Y}$ and $\mathbf{X}$ manually and reproduce the OLS coefficients using matrix operations.

Complete the following code:

```r
Y <- matrix(cars$dist, ncol = 1)

X <- cbind(
  1,
  cars$speed
)

# Replace NULL with your matrix expression
beta_hat <- NULL

beta_hat
```

Then report

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} _____ \\ _____ \end{pmatrix}.$$

Solution 4(b)

```r
Y <- matrix(cars$dist, ncol = 1)
X <- cbind(1, cars$speed)
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat
```

```
           [,1]
[1,] -17.579095
[2,]   3.932409
```

The result agrees with `lm()`:

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} -17.579095 \\ 3.932409 \end{pmatrix} \approx \begin{pmatrix} -17.579 \\ 3.932 \end{pmatrix}.$$

# Assignment 2

**STAT 8561 | Fall 2026**
**Due date:** September 30, 2026
**Total:** 40 points
**Coverage:** Chapter 2 — Least Squares Estimation and Chapter 3 — Distribution Theory of OLS and Inference.

This assignment emphasizes conceptual understanding of least squares, projection, and inference. Read the assumptions in each question carefully.

| Question | Topic | Points |
|---|---|---|
| 1 | Least squares: assumptions and consequences | 6 |
| 2 | Projection geometry and rank | 8 |
| 3 | Sampling properties and the role of normality | 8 |
| 4 | Interpreting confidence and prediction intervals | 6 |
| 5 | Distribution and inference for OLS | 12 |
| **Total** | | **40** |

**i** Instructions

Use **Quarto** or **R Markdown**. Upload **both** files to the **Assignment 2** submission folder on **iCollege**:

1. Your rendered **PDF**, named `A2_LastName_FirstName.pdf`.
2. Its matching source file, named `A2_LastName_FirstName.qmd` or `A2_LastName_FirstName.Rmd`.

The PDF must be generated from the submitted source file. **HTML alone does not satisfy the submission requirement.**

- Put your name and course number at the beginning of the PDF.
- Clearly label every question and subpart.
- For each multiple-choice item, select **exactly one** answer. Enter only the letter.
- For True/False items, enter only `T` or `F`. For matching items, enter one letter per row; letters may be reused.
- Questions 1-4 require **no written explanations, derivations, or R output**. Work through the reasoning before selecting your answers.
- Question 5 requires only the requested choices and numerical answers. Round numerical answers to **three decimal places**, retaining full precision during calculations.
- R is optional for checking calculations. If you use R, include your code after your answers in executable chunks.

- Your source must render from beginning to end without errors. Type mathematical expressions using LaTeX; do not submit screenshots of answers or code.

All necessary information is supplied in the questions. No external data or additional R packages are required.

# Question 1: Least Squares: Assumptions and Consequences [6]

Consider $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{X}$ is fixed, has full column rank, and $n > p$. Select **one answer per part**.

## (a) Changing the distributional assumption [2]

Two students use the same observed $\mathbf{X}$ and $\mathbf{Y}$. One assumes normal errors; the other does not specify an error distribution. Both define their estimate by minimizing

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

Which statement is correct?

A. The second student cannot compute a least squares estimate.

B. The estimates must differ because the distributional assumptions differ.

C. The estimates are the same because they minimize the same criterion for the same data.

D. The estimates are the same only if the observed residuals are all zero.

Solution 1(a)

**C.** Both students minimize the same function of the same observed data. Normality is not needed to compute the least squares estimate; it is used for exact distributional inference.

## (b) What makes the minimizer unique? [2]

Which statement explains why solving the normal equations gives a **unique** least squares minimizer under the stated assumptions?

A. Full column rank makes $\mathbf{X}^\top\mathbf{X}$ positive definite.

B. Normal errors make every system of normal equations invertible.

C. Having more observations than parameters guarantees full column rank for any design matrix.

D. Minimizing the sum of squared residuals forces every residual to be zero.

Solution 1(b)

**A.** Full column rank makes $\mathbf{X}^\top\mathbf{X}$ positive definite because

$$\mathbf{a}^\top\mathbf{X}^\top\mathbf{X}\mathbf{a} = \|\mathbf{X}\mathbf{a}\|^2 > 0 \quad \text{for every } \mathbf{a} \neq \mathbf{0}.$$

The least squares criterion therefore has a unique minimizer. The condition $n > p$ alone does not rule out linearly dependent columns.

### (c) Shifting every response [2]

Now suppose the model includes an intercept. Add 5 to **every observed response**, keep all predictors unchanged, and refit the same model by least squares. What happens?

A. Every slope increases by 5, while the intercept is unchanged.

B. The coefficients are unchanged, and every residual increases by 5.

C. The intercept increases by 5, and every residual increases by 5.

D. The intercept increases by 5, the slopes are unchanged, and the residuals are unchanged.

Solution 1(c)

**D.** Adding 5 to the intercept adds 5 to every fitted response. The slopes are unchanged, and

$$(\mathbf{Y} + 5\mathbf{1}_n) - (\hat{\mathbf{Y}} + 5\mathbf{1}_n) = \mathbf{e},$$

so the residuals are unchanged.

## Question 2: Projection Geometry and Rank [8]

For parts (a) and (c), let $\mathbf{X}$ have full column rank. Define

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{H}, \quad \hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \quad \mathbf{e} = \mathbf{M}\mathbf{Y}.$$

### (a) Projecting a second time [2]

A student applies $\mathbf{H}$ to the fitted vector and to the residual vector. Which pair of results must hold?

A. $\mathbf{H}\hat{\mathbf{Y}} = \mathbf{0}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.

B. $\mathbf{H}\hat{\mathbf{Y}} = \hat{\mathbf{Y}}$ and $\mathbf{H}\mathbf{e} = \mathbf{0}$.

C. $\mathbf{H}\hat{\mathbf{Y}} = \mathbf{Y}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.

D. $\mathbf{H}\hat{\mathbf{Y}} = \hat{\mathbf{Y}}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.

Solution 2(a)

**B.** Since $\mathbf{H}^2 = \mathbf{H}$,

$$\mathbf{H}\hat{\mathbf{Y}} = \mathbf{H}^2\mathbf{Y} = \hat{\mathbf{Y}}, \qquad \mathbf{He} = \mathbf{H}(\mathbf{I}_n - \mathbf{H})\mathbf{Y} = \mathbf{0}.$$

The fitted vector lies in the model space, and the residual vector is orthogonal to it.

## (b) Adding a redundant predictor [2]

Suppose $\mathbf{x}$ is not constant. Compare the least squares fits using

$$\mathbf{X} = [\mathbf{1}_n, \mathbf{x}] \quad \text{and} \quad \mathbf{W} = [\mathbf{1}_n, \mathbf{x}, 2\mathbf{x}]$$

for the **same response vector**. Which statement is correct?

A. The extra column must strictly reduce SSE.

B. The fit using $\mathbf{W}$ does not exist because $\mathbf{W}^\top\mathbf{W}$ is singular.

C. The fit using $\mathbf{W}$ has nonunique coefficients, but its fitted values and SSE equal those from $\mathbf{X}$.

D. The fit using $\mathbf{W}$ has nonunique coefficients and therefore nonunique fitted values.

Solution 2(b)

**C.** Adding $2\mathbf{x}$ does not change the column space. Both models project the response onto the same space, giving identical fitted values, residuals, and SSE. The coefficients for $\mathbf{W}$ are nonunique because its columns are linearly dependent.

## (c) Geometric consequences [4]

Enter `T` or `F`. Each statement is worth **1 point**. Here $\bar{Y} = n^{-1}\sum_i Y_i$.

| Statement | T/F |
|---|---|
| (i) The uncentered identity $\mathbf{Y}^\top\mathbf{Y} = \hat{\mathbf{Y}}^\top\hat{\mathbf{Y}} + \mathbf{e}^\top\mathbf{e}$ holds for the least squares fit. | ___ |
| (ii) The centered identity $\sum_i (Y_i - \bar{Y})^2 = \sum_i (\hat{Y}_i - \bar{Y})^2 + \text{SSE}$ is guaranteed for every least squares model, even one without an intercept. | ___ |
| (iii) If the model includes an intercept, the average fitted value equals the average observed response. | ___ |
| (iv) The residual vector must be orthogonal to the observed response vector $\mathbf{Y}$ itself. | ___ |

Solution 2(c)

**(i) T; (ii) F; (iii) T; (iv) F.**

- **(i)** $\hat{\mathbf{Y}}^\top \mathbf{e} = 0$ gives the uncentered sum-of-squares decomposition.
- **(ii)** The centered decomposition holds when the model includes an intercept, but is not guaranteed without one.
- **(iii)** An intercept gives $\mathbf{1}_n^\top \mathbf{e} = 0$, so the observed and fitted means agree.
- **(iv)** In fact, $\mathbf{e}^\top \mathbf{Y} = \mathbf{e}^\top (\hat{\mathbf{Y}} + \mathbf{e}) = \text{SSE}$, which need not be zero.

## Question 3: Sampling Properties and the Role of Normality [8]

Throughout this question, $\mathbf{X}$ is fixed with full column rank, $n > p$, and $\sigma^2 > 0$.

### (a) Match the result to its assumptions [4]

Choose the **weakest sufficient set of assumptions among A, B, and C** for each result. Each row is worth **1 point**. Letters may be reused.

A. The least squares criterion and full column rank alone; no error moment or distribution assumptions.

B. $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$, without assuming normality.

C. The normal linear model, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

| Result | Letter |
|---|---|
| (i) The least squares coefficient vector is the unique minimizer of $S(\boldsymbol{\beta})$. | ___ |
| (ii) $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$. | ___ |
| (iii) $\text{SSE}/\sigma^2 \sim \chi^2_{n-p}$. | ___ |
| (iv) $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$. | ___ |

Solution 3(a)

**(i) A; (ii) B; (iii) C; (iv) B.**

- **(i)** Uniqueness of the minimizer is an algebraic consequence of full column rank.
- **(ii)** Zero-mean errors give $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$.
- **(iii)** The exact chi-square distribution requires the normal linear model among the listed assumptions.
- **(iv)** The covariance formula follows from $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ without normality.

For (ii), zero-mean errors alone suffice; **B** is the weakest sufficient set *among the choices provided*.

## (b) Why does independence follow? [2]

Under the normal linear model, which argument correctly explains why $\hat{\boldsymbol{\beta}}$ and SSE are independent?

A. $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ are jointly normal with zero covariance, and SSE is a function of $\mathbf{e}$.

B. Any two uncorrelated random vectors are independent, regardless of their distribution.

C. They estimate different parameters, so they must be independent.

D. Each observed residual is independent of every other observed residual in every normal linear model.

Solution 3(b)

**A.** Both $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ are linear transformations of the normal response vector, so they are jointly normal. Also,

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{M} = \mathbf{0}.$$

Joint normality turns zero covariance into independence. Since $\mathrm{SSE} = \mathbf{e}^\top\mathbf{e}$ is a function of $\mathbf{e}$, it is independent of $\hat{\boldsymbol{\beta}}$. Uncorrelatedness alone would not suffice without joint normality.

## (c) Increasing the error variance [2]

Keep $\mathbf{X}$ and $\boldsymbol{\beta}$ fixed. In a second model, the errors still have mean zero, but their covariance is $4\sigma^2\mathbf{I}_n$ instead of $\sigma^2\mathbf{I}_n$. Which statement describes the **sampling distribution's mean and covariance** for the OLS estimator?

A. Its mean and covariance are both multiplied by 4.

B. Its mean is unchanged, and its covariance is multiplied by 16.

C. Its mean and covariance are both unchanged.

D. Its mean is unchanged, and its covariance is multiplied by 4; each coefficient's standard deviation doubles.

Solution 3(c)

**D.** The estimator remains unbiased, while its covariance becomes

$$4\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

Each coefficient's variance is multiplied by 4, so its standard deviation is multiplied by $\sqrt{4} = 2$.

# Question 4: Interpreting Confidence and Prediction Intervals [6]

Assume a correctly specified normal linear model. For prediction, a future observation has an error independent of the original data and with the same error variance.

## (a) Mean response versus one future response [2]

At the **same predictor value**, software reports two intervals at the 95% level:

- Interval I: $(2.493, 3.507)$;
- Interval II: $(1.659, 4.341)$.

One is a confidence interval for the mean response; the other is a prediction interval for one future response. Which interpretation is correct?

A. I is the prediction interval because predicting one observation involves less uncertainty than estimating a mean.

B. Both intervals describe the same target; their different widths are only rounding error.

C. I is the mean-response interval; II is the prediction interval because it also accounts for the new observation's error.

D. II is the mean-response interval because averaging adds the new observation's error variance.

Solution 4(a)

**C.** At the same predictor value, the mean-response interval uses standard error $\hat{\sigma}\sqrt{h_0}$, whereas the prediction interval uses $\hat{\sigma}\sqrt{1+h_0}$, where $h_0 = \mathbf{x}_0^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_0$. The extra 1 accounts for the new observation's error variance. Thus the narrower interval I concerns the mean and the wider interval II concerns one future response.

## (b) Connecting a confidence interval and a test [2]

A 95% confidence interval for $\beta_1$ is $(0.4, 1.6)$. Consider the corresponding two-sided $t$ test of

$$H_0 : \beta_1 = 1 \qquad \text{versus} \qquad H_1 : \beta_1 \neq 1$$

at $\alpha = 0.05$. Which conclusion is correct?

A. Reject $H_0$ because zero is outside the interval.

B. Fail to reject $H_0$ because 1 is inside the interval; this does not prove that $\beta_1 = 1$.

C. Accept $H_0$ as proven true because 1 is inside the interval.

D. No conclusion is possible because a $t$ test can only test a coefficient against zero.

Solution 4(b)

**B.** The hypothesized value 1 lies inside $(0.4, 1.6)$, so the corresponding two-sided test fails to reject $H_0 : \beta_1 = 1$ at the 5% level. This does not prove the null hypothesis. Whether zero lies in the interval is irrelevant to this stated null value.

### (c) What an interval means [2]

Enter `T` or `F`. Each statement is worth **1 point**. Assume a positive estimated error variance and $\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 > 0$.

| Statement | T/F |
|---|---|
| (i) Holding the data and predictor value fixed, changing a mean-response confidence level from 95% to 99% makes the interval wider. | ___ |
| (ii) A 95% confidence interval for the mean response is designed to contain 95% of individual future responses at that predictor value. | ___ |

Solution 4(c)

**(i) T; (ii) F.**

- **(i)** Holding the data fixed, a higher confidence level uses a larger critical value and produces a wider interval.
- **(ii)** A confidence interval for the mean concerns the unknown mean response. It does not describe a range containing 95% of individual future responses; those require a prediction interval.

## Question 5: Distribution and Inference [12]

Suppose a multiple linear regression model has $n = 20$ observations and $p = 3$ parameters: an intercept and two slopes. The first column of $\mathbf{X}$ is $\mathbf{1}_{20}$. Assume

$$\mathbf{Y} \sim N_{20}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_{20}) .$$

Suppose

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} 2.0 \\ 1.5 \\ -0.5 \end{pmatrix}, \qquad (\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} 0.05 & 0 & 0 \\ 0 & 0.05 & 0 \\ 0 & 0 & 0.08 \end{pmatrix},$$

and $\text{SSE} = 34$. Use $t_{0.975,17} = 2.110$.

## (a) [3]

Choose the correct distributions.

For $\hat{\boldsymbol{\beta}}$:

A. $N_3(\mathbf{0}, \sigma^2 \mathbf{I}_3)$

B. $N_3(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$

C. $t_{17}$

D. $\chi^2_{17}$

For $\text{SSE}/\sigma^2$:

A. $N(0, 1)$

B. $t_{17}$

C. $\chi^2_{17}$

D. $F_{3,17}$

Solution 5(a)

**B** for $\hat{\boldsymbol{\beta}}$ and **C** for $\text{SSE}/\sigma^2$:

$$\hat{\boldsymbol{\beta}} \sim N_3(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}), \qquad \frac{\text{SSE}}{\sigma^2} \sim \chi^2_{17}.$$

The residual degrees of freedom are $n - p = 20 - 3 = 17$.

## (b) [2]

Compute

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - p} = _____.$$

Solution 5(b)

$$\hat{\sigma}^2 = \frac{34}{20 - 3} = \boxed{2.000}.$$

## (c) [4]

For testing

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_1 : \beta_1 \neq 0,$$

where $\beta_1$ is the second entry of $\boldsymbol{\beta}$, fill in the following. Use significance level $\alpha = 0.05$.

$$\mathrm{SE}(\hat{\beta}_1) = _____, \qquad t = _____,$$

and choose the decision:

A. Reject $H_0$

B. Fail to reject $H_0$

Solution 5(c)

Use the second diagonal entry of $(\mathbf{X}^\top\mathbf{X})^{-1}$, which is 0.05:

$$\mathrm{SE}(\hat{\beta}_1) = \sqrt{2(0.05)} = 0.3162278 \approx \boxed{0.316},$$

$$t = \frac{1.5 - 0}{\sqrt{0.1}} = 4.7434165 \approx \boxed{4.743}.$$

Since $|t| > 2.110$, choose **A: Reject** $H_0$ at the 5% significance level.

## (d) [3]

Which of the following is the 95% confidence interval for $\beta_1$?

A. $(0.8329,\ 2.1671)$

B. $(1.1838,\ 1.8162)$

C. $(-0.6671,\ 0.6671)$

D. $(0,\ 3.0000)$

Solution 5(d)

**A.** Using the supplied critical value,

$$1.5 \pm 2.110\sqrt{0.1} = (0.832759,\ 2.167241) \approx \boxed{(0.833,\ 2.167)}.$$

These endpoints agree with option A at the requested three decimal places.

# Assignment 3

**Instructions**

Use **Quarto** or **R Markdown**. Upload **both** files to the **Assignment 3** submission folder on **iCollege**:

1. Your rendered **PDF**, named `A3_LastName_FirstName.pdf`.
2. Its matching source file, named `A3_LastName_FirstName.qmd` or `A3_LastName_FirstName.Rmd`.

The PDF must be generated from the submitted source file. **HTML alone does not satisfy the submission requirement.**

- Put your name and course number at the beginning of the PDF.
- Clearly label every question and subpart.
- Include all R code used to obtain your answers in executable chunks.
- Your source must render from beginning to end without errors. Type mathematical expressions using LaTeX; do not submit screenshots of answers or code.

Content: **TBA**.

# Part IV

# Exam

# Exam 1

**Term:** 2026F (Fall 2026)
**Instructor:** Chi-Kuang Yeh
**Date:** October 14, 2026
**Status:** Completed
**Time:** 75 minutes
**Total:** [100] points
**Coverage:** Assignment 1–Assignment 2 and lecture notes through Section 4.2: The Overall F Test; Definition 4.4 and later material are excluded.
**Emphasis:** Model interpretation, least squares geometry, and statistical reasoning.
**Notation:** $p$ is the number of coefficients, including any intercept. Vectors are boldfaced; comma-separated tuples are column vectors.
**Sums of squares:** SST = centered total; SSR = regression; SSE = residual.

| Question | Format | Points |
|---|---|---|
| 1 | Multiple Choice (10 parts, [5] points each) | [50] |
| 2 | True or False (5 parts, [4] points each) | [20] |
| 3 | Conceptual (3 parts, [10] points each) | [30] |
| **Total** | | **[100]** |

**Exam instructions**

1. Work independently. Only a **non-programmable calculator** is permitted as an electronic device. No code execution is required.
2. Record Questions 1–2 on the separate answer sheet: one letter A–D or T/F per part. Only the grid is graded; no explanations are required.
3. For Question 3, use one or two sentences per numbered part on the separate answer sheet.
4. Submit only the answer sheet. You may keep the question paper.

**Answer sheet:** Write all final answers on the separate answer sheet.

# Question 1: Multiple Choice [50]

Each part is worth **[5] points**. Select the **single best answer**: the statement guaranteed by the stated conditions. Record it on the separate answer sheet. No explanation is required.

## 1a)

For fixed predictor values, the mean model is $\mathbb{E}(Y_i) = \beta_1 x_i + \beta_2 z_i$, with no intercept term. At $x_i = z_i = 0$, which statement must hold?

A. The observed response must equal zero.
B. The mean response equals zero, although an individual response may differ from zero.
C. Both regression coefficients must equal zero.
D. The residuals must sum to zero for every data set.

Answer

**Answer: B.** With no intercept, substituting $x_i = z_i = 0$ gives a zero model mean. Random error still permits a nonzero observed response. A zero residual sum is not guaranteed without an intercept.

## 1b)

Each of two measurement errors has variance $\sigma^2 > 0$. Compare the variance of their arithmetic mean when their correlation is positive with the variance when it is zero. Keep the individual variances fixed. Which statement is correct?

A. The average of the uncorrelated errors has greater variance.
B. The two averages have the same variance.
C. The average of the positively correlated errors has zero variance.
D. The average of the positively correlated errors has greater variance.

Answer

**Answer: D.** For errors $\varepsilon_1, \varepsilon_2$, $\text{Var}\{(\varepsilon_1 + \varepsilon_2)/2\} = \sigma^2/2 + \text{Cov}(\varepsilon_1, \varepsilon_2)/2$. Positive covariance increases uncertainty in the average.

## 1c)

Fit a simple linear regression by OLS with an intercept and one nonconstant predictor $x_i$ measured in centimeters. Refit the same data using $x_i^* = x_i/100$, measured in meters. Which relationship must hold?

A. The estimated slope is multiplied by 100; the fitted responses and residuals stay the same.
B. The estimated slope is divided by 100; the fitted responses stay the same.

C. The fitted responses are divided by 100 because the predictor is smaller.
D. The residual sum of squares must decrease because the predictor uses smaller numbers.

Answer

**Answer: A.** Since $x = 100x^*$, $\hat{\beta}_0 + \hat{\beta}_1 x = \hat{\beta}_0 + (100\hat{\beta}_1)x^*$. Changing predictor units reexpresses the same fitted line; it does not improve the fit.

## 1d)

Fit $\mathbf{Y}$ by OLS using a design matrix $X$ with full column rank. Replace the response by $\mathbf{Y}^* = \mathbf{Y} + X\mathbf{a}$, where $\mathbf{a}$ is a fixed vector, and refit by OLS using the same $X$. Which relationship must hold?

A. The fitted vector is unchanged; the residual vector increases by $X\mathbf{a}$.
B. Both the fitted vector and residual vector increase by $X\mathbf{a}$.
C. The new residual vector must be zero.
D. The fitted vector increases by $X\mathbf{a}$; the residual vector is unchanged.

Answer

**Answer: D.** The added vector already lies in the model space. Thus $H(\mathbf{Y} + X\mathbf{a}) = \hat{\mathbf{Y}} + X\mathbf{a}$ and $(I - H)X\mathbf{a} = \mathbf{0}$. The fit absorbs the entire change.

## 1e)

Let $\mathbf{x}$ be nonconstant. Compare OLS using $X = \begin{pmatrix} \mathbf{1}_n & \mathbf{x} \end{pmatrix}$ with OLS using $W = \begin{pmatrix} \mathbf{1}_n & \mathbf{x} & \mathbf{x} + \mathbf{1}_n \end{pmatrix}$ on the same response vector. Which statement must hold?

A. The coefficients for $W$ are nonunique, but its fitted vector and SSE equal those from $X$.
B. The third column adds a new direction, so SSE must strictly decrease.
C. The fit using $W$ does not exist because $W^\top W$ is singular.
D. The coefficients and fitted vector for $W$ are both nonunique.

Answer

**Answer: A.** The added column is a linear combination of the first two. The column space and its unique projection are unchanged, even though different coefficient vectors represent that projection.

## 1f)

For a fixed design matrix $X$ with full column rank, which statement about computing the OLS coefficient estimate is correct?

A. Normally distributed errors are required to define the sum of squared residuals.
B. Normally distributed errors are required to make $X^\top X$ invertible.

C. The OLS estimate can be computed from $X$ and $\mathbf{Y}$ without assuming normally distributed errors.
D. Without normally distributed errors, the OLS estimate can be computed only when all observed residuals are zero.

Answer

**Answer: C.** Full column rank makes $X^\top X$ invertible, so $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{Y}$ can be computed from the observed data without assuming normal errors. Normality is not needed to define or minimize the sum of squared residuals.

## 1g)

In $\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, keep $X$ and $\boldsymbol{\beta}$ fixed, with $X$ of full column rank. Errors have mean zero in both models, but their covariance changes from $\sigma^2 I$ to $4\sigma^2 I$, where $\sigma^2 > 0$. How do the mean and standard deviation of each OLS coefficient estimator change?

A. The mean and standard deviation of each coefficient estimator both double.
B. Each mean is unchanged, and each standard deviation is multiplied by 4.
C. Each mean is unchanged, and each standard deviation doubles.
D. Each mean and each standard deviation are unchanged.

Answer

**Answer: C.** Mean-zero errors preserve $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$. The covariance is multiplied by 4, so standard deviations are multiplied by 2. More noise reduces precision without creating bias.

## 1h)

The usual two-sided 95% $t$ confidence interval for a slope $\beta_1$ is $(-0.2, 0.6)$. What is the conclusion of the corresponding two-sided $t$ test of $H_0 : \beta_1 = 0.5$ at significance level 0.05?

A. Reject $H_0$ because the interval is not centered at 0.5.
B. The interval proves that $\beta_1 = 0.5$.
C. No conclusion is possible because $t$ tests can only test a slope against zero.
D. Fail to reject $H_0$ because 0.5 is inside the interval; this does not prove $H_0$.

Answer

**Answer: D.** A corresponding two-sided test rejects exactly the hypothesized values outside the confidence interval. Failure to reject is not proof of an exact parameter value.

## 1i)

For the same response data, an intercept-only OLS fit has residual sum of squares 80, while an OLS fit with an intercept and predictors has residual sum of squares 30. What does the reduction of 50 represent?

A. The error variance $\sigma^2$ has decreased by 50.
B. The squared prediction error on every future data set will decrease by 50.
C. The reduction in unexplained variation in these observations relative to the constant-mean fit.
D. Each predictor is individually significant because the fit improved.

Answer

**Answer: C.** The intercept-only fit uses $\bar{Y}$, so its SSE is the centered SST of 80. Thus SSR $= 80 - 30 = 50$ is the sample variation explained beyond a constant mean. This improvement does not establish a change in $\sigma^2$, future predictive performance, or individual significance.

## 1j)

In a correctly specified normal linear model with an intercept and three slopes, a valid overall $F$ test rejects $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ at significance level 0.05. Which conclusion is justified?

A. Each of the three slopes must be individually significant at level 0.05.
B. There is evidence that at least one slope is nonzero, without identifying which one.
C. There is evidence that the intercept is nonzero.
D. The model must predict every future response accurately.

Answer

**Answer: B.** The overall test assesses the slopes jointly against an intercept-only model. Rejection provides evidence against all slopes being zero; it does not establish individual significance or accurate predictions.

# Question 2: True or False [20]

Each part is worth **[4] points**. Record **T** or **F** in the answer grid on the separate answer sheet. No explanation is required.

## 2a)

The assumption $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ guarantees that the model errors satisfy $\sum_{i=1}^{n} \varepsilon_i = 0$ in every realized sample.

Answer

**Answer: False.** Mean zero is a statement about expectations across repeated samples. It does not impose a zero sum on a realized error vector. The zero sum of fitted OLS residuals with an intercept is a different, algebraic property.

## 2b)

If the OLS coefficient estimator and the residual vector have zero cross-covariance, they must be independent, regardless of the error distribution.

Answer

**Answer: False.** Uncorrelated random vectors need not be independent. In the normal linear model, joint normality supplies the extra condition that makes the implication valid.

## 2c)

For an OLS fit with an intercept, the centered decomposition $\text{SST} = \text{SSR} + \text{SSE}$ holds even without normally distributed errors.

Answer

**Answer: True.** With an intercept, the centered fitted vector is orthogonal to the residual vector, giving the sum-of-squares identity. Normality is not needed for this algebraic decomposition.

## 2d)

In an OLS model with an intercept, the null hypothesis that all slopes are zero implies that the observed regression sum of squares SSR must equal zero in every sample.

Answer

**Answer: False.** Even when the true mean is constant, random noise can align with predictor directions and produce a nonzero fitted slope and SSR. Under the normal-model null, MSR estimates $\sigma^2$; it is not forced to be zero.

## 2e)

For an OLS model whose design matrix has full column rank, with an intercept and $p$ coefficients in total, the regression degrees of freedom associated with SSR are $p$.

Answer

**Answer: False.** The intercept accounts for the constant-mean direction. The centered regression degrees of freedom are $p-1$; the total and residual degrees of freedom are $n-1$ and $n-p$, respectively.

# Question 3: Conceptual [30]

For each part, write one or two sentences on the separate answer sheet; brief formulas may be used. Equivalent explanations are accepted, with partial credit for correct reasoning.

## 3a) Matrix representation and least squares [20]

### 3a(i) [10]

A regression of exam score (points) on study time (hours) gives the fitted line $\hat{y} = 60 + 2x$. Interpret the slope. Does the model require a student who studies one hour more than another to score exactly 2 points higher? Explain briefly.

Answer

**Answer:** A student who studies one hour more has a fitted mean exam score that is 2 points higher (mean [3], response units [2], one-unit change [2]). The two observed scores need not differ by exactly 2 points because each can deviate from its fitted mean [3]. Do not require or award a causal interpretation.

### 3a(ii) [10]

For simple linear regression with an intercept and $\mathbf{x} = (0, 1, 2, 3)$, a student proposes residuals $\mathbf{e} = (1, -1, 1, -1)$ because $\sum_i e_i = 0$. Is zero sum sufficient? Evaluate the other normal equation and decide whether these can be OLS residuals.

Answer

**Answer:** OLS also requires $\mathbf{x}^\top \mathbf{e} = 0$, so a zero residual sum alone is insufficient [5]. Here $0(1) + 1(-1) + 2(1) + 3(-1) = -2$ [2], so these cannot be the residuals of the claimed OLS fit [3].
Full-credit short answer: "No: $\mathbf{x}^\top \mathbf{e} = -2 \neq 0$ violates the OLS normal equations."

## 3b) Interpreting uncertainty [10]

### 3b(i) [10]

From the same fitted normal linear model and data, compare the usual 95% confidence interval for the mean with the 95% prediction interval for one independent future response at the same predictor value. Assume positive estimated error variance. Which is wider? Why would individual prediction remain uncertain even if the true mean were known exactly?

Answer

**Answer:** The prediction interval is wider [3] because it includes additional random error from the independent future response [4]. That individual variation remains even if the true mean response is known exactly [3]. This explanation earns full credit; no formula or separate statement about mean-estimation uncertainty is required.

# Answer Key and Grading Rubric

## Question 1: Multiple Choice [50]

| Part | 1a | 1b | 1c | 1d | 1e | 1f | 1g | 1h | 1i | 1j |
|---|---|---|---|---|---|---|---|---|---|---|
| Answer | B | D | A | D | A | C | C | D | C | B |

Award **[5] points** for each correct answer and **[0] points** otherwise. No explanations are required.

## Question 2: True or False [20]

| Part | 2a | 2b | 2c | 2d | 2e |
|---|---|---|---|---|---|
| Answer | False | False | True | False | False |

Award **[4] points** for each correct answer and **[0] points** otherwise. No explanations are required.

## Question 3: Conceptual [30]

Accept equivalent concise reasoning, including equations. Apply the allocations below for partial credit; deduct an arithmetic error once and retain credit for otherwise correct reasoning.

| Part | Grading criteria | Points |
|---|---|---|
| 3a(i) | Fitted/estimated mean [3]; response units [2]; per one-unit predictor increase [2]; individual outcomes still have error variation [3]. | [10] |
| 3a(ii) | Identify the additional OLS requirement $\mathbf{x}^\top \mathbf{e} = 0$ and show why the residual-sum check is insufficient [5]; compute $\mathbf{x}^\top \mathbf{e} = -2$ [2]; reject the proposed OLS fit [3]. | [10] |
| 3b(i) | Prediction interval is wider [3]; it includes additional random error from the independent future response [4]; that variation remains even if the true mean is known exactly [3]. | [10] |
| **Total** | | **[30]** |

# Part V

# Project

# Instruction

**STAT 8561 | Fall 2026**
**Due on December 13, 2026**
**Total:** 100 points
**Course-grade weight:** 20% (see Grade Distribution)
**MS students:** 1-4 students per group
**PhD students:** Complete the project individually
**Project:** Linear Regression Data Analysis and Machine Learning Comparison

Analyze a **real dataset with a continuous response** using multiple linear regression. Explain the findings in context, and compare predictions with **one machine learning regression algorithm of your choice**.

## Group Members

| Group | Members | Topic |
|---|---|---|
| 1 | Xiaorui Bao | TBD |
| 2 | Tommy Stell | TBD |
| 3 | Xinning Wang | TBD |
| TBD | TBD | TBD |

## Project Overview

| Criterion | Topic | Points |
|---|---|---|
| 1 | Question and data | 10 |
| 2 | Linear model and interpretation | 25 |
| 3 | Model comparison and diagnostics | 20 |
| 4 | Machine learning method | 10 |
| 5 | Fair predictive evaluation | 15 |
| 6 | Findings and critical discussion | 10 |
| 7 | Communication and reproducibility | 10 |
| **Total** | | **100** |

Instruction

Use **Quarto** or **R Markdown**. Upload **both** files to the **project submission area on iCollege**:

1. Your rendered **PDF**, named `Project_Report.pdf`.
2. Its matching source file, named `Project_Report.qmd` or `Project_Report.Rmd`.

Write **8-10 pages of main text**, excluding the title page, references, and appendices. List all group members and their contributions.
The submitted source and supporting files must reproduce the PDF and results. **HTML alone does not satisfy the requirement.**
**Python or Jupyter Notebook:** Discuss the format with the instructor in advance; a PDF and reproducible source are still required.
See Report requirements and submission for the full instructions. The submission folder will be announced on iCollege.

## Assessment criteria

The project totals **100 points** and is worth **20% of the course grade** (Grade Distribution). Each weight below is a percentage of the **project grade**: 25% equals 25 points. A score of 90/100 contributes 18 percentage points to the course grade.

| Criterion | Weight | Full-credit evidence (points) |
|---|---|---|
| Question and data | 10% | Clear question, observational unit, and outcome units (3); cited source and variable definitions (3); justified cleaning, missing-data handling, and dependence assessment (4). |
| Linear model and interpretation | 25% | Correct mean model and predictor coding (5); correct interpretations of at least two coefficients or contrasts and the intercept, including units and adjustment (12); correct 95% confidence intervals and uncertainty discussion (8). |
| Model comparison and diagnostics | 20% | Valid nested-model restriction and test, with assumptions addressed (6); explanation of retained coefficients across models (4); informative residual, Q-Q, and influence checks with justified responses to problems (10). |
| Machine learning method | 10% | Appropriate regression method and accurate explanation (4); documented implementation, preprocessing, and final settings (6). |
| Fair predictive evaluation | 15% | Appropriate common test set and no data leakage (6); documented training-only validation and tuning (4); correct baseline, test RMSE/MAE, and prediction plots for both models (5). |

| Criterion | Weight | Full-credit evidence (points) |
|---|---|---|
| Findings and critical discussion | 10% | Contextual conclusions about effect size and uncertainty (4); reasoned prediction comparison and model recommendation (3); meaningful limitations and next step (3). |
| Communication and reproducibility | 10% | Organized report and readable, discussed figures and tables (3); runnable source that reproduces results with documented data, seed, and environment (5); citations and each member's contribution (2). |
| **Total** | **100%** | **100 points** |

Subitem points appear in parentheses. Partial credit reflects accuracy, completeness, and justification; omissions earn zero. Full credit does not require machine learning to outperform linear regression.

## How to get started

1. **Choose a question and data.** Identify a continuous outcome, its units, the observational unit, and plausible predictors. Cite the original data source.
2. **Plan the comparison.** Choose one machine learning method for regression, such as a regression tree, random forest, gradient boosting, or support vector regression. Decide how to reserve test observations before fitting or tuning models.
3. **Analyze and explain.** Fit and diagnose the linear regression, interpret its coefficients and uncertainty, and compare a scientifically motivated pair of nested linear models.
4. **Evaluate and communicate.** Compare both methods on the same held-out observations using RMSE and MAE. Explain what each model reveals, where it performs poorly, and which you would use for your stated purpose.

## A few possible questions

- How is a property's sale price associated with floor area and other recorded characteristics, and does a random forest improve prediction?
- How is a building's energy use associated with its characteristics, and does gradient boosting capture patterns missed by the linear model?
- How is a measured environmental outcome associated with recorded conditions, and how does a regression tree compare with linear regression?

These are examples of scope; you may choose another application and another suitable regression algorithm.

> **i** What makes a strong project?
>
> A strong project gives a defensible linear regression analysis, clear interpretations, a fair prediction comparison, and reproducible evidence. A machine learning model does **not** have to outperform linear regression. Explain the result you obtain, including its limitations.

> Use existing software to fit the models, and explain the method, important settings, and analysis decisions. Record **8561** as the seed for randomized procedures. In R, use `set.seed(8561)` and prefer **ggplot2** for figures.

**Next:** Report requirements, assessment criteria, and submission.

# Report

**STAT 8561 | Fall 2026**
**Due on December 13, 2026**
**Course-grade weight:** 20% (see Grade Distribution)
**Points:** 100
**MS students:** 1-4 students per group
**PhD students:** Complete the project individually

Use a real dataset to explain an outcome with **multiple linear regression**, and compare its predictive performance with **one machine learning regression algorithm of your choice**. Begin with the project overview.

## Research question and data

Choose a continuous response and at least two scientifically meaningful candidate predictors. State what one row represents, the response's units, the population or setting of interest, and the question your analysis will answer. Choose a dataset large enough to support the proposed models and a meaningful evaluation on held-out observations.

- Link to and cite the original data source. Describe how the observations were collected, the sample size, and the variable definitions.
- Explain exclusions, missing values, unusual observations, and any transformations. Give the number of observations retained for analysis.
- Distinguish variables available when a prediction would be made from information recorded afterward. Use predictors that would actually be available in the intended application.
- Explain whether observations are independent. If rows share a subject, site, or time sequence, use a splitting strategy that respects that structure and state the implications for inference.

Before choosing models, write one question about **association or explanation** and one about **prediction**. For example, an adjusted association between size and price is a different question from predicting the price of a new property.

## Linear regression analysis

Use the ideas from least squares estimation, OLS inference, model comparison, and multiple regression and categorical predictors.

1. **Specify the model.** Write the mean model, define its variables and units, and explain why the predictors belong in the analysis. State reference categories and justify any interactions or transformed terms.

2. **Explain the coefficients.** Interpret at least two scientifically relevant coefficients or contrasts in context, stating the predictor change, response units, and other variables held fixed. Explain the intercept's baseline and whether it is meaningful for the observed data. With an interaction, make clear which coefficient interpretation depends on the other variable's value.
3. **Report uncertainty.** Give estimates and 95% confidence intervals for the coefficients or contrasts you interpret. State the assumptions supporting the inference. Distinguish an interval for a model parameter from a prediction interval for an individual outcome. Do not equate a small p-value with a large practical effect.
4. **Compare nested models.** Compare one reduced and one full linear model on the same training observations and response scale. State the restriction, such as an added coefficient or group of coefficients being zero. Use an appropriate partial F test when its assumptions are defensible; otherwise explain the limitation of that test. Explain whether a retained coefficient's meaning or estimate changes when the adjustment set changes.
5. **Check the model.** Use residual-versus-fitted and normal Q-Q plots, and examine influential observations. Discuss nonlinearity, unequal variance, unusual observations, and the plausibility of independence. Justify any remedial action; do not remove an observation solely to improve a result. See diagnostics and model adequacy (not included in this edition).

Use the training data for these modeling decisions. If you select terms or transformations after examining the data, report that process and treat ordinary confidence intervals and p-values as exploratory: standard fixed-model formulas do not account for data-driven selection. Treat findings from observational data as associations unless the study design justifies a causal interpretation.

## Machine learning comparison

Choose **one** method suited to predicting a continuous response, for example a regression tree, random forest, gradient boosting, or support vector regression. Explain how it produces predictions, why it is appropriate for your data, and which important tuning parameters you choose. Cite the method and software documentation.

Existing implementations are appropriate. Identify the functions, software versions, preprocessing, and settings needed to reproduce the fit. If you report a feature-importance measure or an effect plot, explain what it measures; it is not automatically an adjusted regression coefficient or a causal effect.

### A fair evaluation

1. **Reserve a test set first.** State the split and its rationale. For independent observations, an 80% training / 20% test split is a reasonable starting point. Use a group-aware or chronological split when appropriate. Conduct model-building exploration on the training data.
2. **Use the same comparison.** Both methods predict the same response for the same test observations and use the same available candidate predictors. Explain any model-specific encoding, transformations, or training-based variable selection.
3. **Keep learning within training data.** Estimate imputation, scaling, encoding rules, and other learned preprocessing using training data. If you use cross-validation to select a model

or tune parameters, repeat all learned preprocessing inside each training fold. Apply the fitted transformations to validation or test observations.

4. **Tune without the test outcomes.** Choose linear-model terms and machine learning settings using training data, with cross-validation or a separate validation subset as appropriate. Report the candidate settings, validation design, and selection metric. Finalize the models before evaluating them on the test set; do not revise them after seeing test performance.
5. **Report the comparison.** Evaluate a training-mean baseline, the final linear regression, and the machine learning model on the same test set. Report RMSE and MAE in the original response units. Explain how any response transformation is reversed and how any correction is estimated using training data.

For $m$ test observations, calculate

$$\text{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}, \qquad \text{MAE} = \frac{1}{m}\sum_{i=1}^{m}|y_i - \hat{y}_i|.$$

The baseline predicts the **training-set mean** for every test observation. Smaller RMSE and MAE indicate smaller prediction errors. Training $R^2$ alone does not establish which model predicts new observations better. See model assessment and prediction (not included in this edition) and the tidymodels resampling guide.

Use this format for the main results table; fill it with your own results:

| Model | Test RMSE | Test MAE |
|---|---|---|
| Training-mean baseline | Your result | Your result |
| Multiple linear regression | Your result | Your result |
| Chosen machine learning method | Your result | Your result |

Include an observed-versus-predicted plot for both fitted models and discuss where errors are largest. Differences on one test set are estimates, not proof of universal superiority; discuss how the sample size, split, and application limit the comparison.

For implementation guidance, consult the scikit-learn guide to avoiding data leakage and the official ensemble-methods documentation when relevant to your chosen method.

## Explain the findings

Your conclusion should answer the research question in ordinary language and address all of the following:

- **Substantive findings:** Which adjusted relationships are supported, how large are they in meaningful units, and how uncertain are they?
- **Model comparison:** Did the machine learning method improve test prediction, and is the difference useful in this application? Explain why a flexible model might or might not improve performance.
- **Interpretation:** What does the linear model make easy to explain? What additional patterns, if any, does the machine learning analysis suggest? Distinguish coefficient estimates from prediction-based importance measures.
- **Recommendation and limitations:** Which model would you use for the stated purpose, and why? Discuss assumptions, data quality, the scope of generalization, and one useful next step.

## Report format and reproducibility

Write **8-10 pages of main text**, excluding the title page, references, and appendices. Use Quarto or R Markdown to produce a PDF with readable 11- or 12-point text, standard margins, and numbered sections. For a Python or Jupyter Notebook workflow, follow the prior-discussion requirement in the submission note.

Organize the report as follows:

1. **Title and summary:** Names, course, project title, research question, and a short statement of the main finding.
2. **Data and study design:** Source, definitions, cleaning, training-data exploration, and evaluation plan.

3. **Linear regression analysis:** Model formulation, interpreted estimates and intervals, nested-model comparison, diagnostics, and limitations.
4. **Machine learning method:** Main idea, preprocessing, tuning procedure, and final settings.
5. **Prediction comparison:** Test-set results table, plots, and interpretation.
6. **Discussion and conclusion:** Substantive findings, model recommendation, uncertainty, and limitations.
7. **References and contributions:** Data, methods, software, and adapted code, plus a brief statement of each author's contribution.

Label and caption every figure and table, and discuss it in the text. Place long code listings and supplementary output in an appendix or supporting files. Include all code needed to reproduce the reported numbers and figures.

Set the seed to **8561** for randomized splitting, resampling, and model fitting; in Python, set the relevant `random_state` or generator seed to `8561`. Save the split identifiers and record the software environment. Prefer **ggplot2** for R figures. Use relative paths and provide instructions that work from a fresh session.

# Assessment criteria

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| **Total** | **100%** | **100 points** |

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# Submission

> Report submission checklist

**MS students** work in groups of **1-4 students**. **PhD students** must complete the project **individually**. Upload the following to the **iCollege project submission area** by **December 13, 2026**. The submission folder will be announced on iCollege:

- `Project_Report.pdf`.
- Its matching `Project_Report.qmd` or `Project_Report.Rmd` source (the default format; see the Python/Jupyter Notebook note below).

If the source depends on supporting files, include them with the source in a ZIP. Provide any required scripts, data that can be redistributed or precise data-access instructions, split identifiers, and software versions and rendering instructions.
Put all author names and **STAT 8561** on the report. The PDF must be generated from the submitted source file; **HTML alone does not satisfy the submission requirement**. Check that the submitted files reproduce the figures, tables, and model-comparison results from beginning to end.

> Python and Jupyter Notebook submissions

If you would like to use **Python** or submit a **Jupyter Notebook (`.ipynb`)**, discuss your plan with the instructor **in advance** and agree on the submission format. Python/Jupyter Notebook submissions are subject to that discussion; they are not an automatic alternative to the default `.qmd`/`.Rmd` requirement.
Any agreed alternative must still include a **PDF report**, the **reproducible source** (including the notebook if applicable), and the supporting files needed to regenerate the analysis. The same report requirements and grading criteria apply.

**Related:** Project Instruction.